

Algebra I



Lecture Notes & Transcription

Course: **Algebra I**

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Wednesday 16 September 2026

Gaussian Integers, Ring Axioms, Ring Homomorphisms, and Ideals

Course Content Overview:

Gaussian Integers, Ring Axioms, Homomorphisms & Ideals

In this lecture, we bridge arithmetic and abstract algebra by exploring the following core themes:

- **Gaussian Integers and Primes:** Comparing factorization and units in $\mathbb{Z}$ and $\mathbb{Z}[i]$, and establishing how the primality of an ordinary integer prime in $\mathbb{Z}[i]$ connects directly to Fermat's two-square theorem ($p = a^2 + b^2 \iff p \equiv 1 \pmod{4}$).
- **Axioms and Elementary Properties of Rings:** Formalizing the ring axioms $(R, +, \cdot, 0, 1)$, deriving fundamental arithmetic identities such as $0 \cdot a = 0$ and $(-1) \cdot a = -a$, analyzing the degenerate zero ring ($1 = 0$), and defining scalar multiples by integers.
- **Ring Homomorphisms:** Defining structure-preserving maps between rings, explaining why identity preservation ($f(1) = 1$) must be explicitly required while $f(0) = 0$ is automatic, and exploring canonical examples including standard subring inclusions, modular reduction $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, and the initial evaluation map $\mathbb{Z} \rightarrow R$.
- **Ideals and Field Structure:** Introducing the kernel $\ker(f)$, defining ideals $I \trianglelefteq R$ via additive closure and multiplicative absorption, proving that every kernel is an ideal, and classifying the ideals of a field (establishing that fields possess only $\{0\}$ and F).
- **Outlook:** Connecting the existence of bases in infinite-dimensional vector spaces to Zorn's lemma and the Axiom of Choice.

III Gaussian Integers and Primes

III Units and Primes in $\mathbb{Z}$ and $\mathbb{Z}[i]$

Spoken Transcript [00:00:00 - 00:00:53]

Okay, so we'll try to start. Maybe you could sit down.

So, on the practical aspects, there was some progress. The website is more or less

correct now, I think, so you can look for the information on the website.

The second issue is that people asked, and now the lectures will be recorded. You can see yourself, or maybe you can only see me, but anyway — the lectures will be recorded, so you can find them later on some website somewhere.

At the beginning I said that the website is more or less in reasonable form; in particular, the assignment for September 25th is there. Some of the technical parts we're still working on, so we'll get back to you on that, but...

Administrative Opening

The lecturer makes brief administrative announcements regarding the course website, the availability of the first homework assignment due on September 25th, and lecture video recordings.

Spoken Transcript [00:00:53 - 00:02:48]

So, if you spent last night thinking about the Gaussian integers, then... well, maybe for the people who didn't spend the whole night thinking about the Gaussian integers, I'll say a few things here.

As I said in the previous lecture, we discussed $\mathbb{Z}$ and its factorization properties. And I said you can discuss the same thing for the Gaussian integers. It will be part of the theory of the course, but it's nice to think about the example.

So, the first thing is: for $\mathbb{Z}$, there were the units, that's ± 1 . And those confused our multiplication always, because you could always put in as many -1 's and $+1$'s as you want.

So, what's the analogue here? What are the units, what are the invertible elements in here? Well, of course, there's still ± 1 , but there's also $\pm i$. These are also invertible, because $i^2 = -1$ and then you can multiply it by 1 . Is it clear that these four objects play in some sense the role of the two there? Is that clear?

So, these are the units. It means if you write down any multiplication, like any unique factorization into primes, before you had to worry about ± 1 ; now you have to worry about more, okay?

In particular, the notion of prime here was that p is prime if the factors are ± 1 and $\pm p$. That's on the $\mathbb{Z}$ side. Here, if we have some element — maybe I call it q — is prime, then we can consider factorizations here, but now there's a lot more: there's four choices, there could be any one of these. ± 1 and $\pm i$ are factors, and also $\pm q$ and $\pm iq$; these are all factors. That's the analogous statement.

This is all some kind of nonsense. It's just saying that, you know, this is more confusion with these extra i 's.

Units and Primes in the Gaussian Integers

 **Definition 1.1** (*Gaussian Integers*): The ring of “*Gaussian integers*”, denoted by $\mathbb{Z}[i]$,

is the subring of the complex numbers $\mathbb{C}$ given by:

$$\mathbb{Z}[i] := \{a + bi \in \mathbb{C} \mid a, b \in \mathbb{Z}\}, \quad \text{where } i^2 = -1.$$

Comparison of Units and Prime Factorizations:

- **In the Integers $\mathbb{Z}$:**

- **Units (Invertible Elements):** $\mathbb{Z}^\times = \{1, -1\} = \{\pm 1\}$.
- **Prime Elements:** An integer $p \in \mathbb{Z} \setminus \{0, \pm 1\}$ is prime if its only divisors are the trivial factors (the units and its associates):

$$\text{Divisors of } p = \{\pm 1, \pm p\}.$$

- **In the Gaussian Integers $\mathbb{Z}[i]$:**

- **Units (Invertible Elements):** $\mathbb{Z}[i]^\times = \{1, -1, i, -i\} = \{\pm 1, \pm i\}$.
- **Prime Elements:** An element $q \in \mathbb{Z}[i]$ is prime if its only divisors are units or associates of q :

$$\text{Divisors of } q = \{\pm 1, \pm i, \pm q, \pm iq\}.$$

≡ **Algebraic Intuition: Units and Associates:** In a unique factorization domain (such as $\mathbb{Z}$ or $\mathbb{Z}[i]$), factorizations into irreducible elements are unique up to multiplication by invertible elements (“units”) and order. In $\mathbb{Z}$, the only units are 1 and -1 , meaning every prime p has two associates: p and $-p$. In $\mathbb{Z}[i]$, since $i \cdot (-i) = 1$, the imaginary unit i and $-i$ are also units. Consequently, every Gaussian integer q has four associates:

$$q, \quad -q, \quad iq, \quad -iq.$$

III When Does a Rational Prime Remain Prime in $\mathbb{Z}[i]$?

Spoken Transcript [00:02:48 - 00:04:25]

So, let’s do something that’s not nonsense. A math question you can ask — which is a very nice question — is: if I take a prime, an actual prime in the integers, and I view that prime, that p , as an element of the Gaussian integers, which it is... Because remember, the Gaussian integers are the set of integers plus i times an integer. So, the integers naturally sit inside the Gaussian integers by just an integer going here. Is that clear?

So, if I take a prime, an honest prime in the integers — so this would be, you know, like 2, 3, 5, 7, 11, 13, and so on — take one of them, and I consider it in the Gaussian integers, does it stay prime? Or maybe say: when does p stay prime? Meaning it’s born prime in the integers: when does it stay prime when I consider it as a Gaussian integer? That’s a real question, and we had it in an example in the class last time.

But now we try to think about it mathematically. So, we have p , and we have to consider whether it’s divisible in the Gaussian integers. So, we think: okay, maybe it’s divisible. How do I do that? Well, that means $a + bi$ divides p in the Gaussian integers. Let’s try to think what that means.

Embedding of $\mathbb{Z}$ and the Prime Divisibility Question

The ring of integers embeds naturally into the Gaussian integers via the inclusion map:

$$\mathbb{Z} \hookrightarrow \mathbb{Z}[i], \quad n \mapsto n + 0i.$$

Core Question: Let $p \in \{2, 3, 5, 7, 11, 13, \dots\}$ be an ordinary prime in $\mathbb{Z}$. When viewed as an element of $\mathbb{Z}[i]$, does p remain prime (irreducible)?

To investigate whether p factors non-trivially in $\mathbb{Z}[i]$, suppose there exists a non-unit divisor:

$$(a + bi) \mid p \quad \text{in } \mathbb{Z}[i] \quad (a, b \in \mathbb{Z}). \quad (1.1)$$

III Complex Conjugation and the Sum of Two Squares

Spoken Transcript [00:04:25 - 00:05:18]

Now, if $a + bi$ divides p , that means $a + bi$ times something equals p ; that's the definition of divisibility.

So, if that happened, I could just complex conjugate the whole thing. If this happens — you have to think about this —, then $a - bi$ also divides p .

Do you believe that statement? This has to do with complex conjugation, and it depends on what your feeling is for the complex numbers. If you don't understand that statement, you should think about it, because what does this mean? This means p is equal to something something something times $a + bi$; that's what divisibility means.

And now I just take the complex conjugate of everything. But the complex conjugate of p is p , and all the something something something gets... I don't care, but at the end, this one becomes that. So, that means that that also happens, right?

Complex Conjugation of Divisors

 Proposition 1.2 (Conjugate Divisor Property): Let $p \in \mathbb{Z}$. If $(a + bi) \mid p$ in $\mathbb{Z}[i]$, then $(a - bi) \mid p$ in $\mathbb{Z}[i]$.

Proof

By definition of divisibility in $\mathbb{Z}[i]$, $(a + bi) \mid p$ means there exists an element $z \in \mathbb{Z}[i]$ such that:

$$p = z \cdot (a + bi).$$

Applying the complex conjugation automorphism to both sides yields:

$$\bar{p} = \overline{z \cdot (a + bi)} = \bar{z} \cdot \overline{(a + bi)}.$$

Since $p \in \mathbb{Z}$ is real, $\bar{p} = p$. Furthermore, $\overline{a + bi} = a - bi$, and $\bar{z} \in \mathbb{Z}[i]$. Thus,

$$p = \bar{z} \cdot (a - bi),$$

which proves that $(a - bi) \mid p$ in $\mathbb{Z}[i]$.

Q.E.D.

Spoken Transcript [00:05:18 - 00:06:08]

So, then you might think then — and this is the kind of thing we will make precise in this class — that actually then both of them divide. That already requires some statement about unique factorization.

And then this is now a nice integer: $a^2 + b^2$ divides p . And if it's an integer, then p has to be of the form $a^2 + b^2$. So, it has to be a prime that's a sum of two squares.

So, this argument will show that if you want p to stay prime... so if p factors, then it's a sum of two squares. And this now connects to this world of trying to express primes as a sum of two squares, and there's facts about this.

Reduction to the Sum of Two Squares

Multiplying the two conjugate factors together yields an ordinary integer:

$$(a + bi)(a - bi) = a^2 + b^2 \in \mathbb{Z}.$$

Using the field norm $N(x + yi) = x^2 + y^2 = (x + yi)(x - yi)$, the equation $p = (a + bi)z$ implies:

$$N(p) = p^2 = N(a + bi) \cdot N(z) = (a^2 + b^2) \cdot N(z).$$

If $a + bi$ is neither a unit (which would require $a^2 + b^2 = 1$) nor an associate of p (which would require $a^2 + b^2 = p^2$), then because p is a prime in $\mathbb{Z}$, the only possible non-trivial integer divisor of p^2 is p itself:

$$a^2 + b^2 = p. \tag{1.2}$$

≡ Connection Between Gaussian Primes and Sums of Squares: This algebraic deduction establishes a profound bridge:

- If an integer prime p factors non-trivially in $\mathbb{Z}[i]$ as $p = (a + bi)(a - bi)$, then p must be expressible as the sum of two integer squares: $p = a^2 + b^2$.
- By contrapositive, if p cannot be expressed as the sum of two squares, it cannot factor non-trivially into non-unit Gaussian integers, and hence p must remain prime in $\mathbb{Z}[i]$.

III Fermat's Theorem on Sums of Two Squares

Spoken Transcript [00:06:08 - 00:08:12]

The simple fact is that if $p \equiv 3 \pmod{4}$, this is impossible. You can never write a prime that's $3 \pmod{4}$ as a sum of two squares. That's easy to prove; I think it's part of the exercise, I'm not sure.

The one that's tricky is: if $p \equiv 1 \pmod{4}$, then you can. You can solve $p = a^2 + b^2$ in the integers. This is a theorem that's due to Fermat, and this is one of the problems on the problem set.

And so the answer to this question, finally, is that if you have a prime and you want to know whether it stays prime in the Gaussian integers, the whole thing that matters is

whether p is equal to 1 or 3 (mod 4). So, it's kind of an interesting question.

And as I said, we will return to these matters in the class, but it already shows us how the very naive idea of thinking about how things factor leads to interesting mathematics, and in this case, it goes in a number-theoretic direction. That's this question about whether a prime can be expressed as a sum of two squares — really a basic question.

We had an example yesterday, I remind you; the answer yesterday had to do with $5 = 2^2 + 1^2$. So, 5 is a prime in the integers, $5 \equiv 1 \pmod{4}$, and you can express it as a sum of two squares.

So, the thing that's tricky on the board here is this statement: that if $p \equiv 1 \pmod{4}$, then you can solve this equation. And that is the exercise on the homework, on the first homework set. There's a way to go through that, a very clever way to go through that — not the way that Fermat used.

Fermat's Two-Square Theorem and Gaussian Primes

 **Theorem 1.3 (Fermat's Two-Square Theorem):** Let p be an odd integer prime ($p > 2$).

- (i) If $p \equiv 3 \pmod{4}$, then p cannot be written as the sum of two squares ($p \neq a^2 + b^2$). Consequently, p remains prime in $\mathbb{Z}[i]$.
- (ii) If $p \equiv 1 \pmod{4}$, then p can be written uniquely (up to sign and order) as the sum of two squares:

$$p = a^2 + b^2 = (a + bi)(a - bi) \quad (a, b \in \mathbb{Z}).$$

Consequently, p factors in $\mathbb{Z}[i]$ and does not remain prime.

 **Example 1.4 (Decomposition of 5 vs. Primality of 3 in $\mathbb{Z}[i]$):**

- **The prime 5:** Since $5 \equiv 1 \pmod{4}$, we have:

$$5 = 2^2 + 1^2 = (2 + i)(2 - i).$$

Thus, 5 is composite in $\mathbb{Z}[i]$.

- **The prime 3:** Since $3 \equiv 3 \pmod{4}$, and squares modulo 4 can only be 0 or 1 ($0^2 \equiv 0$, $1^2 \equiv 1$, $2^2 \equiv 0$, $3^2 \equiv 1$), the sum of two squares can only be 0, 1, or 2 (mod 4). Hence, $a^2 + b^2 \equiv 3 \pmod{4}$ has no integer solutions, and 3 remains prime in $\mathbb{Z}[i]$.

Board Transition

The lecturer erases the center and right chalkboards with a sponge to prepare for reviewing the abstract definition and axioms of a ring.

||| The Axioms and Basic Properties of Rings

||| Definition of a Ring

Spoken Transcript [00:08:15 - 00:09:45]

All right, so then the main direction of the lecture last time was about rings, which we revisit now.

Okay, so we go back to rings and the foundational material. I had defined a ring R with multiplication, 0 , and 1 . This is a ring, and the axioms, if we rewrite them now differently: there's the axioms for $+$, and this axiom for $+$ we can say just that $(R, +, 0)$ is an abelian group. This is just the same axioms written differently; those are just the group axioms. Is that clear?

So, then there's the axiom for multiplication, and these were — now I don't have a more efficient way to say this — it's associative, and then there's the axiom of the unit: $1 \cdot a = a \cdot 1 = a$.

And then there's the axioms together, and these are two times the distributive law, which I don't write, okay?

So, this is just a definition which we've seen and you've seen before, and the only thing I've done now is abbreviated the $+$ by saying that if I look at the additive structure, it's an abelian group, because that's what the axioms were.

Axiomatic Definition of a Ring

 **Definition 1.5 (Ring):** A “ring” is a tuple $(R, +, \cdot, 0, 1)$ consisting of a set R , two binary operations $+$ (addition) and $\cdot$ (multiplication), and two distinguished elements $0, 1 \in R$ satisfying the following axioms:

- (1) **Additive Abelian Group Axioms:** $(R, +, 0)$ is an abelian group:
 - **Associativity:** $\forall a, b, c \in R: (a + b) + c = a + (b + c)$.
 - **Identity:** $\forall a \in R: a + 0 = 0 + a = a$.
 - **Inverse:** $\forall a \in R \exists (-a) \in R: a + (-a) = (-a) + a = 0$.
 - **Commutativity:** $\forall a, b \in R: a + b = b + a$.
- (2) **Multiplicative Monoid Axioms:** Multiplication is associative and possesses a unit element 1 :
 - **Associativity:** $\forall a, b, c \in R: (a \cdot b) \cdot c = a \cdot (b \cdot c)$.
 - **Identity (Unit):** $\forall a \in R: 1 \cdot a = a \cdot 1 = a$.
- (3) **Distributivity Axioms:** Multiplication distributes over addition from both the left and the right:
 - **Left Distributivity:** $\forall a, b, c \in R: a \cdot (b + c) = (a \cdot b) + (a \cdot c)$.
 - **Right Distributivity:** $\forall a, b, c \in R: (a + b) \cdot c = (a \cdot c) + (b \cdot c)$.

 **Structural Synthesis of the Ring Axioms:** Summarizing the additive structure compactly as an abelian group $(R, +, 0)$ packages four standard axioms (associativity, commutativity, neutral element, and inverses) into a single standard algebraic concept. The distributive laws act as the essential bridge coupling the additive abelian group structure with the multiplicative monoid structure.

III Elementary Property: $0 \cdot a = 0$

Spoken Transcript [00:09:45 - 00:10:40]

So, then there's a bunch of things to say at the beginning — some basic properties, and these are all in the books. But in these axioms, there are lots of things that you're familiar with with the integers and arithmetic that are not in the axioms. Like, one question is: what happens if I take $0 \cdot a$? Because 0 is an object that lives on the additive side, and I'm multiplying it, so it's on the multiplicative side. You can wonder what this is, and you can imagine what the answer is going to be.

If we're going to formally prove it, you of course have to use the distributive law. The distributive law is the only thing that relates the additive to the multiplication. So, we're going to try to prove this, and to prove this, we know to start we have to use the distributive law somewhere.

Spoken Transcript [00:10:40 - 00:11:31]

So, one way to start is we can write this as $(0 + 0) \cdot a$. We have not used the distributive law yet; this uses the fact that $0 + 0 = 0$, which is the neutral part of the group law.

And now we use the distributive law: $(0 \cdot a) + (0 \cdot a)$.

Okay, and now we're happy, because we have this equation, and then we go back to using the additive properties, which is the left cancellation, the additive inverse, okay?

So, now we can cancel, and we happily see that $0 = 0 \cdot a$, which is, of course, a basic law of arithmetic. It's not an axiom, but it follows immediately from the axioms, which is why it's not an axiom. You could actually just put it as an axiom, it wouldn't hurt anybody, but... okay.

Derivation of Multiplication by Zero

 **Proposition 1.6 (Multiplication by Zero):** In any ring R , for every element $a \in R$:

$$0 \cdot a = 0 \quad \text{and} \quad a \cdot 0 = 0. \quad (1.3)$$

Proof

We prove $0 \cdot a = 0$ using the additive identity property and distributivity:

$$\begin{aligned} 0 \cdot a &= (0 + 0) \cdot a && \text{(since } 0 + 0 = 0 \text{ in } (R, +)) \\ &= (0 \cdot a) + (0 \cdot a) && \text{(by right distributivity).} \end{aligned} \quad (1.4)$$

Since $(R, +, 0)$ is an abelian group, the element $0 \cdot a \in R$ possesses a unique additive inverse $-(0 \cdot a) \in R$. Adding this inverse to both sides of (1.4):

$$\begin{aligned} -(0 \cdot a) + (0 \cdot a) &= -(0 \cdot a) + ((0 \cdot a) + (0 \cdot a)) \\ 0 &= (-(0 \cdot a) + (0 \cdot a)) + (0 \cdot a) && \text{(by associativity)} \\ 0 &= 0 + (0 \cdot a) \\ 0 &= 0 \cdot a. \end{aligned}$$

The proof for $a \cdot 0 = 0$ follows identically using left distributivity.

Q.E.D.

≡ **Why Distributivity is Indispensable:** The zero element 0 is defined exclusively through addition ($a + 0 = a$). To deduce its behavior under multiplication ($0 \cdot a = 0$), one must invoke the distributive law, as it provides the only axiomatic connection linking addition to multiplication.

III The Zero Ring ($1 = 0$)

Spoken Transcript [00:11:31 - 00:12:45]

In particular, if $1 = 0$ — because I said for a ring we allow this extremely unpleasant boundary case where $1 = 0$, degenerate case — if that happens, so if we have a ring where $1 = 0$, then we have a , for any a , is equal to $a \cdot 1$, which was the multiplication axiom. And then if we use $1 = 0$, then this is equal to $a \cdot 0$, and then this is 0 .

So, it's a rather degenerate case. If you have a ring where $1 = 0$, then all elements are equal to 0 . So, it's not a super exciting ring; there's only one element.

If I say this formally: if you have $0 = 1$, this implies your ring is a set consisting of just one element: $R = \{0\}$.

Anyway, it's a degenerate case. It's unpleasant to think about, but we carry it around because it fits well in certain statements of theorems.

The Degenerate Zero Ring

💎 **Proposition 1.7 (Characterization of the Zero Ring):** Let R be a ring. If $1 = 0$, then R is the trivial ring containing only a single element:

$$R = \{0\}.$$

Proof

Let $a \in R$ be an arbitrary element. Using the unit axiom ($a = a \cdot 1$), the hypothesis $1 = 0$, and [Proposition 1.6](#):

$$a = a \cdot 1 = a \cdot 0 = 0.$$

Since every element $a \in R$ is equal to 0 , it follows that $R = \{0\}$.

Q.E.D.

📌 **Remark 1.8 (The Zero Ring in Abstract Algebra):** The trivial ring $R = \{0\}$ is often called the “*zero ring*”. While fields strictly require $1 \neq 0$, the general definition of a ring intentionally admits $1 = 0$ so that the category of rings has a terminal object (the zero ring) and quotient rings R/I remain rings even when $I = R$.

III Additive Inverses and the Element $-a$

Spoken Transcript [00:12:50 - 00:15:05]

Other things that are kind of convention is that if you take -1 ... So, 1 is in the ring because it's part of the data. Because of the additive structure, the group axioms say that there exists an additive inverse. So, by convention, -1 is by definition the additive inverse of 1 . And you can prove that's unique — maybe I leave that as an exercise. Given the ring, there's 1 and there's an additive inverse. The additive inverses are unique, you can prove that. So, that's -1 .

And then $-a$, for any element, is defined to be $(-1) \cdot a$. It's not a surprising definition.

And then the little lemma — maybe it doesn't even get to be called a lemma — is that $-a$ is the additive inverse of a , okay? That's very convenient; it would be really unpleasant if that weren't the case.

So, the proof of this is we should start with $0 = (1 + (-1)) \cdot a$. That's 0 . And we've already proven that that's that, okay? I've used two steps here. So, here on this side, this thing is 0 , and we've already proved $0 \cdot a = 0$, so... okay?

And then I use the distributive law, and distribute this side as a , and on this side as $+(-1) \cdot a$. And that exhibits $(-1) \cdot a$ as the additive inverse of a .

So, there's a number of things. I think we spent enough time on this, but the way the axioms work, they work so that it fits with your intuition about how multiplication works.

So, are you happy with this? You can also look in the book; there'll be a couple of pages of playing around with these. It's not deep, but it's good to be familiar with it.

Additive Inverses and Multiplication by -1

 **Definition 1.9** (*The Element -1 and the Notation $-a$*): Let R be a ring.

- (i) We denote by $-1 \in R$ the unique additive inverse of the multiplicative identity $1 \in R$, so that:

$$1 + (-1) = 0.$$

- (ii) For any element $a \in R$, we define:

$$-a := (-1) \cdot a. \quad (1.5)$$

 **Lemma 1.10** (*$-a$ is the Additive Inverse of a*): For any element $a \in R$, the element $-a = (-1) \cdot a$ is the additive inverse of a , that is:

$$a + (-a) = 0.$$

Proof

Using [Proposition 1.6](#) ($0 \cdot a = 0$), [Definition 1.9](#) ($1 + (-1) = 0$), and distributivity:

$$\begin{aligned} 0 &= 0 \cdot a \\ &= (1 + (-1)) \cdot a \\ &= (1 \cdot a) + ((-1) \cdot a) && \text{(by distributivity)} \\ &= a + (-a) && \text{(since } 1 \cdot a = a \text{ and } (-1) \cdot a = -a). \end{aligned}$$

Since additive inverses in an abelian group are unique, this confirms that $-a = (-1) \cdot a$ is indeed the unique additive inverse of a . Q.E.D.

Integer Multiples and the Canonical Map $\mathbb{Z} \rightarrow R$

Spoken Transcript [00:15:05 - 00:17:10]

Okay, so another convention about a ring is: as I said, if I have a as an element of the ring, we just write $-a$, and by definition $-a$ is just $(-1) \cdot a$, and it's a canonical element of the ring.

There's something else: we can also look at $a + a$, and this is defined as $2a$. You can take $a + a + a$, n times. This is how we define this term na , that's what we defined it to be, okay?

So, in any ring, if I have an element a in a ring, I can take 17 times that element; that is an element of the ring, and I just mean you add a to itself 17 times. It's well-defined, just using the ring structure. You have no objections?

And I can also do -17 times, because I just either add $-a$ to itself 17 times, or I add a to itself 17 times and take the minus, and they're the same, all right?

So, in particular, there's a map from $\mathbb{Z}$ to the ring. And this map requires no choices: this map sends 0 to 0, sends 1 to 1, sends 2 to $1 + 1$, sends 3 to $1 + 1 + 1$, whatever, okay? And it sends -1 to the additive inverse.

So, if you give me a ring, I know how to map $\mathbb{Z}$ to that ring. There's nothing going on with this map; it's just I map 1 to 1, and then everything else I just add up.

Scalar Multiplication by Integers and the Canonical Map $\mathbb{Z} \rightarrow R$

 **Definition 1.11 (Multiples of Ring Elements by Integers):** Let R be a ring and $a \in R$. For any integer $n \in \mathbb{Z}$, the element $n \cdot a \in R$ is defined inductively by:

$$\begin{aligned} 0 \cdot a &:= 0_R, \\ (n+1) \cdot a &:= (n \cdot a) + a \quad \text{for } n \geq 0, \\ (-n) \cdot a &:= -(n \cdot a) = n \cdot (-a) \quad \text{for } n > 0. \end{aligned}$$

In particular, for $n \in \mathbb{N}_{>0}$:

$$n \cdot a = \underbrace{a + a + \cdots + a}_{n \text{ times}}.$$

 **Definition 1.12 (Canonical Map from $\mathbb{Z}$ into a Ring):** For every ring R , there exists a canonical map $\phi: \mathbb{Z} \rightarrow R$ defined by evaluating integer multiples on the multiplicative unit 1_R :

$$\phi(n) := n \cdot 1_R.$$

Explicitly, the map sends:

$$\begin{aligned} 0 &\mapsto 0_R, \\ 1 &\mapsto 1_R, \\ 2 &\mapsto 1_R + 1_R, \\ 3 &\mapsto 1_R + 1_R + 1_R, \\ -1 &\mapsto -1_R, \quad \dots \end{aligned}$$

≡ **Universal Initial Property of the Integers $\mathbb{Z}$:** This map $\phi: \mathbb{Z} \rightarrow R$ requires absolutely no choices: it is entirely determined by the ring structure of R . As we will see, this map is not merely a set-theoretic function, but the unique ring homomorphism from $\mathbb{Z}$ to R , establishing that $\mathbb{Z}$ is the “*initial object*” in the category of rings with unit.

Ring Homomorphisms

Standing Convention: Commutativity and Units

Spoken Transcript [00:17:10 - 00:18:44]

This is not just any map. It turns out this is what’s called a ring homomorphism. So, we have to define this.

So, let R_1 and R_2 be two rings. And now maybe I say a little bit more about these rings.

For this class, as I said, if I consider totally abstract rings — like totally arbitrary rings here — I have not assumed the multiplication is commutative. And for a lot of things, we don’t, and sometimes we do. But when I say rings in this class from now on, I will mean this ring that has commutative multiplication, and it always has a unit, 1.

So, if you look in the literature about what the definition of a ring is, of course, a ring does not require the multiplication to be commutative, and that’s reasonable. So, if you get off the plane in Tokyo and you meet a mathematician, and you say, “*I’m working on a ring,*” that person will not know whether it’s commutative or not. You have to tell them.

And also, you can define rings without units. So, the fancy word for this would be a **commutative ring with unit**. That’s just telling a person that that’s the version of ring you’re signing up for. You can study anything you want, but in this class, we will be considering, for the most part, commutative rings with unit. I just wanted to say that.

Convention: Commutative Rings with Unit

i Remark 1.13 (Standing Convention for Rings in this Course): In the general mathematical literature, rings are not necessarily required to have commutative multiplication, nor are they always required to possess a multiplicative identity 1 (sometimes called “*rngs*” or pseudo-rings).

Standing Convention: In this course, unless explicitly stated otherwise, the term “*ring*”

will always designate a **commutative ring with unit** ($1 \in R$ such that $a \cdot b = b \cdot a$ for all $a, b \in R$).

Terminology Across the Literature: Rings vs. Commutative Rings

Mathematical literature reflects varying conventions:

- **Commutative Algebra & Algebraic Geometry (Atiyah–Macdonald, Eisenbud):** A “ring” implicitly means a commutative ring with multiplicative identity 1.
- **General Algebra (Bourbaki, Lang, Artin):** A “ring” possesses a multiplicative identity 1, but multiplication is not required to be commutative.
- **Functional Analysis & Operator Algebras:** Rings often lack a multiplicative identity 1 (sometimes called “rngs” or pseudo-rings).

In this course, Prof. Pandharipande adheres to the commutative algebra convention: unless explicitly stated otherwise, all rings are commutative and unital ($1 \neq 0$).

III Definition of a Ring Homomorphism

Spoken Transcript [00:18:44 - 00:20:03]

Okay, so the definition of a homomorphism: a ring homomorphism f from R_1 to R_2 is defined to satisfy some properties.

The first is that $f(a + b) = f(a) + f(b)$. So, this is saying that this map respects the linear structure, the additive structure of the two rings. This should remind you from linear algebra class: there’s an idea of a linear homomorphism of vector spaces, and it has this linear condition, and that’s the linear condition.

The second condition is that it respects the multiplication. So, you can imagine what’s going to be written here: $f(ab) = f(a)f(b)$, okay?

And the third condition is it respects the identity: $f(1) = 1$.

Those are the three conditions. That’s the definition of a homomorphism of rings.

Definition of a Ring Homomorphism

Definition 1.14 (Ring Homomorphism): Let R_1 and R_2 be rings. A mapping $f: R_1 \rightarrow R_2$ is called a “ring homomorphism” if it satisfies the following three conditions for all $a, b \in R_1$:

(i) **Additive Compatibility:**

$$f(a + b) = f(a) + f(b). \quad (1.6)$$

(ii) **Multiplicative Compatibility:**

$$f(a \cdot b) = f(a) \cdot f(b). \quad (1.7)$$

(iii) **Identity Preservation:**

$$f(1_{R_1}) = 1_{R_2}. \quad (1.8)$$

≡ **Comparison to Linear Algebra and Group Theory:** Condition (i) asserts that f is a group homomorphism between the additive groups $(R_1, +)$ and $(R_2, +)$, mirroring the linearity condition $T(u + v) = T(u) + T(v)$ from linear algebra. Condition (ii) ensures compatibility with multiplication. Condition (iii) explicitly guarantees that the multiplicative unit of the domain maps to the multiplicative unit of the codomain.

III Preservation of the Additive Identity: $f(0) = 0$

Spoken Transcript [00:20:03 - 00:22:25]

So, there's some puzzling aspects about this definition. You could say, "Well, why don't I say it respects the zero?" That's a very reasonable question. You could say, "What about $f(0)$?"

Well, we can write $f(0 + 0)$ — I want to use this one — $f(0 + 0) = f(0) + f(0)$, right? That's just applying condition (i) of the homomorphism.

But now, $0 + 0 = 0$, we know that, because 0 is the neutral element for addition. So, I can rewrite this as $f(0) = f(0) + f(0)$, okay?

And then I can use the left cancellation by taking the inverse. I've said that several times, and I hope you know what I mean by that: if I take such an equation... let me just do it once formally, because I said it several times.

I have one $f(0)$ here and two $f(0)$'s here, and I want to get rid of the $f(0)$ on both sides. So, what I do — and this is what I mean by left cancellation — is that there exists some $b \in R_2$ such that $b + f(0) = 0$. That's the group property on R_2 .

And then I just add b to both sides of this, so I get:

$$b + f(0) = b + f(0) + f(0).$$

So, I use this cancellation, I use the additive inverse of $f(0)$ on both sides, and then since that equals 0, this is 0, that's 0, and finally I conclude that $f(0) = 0$, okay?

So, it is true that $f(0) = 0$, but I don't have to put it in the definition. It's traditional not to. You could put it here; it wouldn't make any difference.

Automatic Preservation of the Additive Identity

💎 **Proposition 1.15 (Preservation of Zero):** Let $f: R_1 \rightarrow R_2$ be a ring homomorphism. Then:

$$f(0_{R_1}) = 0_{R_2}. \quad (1.9)$$

Proof

Applying condition (i) of Definition 2.19 to $0_{R_1} = 0_{R_1} + 0_{R_1}$:

$$f(0_{R_1}) = f(0_{R_1} + 0_{R_1}) = f(0_{R_1}) + f(0_{R_1}). \quad (1.10)$$

Since $(R_2, +, 0_{R_2})$ is a group, the element $f(0_{R_1}) \in R_2$ possesses an additive inverse $b \in R_2$ such that $b + f(0_{R_1}) = 0_{R_2}$. Adding b to both sides of (1.10):

$$\begin{aligned} b + f(0_{R_1}) &= b + (f(0_{R_1}) + f(0_{R_1})) \\ 0_{R_2} &= (b + f(0_{R_1})) + f(0_{R_1}) && \text{(by associativity in } R_2) \\ 0_{R_2} &= 0_{R_2} + f(0_{R_1}) \\ 0_{R_2} &= f(0_{R_1}). \end{aligned}$$

Thus, $f(0_{R_1}) = 0_{R_2}$.

Q.E.D.

III Why Identity Preservation Must Be Axiomatized

Spoken Transcript [00:22:25 - 00:24:16]

So, then maybe the question is: well, then why do I put $f(1) = 1$? That's a reasonable question; they're treated not symmetrically.

And the answer to that... *[erases the cancellation calculation for $f(0)$ on the board]*

Yeah, it has to do with the fact that I cannot... I mean, maybe this is what you're hinting at, but I don't have this cancellation property in multiplication. I have this cancellation — the existence of an additive inverse — that is an axiom of the ring, so I can use it. But there is not an existence of a multiplicative inverse, so I cannot just follow these steps.

So, I have to assume it by hand. This is about the definitions, and as I said, the strange thing about mathematics is the definitions: when it's taught, the definitions are first, but somehow when it's understood, the definitions are last. That's somehow a constant difficulty. *[erases more board space and begins writing]*

Yeah, so the point is that we don't always have multiplicative inverses in R , so we can't use that argument — the one that's in the corner there.

So, to fix that, we just say at the beginning: $f(1) = 1$. We don't leave it to chance.

You could hope maybe that even though we couldn't use this argument, maybe you could find some different argument. You could hope that, but it's not true.

The Asymmetry Between Addition and Multiplication

Board State Note:

"We don't have mult. inverses in R_2 , so we can't use this argument."

≡ **Why $f(1) = 1$ Does Not Follow from $f(ab) = f(a)f(b)$:** One might wonder why $f(1) = 1$ is required as an explicit axiom in Definition 2.19, whereas $f(0) = 0$ is a provable theorem.

- **Additive Case:** In the additive group $(R_2, +)$, every element has an additive inverse. Hence, from $f(0) = f(0) + f(0)$, we can subtract $f(0)$ to obtain $f(0) = 0$.
- **Multiplicative Case:** Using multiplicativity, we know that $f(1) = f(1 \cdot 1) =$

$f(1) \cdot f(1)$. However, in a general ring R_2 , $f(1)$ does not necessarily possess a multiplicative inverse! Thus, we cannot “divide by $f(1)$ ” to conclude $f(1) = 1$.

Spoken Transcript [00:24:16 - 00:25:08]

For example, I could take a bad example. *[writes “A bad example:” on the board]*

I could take R and map it to... sorry, I could map it so everything goes to zero. So, here’s R_1, R_2 ; say they’re both healthy rings, in fact they could both be $\mathbb{Z}$. I could map every element to zero. *[writes $R_1 \xrightarrow{f} R_2, a \mapsto 0$]*

If you map every element to zero, you certainly are not mapping 1 to 1. But you won’t get a contradiction with every element being zero; all the other properties will be satisfied.

Okay, so maybe it’s not a good idea to think about this, but the fact of the matter is we need to just assume it. *[erases the bad example]* Okay.

Counterexample: The Non-Unital Zero Map

Example 1.16 (The Trivial Zero Map Fails Condition (iii)): Let $R_1 = R_2 = \mathbb{Z}$ (or any non-zero rings). Consider the zero map:

$$f: R_1 \rightarrow R_2, \quad f(a) := 0 \quad \forall a \in R_1.$$

This map satisfies conditions (i) and (ii):

- **Addition:** $f(a + b) = 0 = 0 + 0 = f(a) + f(b)$,
- **Multiplication:** $f(a \cdot b) = 0 = 0 \cdot 0 = f(a) \cdot f(b)$.

However, $f(1_{R_1}) = 0 \neq 1_{R_2}$. Without demanding condition (iii) ($f(1) = 1$), the zero map would be considered a ring homomorphism between non-zero rings.

III Examples of Ring Homomorphisms: The Canonical Map $\mathbb{Z} \rightarrow R$

Spoken Transcript [00:25:08 - 00:27:20]

So, then there’s lots of examples of these ring homomorphisms, and it’s the main way we compare rings: by homomorphism.

For example, the thing I said here: $\mathbb{Z} \rightarrow R$, this basic map ϕ , this is a ring homomorphism. *[writes on the board: “Examples: $\mathbb{Z} \xrightarrow{\phi} R$, This is a ring homom.”]*

Not only does $\mathbb{Z}$ map to R by just seeing where 1 goes, it’s actually a homomorphism. And to check it’s a homomorphism, you have to do something.

I’m not going to do this for you; you have to do it, because it’s some kind of internal tautology about addition. So, where does 2 go? We said 2 goes to $1 + 1$. *[writes $2 \mapsto 1 + 1$]*

Where does 3 go? 3 goes to $1 + 1 + 1$. *[writes $3 \mapsto 1 + 1 + 1$]*

So, then we’d have to check that if I add these two, 5 goes to five 1’s, which looks

pretty good. [*writes* $2 + 3 = 5 \mapsto (1 + 1) + (1 + 1 + 1)$]

And then you'd also have to check: where does 6 go? $6 = 2 \times 3$, and this goes to $(1 + 1) \times (1 + 1 + 1)$. [*writes* $6 = 2 \cdot 3 \mapsto (1 + 1)(1 + 1 + 1)$]

And now you'd use the distributive law a lot of times. If you use the distributive law a lot of times, you'll get this to be $1 + 1 + 1 + 1 + 1 + 1$, [*writes* $= 1 + 1 + 1 + 1 + 1 + 1$] which is the definition of where 6 goes, okay?

So, if you do this, you'll see that there's essentially nothing going on, except understanding how the distributive law is making this a homomorphism.

This kind of mathematics that's on this foundational, definitional level, you pretty much have to do for yourself to understand, you know? Because it has a certain kind of logic, and I hope this is clear. But there is some content — a very small amount of content — about how this definition of 2 as $1 + 1$ interacts with the distributive law.

And as I say, I think most people find this not so controversial, but if you want to understand it, you have to go through the mental effort of checking the things yourself. On the board, I could convince you of anything, probably.

The Canonical Homomorphism $\mathbb{Z} \rightarrow R$

 **Example 1.17 (Canonical Homomorphism from the Integers):** For any ring R , the canonical evaluation map:

$$\phi: \mathbb{Z} \rightarrow R, \quad \phi(n) := n \cdot 1_R$$

is a ring homomorphism.

Verification of Homomorphism Axioms:

- **Preservation of Unit:** $\phi(1) = 1 \cdot 1_R = 1_R$.
- **Additive Compatibility (e.g., $2 + 3 = 5$):**
 $\phi(2) + \phi(3) = (1_R + 1_R) + (1_R + 1_R + 1_R) = 1_R + 1_R + 1_R + 1_R + 1_R = \phi(5) = \phi(2 + 3)$.
- **Multiplicative Compatibility (e.g., $2 \cdot 3 = 6$):** Using repeated applications of the distributive laws:

$$\begin{aligned} \phi(2) \cdot \phi(3) &= (1_R + 1_R)(1_R + 1_R + 1_R) \\ &= 1_R(1_R + 1_R + 1_R) + 1_R(1_R + 1_R + 1_R) \\ &= (1_R + 1_R + 1_R) + (1_R + 1_R + 1_R) \\ &= 1_R + 1_R + 1_R + 1_R + 1_R + 1_R = \phi(6) = \phi(2 \cdot 3). \end{aligned}$$

 **Universal Initial Property of $\mathbb{Z}$:** In category theory, this property establishes that $\mathbb{Z}$ is the “*initial object*” in the category of rings: for every ring R , there exists a **unique** ring homomorphism from $\mathbb{Z}$ to R .

III Inclusion Homomorphisms and Subrings

Spoken Transcript [00:27:20 - 00:28:24]

Okay, so that's a very simple example of a homomorphism, but in fact, there's lots of them — all the ones we've seen so far. If you take $\mathbb{Z}$ inside $\mathbb{Q}$, [writes $\mathbb{Z} \subseteq \mathbb{Q}$] if you view this as a map, just the embedding map — if you view an integer as a rational number —, then this gives me a homomorphism. And that's a homomorphism inside $\mathbb{R}$, [writes $\subseteq \mathbb{R}$] and inside $\mathbb{C}$, [writes $\subseteq \mathbb{C}$] so these inclusions are all homomorphisms.

And I can also include this inside the ring of polynomials: [writes $\mathbb{Z}[x] \subseteq \mathbb{Q}[x] \subseteq \mathbb{R}[x] \subseteq \mathbb{C}[x]$] polynomials in one variable.

These are many of the rings that we will study, but if you start at the bottom, $\mathbb{Z}$, they just include, and all these inclusions are homomorphisms. Whenever I include, I'm not changing the definition of addition or multiplication for the smaller object, and 1 is always going to 1. So, this is an example.

Inclusion Homomorphisms of Standard Rings

 **Example 1.18 (Canonical Inclusions):** The canonical inclusion embeddings of the standard number systems:

$$\mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R} \hookrightarrow \mathbb{C} \quad (1.11)$$

and their single-variable polynomial rings:

$$\mathbb{Z}[x] \hookrightarrow \mathbb{Q}[x] \hookrightarrow \mathbb{R}[x] \hookrightarrow \mathbb{C}[x] \quad (1.12)$$

are all injective ring homomorphisms. For any subring $S \subseteq R$, the inclusion map $\iota: S \hookrightarrow R$ defined by $\iota(s) = s$ is a ring homomorphism because S inherits the operations and the unit $1_S = 1_R$ directly from R .

III The Ring of Residue Classes $\mathbb{Z}/n\mathbb{Z}$ and the Projection Homomorphism

Spoken Transcript [00:28:24 - 00:30:48]

So, let's try some different kinds of examples. One example is that you've studied this ring $\mathbb{Z}/n\mathbb{Z}$. [writes $\mathbb{Z}/n\mathbb{Z}$] Well, if we haven't, let me at least say what it is, because this also plays a role in the class.

So, this ring here, this set here... [writes $(\mathbb{Z}/n\mathbb{Z}, +, \cdot, 0, 1)$] This is a standard name — probably in other courses it was also given other names —, but the words are “ $\mathbb{Z} \bmod n\mathbb{Z}$ ”, and this is the set of residue classes mod n . [writes $\mathbb{Z}/n\mathbb{Z} = \{\text{residue classes mod } n\}$]

If you want to be explicit, this is equal to the set $\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}$, because n is zero. [writes $= \{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}\}$] Is this notation reasonable notation? It's kind of semi-standard mathematical notation, yeah. So, there's exactly n elements of this: it's just the residue classes mod n .

And you can add residue classes, you can multiply them, there's the zero, and there's the one. So, if I take this whole data together, this is exactly a commutative ring with unit. [writes Ring!]

So, it's a ring for us. It's a ring of a slightly different nature than the other ones because it has finitely many elements. That's an interesting ring to think about.

In some sense, one of the simplest and most confusing ones is this one: *[writes $\mathbb{Z}/2\mathbb{Z} = \{\bar{0}, \bar{1}\}$]* This is a ring with two elements: 0 and 1. That's kind of a fun ring, but it's a full-fledged ring — a ring is a ring no matter how small.

And we get here $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$; *[writes $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$]* this is also a homomorphism. It's a ring homomorphism where for any integer a , I take $f(a) = \bar{a}$. *[writes $f(a) = \bar{a}$]*

Okay? So, that's another example.

Residue Class Rings and the Reduction Modulo n

 **Definition 1.19 (Ring of Residue Classes Modulo n):** For any integer $n \geq 1$, the ring of “*residue classes modulo n* ” (denoted $\mathbb{Z}/n\mathbb{Z}$ or $\mathbb{Z}_n$) is the set:

$$\mathbb{Z}/n\mathbb{Z} := \{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}\}, \quad (1.13)$$

equipped with modular addition and multiplication:

$$\bar{a} + \bar{b} := \overline{a+b}, \quad \bar{a} \cdot \bar{b} := \overline{a \cdot b},$$

with additive identity $\bar{0}$ and multiplicative identity $\bar{1}$.

 **Example 1.20 (The Two-Element Field $\mathbb{Z}/2\mathbb{Z}$):** For $n = 2$, $\mathbb{Z}/2\mathbb{Z} = \{\bar{0}, \bar{1}\}$ is the smallest non-trivial commutative ring with unit ($1 \neq 0$), having addition $\bar{1} + \bar{1} = \bar{0}$ and multiplication $\bar{1} \cdot \bar{1} = \bar{1}$.

 **Proposition 1.21 (Canonical Projection Homomorphism):** The reduction map modulo n :

$$\pi: \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}, \quad a \mapsto \bar{a} := a \bmod n \quad (1.14)$$

is a surjective ring homomorphism.

The Kernel of a Ring Homomorphism and Ideals

Definition of the Kernel

Spoken Transcript [00:31:02 - 00:32:56]

There's lots of interesting examples; I'm not going to get to them all today, but I want to say that... *[pushes up the lower-right blackboard]* For homomorphisms, like for vector spaces, whenever you have one, you can consider the kernel and the image. And it turns out, in maybe a surprising way, for ring theory the notion of the kernel turns out to be much more important than the notion of the image.

In linear algebra, if you have a homomorphism between two vector spaces — did you use the word homomorphism? You can also say linear transformation, both those words are used — if you have a linear transformation between two vector spaces, there's a notion of its kernel (those are the elements that go to zero) and the notion of its image. And they're both sub-vector spaces. In linear algebra, these things are kind of on similar footing.

But in ring theory, when you have a homomorphism, there's again the kernel and the image. We'll start with the kernel first; probably next time we'll say a little bit about

the image. So, this is an interesting object. *[writes on the lower board]*

Here we have: let $f: R_1 \rightarrow R_2$ be a homomorphism of rings. Then we define the kernel of f . It's those elements in the first ring — because the kernel lives in the first ring — such that $f(a) = 0$, okay?

I hope this definition is clear: it's everybody that goes to zero.

In particular, you could say: what is the kernel of this? *[writes $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$]* What is the kernel here?

Student Answer

Multiples of n .

Spoken Transcript [00:32:57 - 00:33:17]

Yes, so the kernel here is exactly all multiples of n , and this is often denoted $n\mathbb{Z}$. *[writes $\ker(\pi) = n\mathbb{Z}$]* This is equal to the set of multiples of n . That's the kernel.

So, the kernel has very specific properties.

Definition of the Kernel of a Ring Homomorphism

Definition 1.22 (Kernel of a Ring Homomorphism): Let R_1, R_2 be rings and let $f: R_1 \rightarrow R_2$ be a ring homomorphism. The “kernel” of f , denoted by $\ker(f)$, is the pre-image of the zero element 0_{R_2} :

$$\ker(f) := \{a \in R_1 \mid f(a) = 0_{R_2}\} \subseteq R_1. \quad (1.15)$$

Example 1.23 (Kernel of the Modular Reduction Homomorphism): For the canonical projection $\pi: \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ given by $\pi(a) = \bar{a}$, an integer a maps to $\bar{0}$ if and only if a is a multiple of n . Hence:

$$\ker(\pi) = n\mathbb{Z} := \{k \cdot n \mid k \in \mathbb{Z}\}. \quad (1.16)$$

III Properties of the Kernel

Spoken Transcript [00:33:17 - 00:34:49]

So, properties of the kernel: one is that it's closed under addition. *[writes “Properties of the $\ker(f)$ ”]* If we have a and b in the kernel of f , this implies $a + b$ is in the kernel. *[writes $a, b \in \ker(f) \implies (a + b) \in \ker(f)$]*

This is the linear algebra property, and all of these things are easy to prove. To check whether $a + b$ is in the kernel, I have to calculate $f(a + b)$. But using the definition of homomorphism, this is $f(a) + f(b)$. Then using the fact that a and b are both in the kernel, well, this one is zero and that one is zero, so the whole thing is zero, so it's in the kernel, right? *[writes $f(a + b) = f(a) + f(b) = 0 + 0 = 0$]*

I should have said even earlier: we already know that zero is in the kernel. *[writes*

$$0 \in \ker(f)]$$

If a is in the kernel, and r is any element of the ring, then $r \cdot a$ is in the kernel. [writes:
if $a \in \ker(f), r \in R_1 \implies ra \in \ker(f)$]

Spoken Transcript [00:34:49 - 00:36:35]

So, that's a wider property. These are the linear properties — we'll say more about them in a moment —, but this property is about the multiplication. It says that if I take an element of the kernel, I can then go pick any element of the ring and multiply it, and it'll be in the kernel. So, it's a more interesting property.

You can see that here: say if you have an element of the kernel, it's n times something. If I multiply it by any integer, it's still n times something.

This is not hard to prove either: we have to calculate r times a , and then of course we use the multiplicative property. And now this is the point, because in the multiplicative property, in order to deduce that this product is zero, we only need one of the factors to be zero. [writes $f(ra) = f(r)f(a) = f(r) \cdot 0 = 0$]

For the addition we needed both, but for the multiplication we only need to know that one of the factors is zero, because that'll kill the product. And that's already there because a is in the kernel, so this is zero, so the whole product is zero, okay?

In particular, if a is in the kernel, $-a$ is in the kernel. [writes $a \in \ker(f) \implies -a \in \ker(f)$] That's the additive inverse, because for $-a$, $r = -1$.

Those are all properties enjoyed by the kernel. And those properties can be said in a more concise way — which is a more natural way to say it — and it comes with a definition: the definition of an ideal.

Algebraic Properties of the Kernel

 **Proposition 1.24 (Axiomatic Properties of the Kernel):** Let $f: R_1 \rightarrow R_2$ be a ring homomorphism. The kernel $\ker(f) \subseteq R_1$ satisfies the following properties:

- (1) **Identity:** $0_{R_1} \in \ker(f)$.
- (2) **Additive Closure:** $\forall a, b \in \ker(f) \implies a + b \in \ker(f)$.
- (3) **Inverses:** $\forall a \in \ker(f) \implies -a \in \ker(f)$.
- (4) **Multiplicative Absorption (Ideal Property):** $\forall a \in \ker(f), \forall r \in R_1 \implies r \cdot a \in \ker(f)$.

Proof

We verify each property:

- **Property (1):** Follows directly from [Proposition 1.15](#), since $f(0_{R_1}) = 0_{R_2}$.
- **Property (2):** If $a, b \in \ker(f)$, then $f(a) = 0$ and $f(b) = 0$. By condition (i) of homomorphisms:

$$f(a + b) = f(a) + f(b) = 0 + 0 = 0 \implies a + b \in \ker(f).$$

- **Property (4):** Let $a \in \ker(f)$ and $r \in R_1$. Using condition (ii) and [Proposition 1.6](#):

$$f(r \cdot a) = f(r) \cdot f(a) = f(r) \cdot 0 = 0 \implies r \cdot a \in \ker(f).$$

- **Property (3):** Setting $r = -1 \in R_1$ in Property (4), we obtain:

$$-a = (-1) \cdot a \in \ker(f).$$

Q.E.D.

≡ **Additive vs. Multiplicative Absorption:** In Property (2), both terms must evaluate to zero ($0 + 0 = 0$) to keep the sum inside the kernel. In contrast, for Property (4), multiplication by zero ($f(r) \cdot 0 = 0$) absorbs **any** element $r \in R_1$ from the ambient ring. This strong absorption property distinguishes ring kernels from mere subrings.

III Ideals in a Ring

Spoken Transcript [00:36:35 - 00:38:07]

So, an ideal... *[erases the lower-left board]*

This is the definition of an ideal, $I \subseteq R$. *[writes "Def: Ideal $I \subseteq R$ "]*

The properties are: 0 must be in I . *[writes $0 \in I$]* And if I take I , the $+$ from R , and the 0 , this is an abelian group. *[writes $(I, +, 0)$ abelian group]*

Remember, whenever I write R , I always mean R with the two operations and this data: $(R, +, \cdot, 0, 1)$. *[writes $R = (R, +, \cdot, 0, 1)$]*

So, to be an ideal, you have to have zero, and when I consider this subset with $+$ and zero should be an abelian group, this means it's closed under addition, and it's closed under additive inverses, okay?

And the second thing is that for every $a \in I$ and for every $r \in R$, ra remains in I , okay? *[writes $\forall a \in I, \forall r \in R \implies ra \in I$]* That's the ideal condition.

So, this is an extremely important definition for the class. It tells you various subsets of R are ideals: they have this linear property (which is kind of like linear algebra), but then they also have this multiplicative property (which is not like linear algebra). And the role of an ideal will play a little bit like the role of a subspace in linear algebra.

Definition of an Ideal

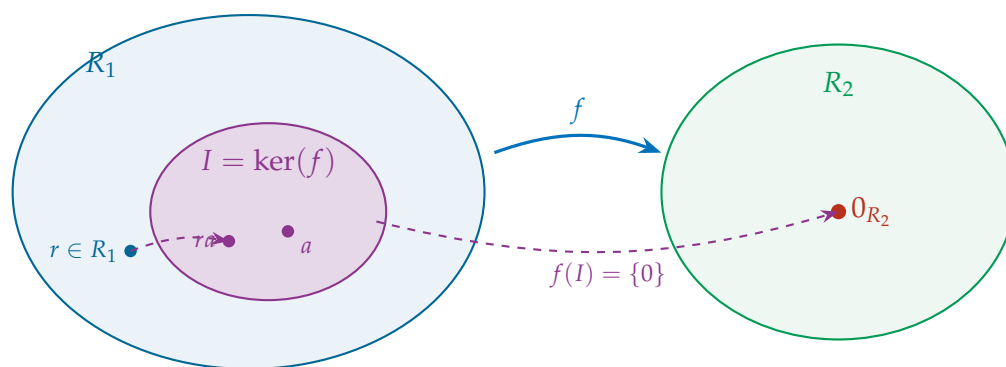
 **Definition 1.25 (Ideal):** Let $R = (R, +, \cdot, 0, 1)$ be a commutative ring with unit. A subset $I \subseteq R$ is called an "**ideal**" of R (denoted $I \trianglelefteq R$) if it satisfies:

- (1) **Additive Subgroup:** $(I, +, 0)$ is an abelian subgroup of $(R, +, 0)$:

- $0 \in I$,
- $\forall a, b \in I \implies a + b \in I$,
- $\forall a \in I \implies -a \in I$.

- (2) **Multiplicative Absorption (Ideal Condition):**

$$\forall a \in I, \forall r \in R \implies r \cdot a \in I. \quad (1.17)$$



≡ **Ideals as the Ring-Theoretic Analogue of Normal Subgroups and Subspaces:** In linear algebra, kernels of linear maps are vector subspaces ($T(u + v) = T(u) + T(v)$ and $T(cv) = cT(v)$). In group theory, kernels of group homomorphisms are normal subgroups ($gNg^{-1} \subseteq N$). In ring theory, kernels of ring homomorphisms are “*ideals*”: they are closed under addition and absorb multiplication from the entire ambient ring R .

Spoken Transcript [00:38:07 - 00:39:13]

And the proposition we’ve proven with this is: the kernel of a homomorphism $f: R_1 \rightarrow R_2$ is an ideal in R_1 . [writes: “Prop: The ker of a hom $f: R_1 \rightarrow R_2$ is an ideal in R_1 ”]

That’s what we’ve proven on the board up there: with a kernel, you always get an ideal.

And rings can have... we will do some examples next time, I’m running out of time here, but in $\mathbb{Z}$, if I take $n\mathbb{Z}$, that’s an ideal. [writes $\mathbb{Z} \supseteq n\mathbb{Z} = I$] Rings can have lots of ideals.

What about a field? I’ll just say one more thing about this. Is that clear?

We have lots of examples of rings and lots of examples of ideals. But suppose F is a field. [writes: $F = \text{field}, (F, +, \cdot, 0, 1)$, “What are its ideals?”]

The Kernel is an Ideal

💎 **Proposition 1.26 (Kernel as an Ideal):** Let $f: R_1 \rightarrow R_2$ be a ring homomorphism. Then $\ker(f)$ is an ideal of R_1 :

$$\ker(f) \trianglelefteq R_1.$$

Proof

This is the synthesis of Proposition 1.24:

- (1) $(\ker(f), +, 0)$ is an abelian subgroup of $(R_1, +, 0)$ because $0 \in \ker(f)$, $a + b \in \ker(f)$ for all $a, b \in \ker(f)$, and $-a \in \ker(f)$ for all $a \in \ker(f)$.
- (2) Multiplicative absorption holds because for all $a \in \ker(f)$ and all $r \in R_1$:

$$f(r \cdot a) = f(r) \cdot f(a) = f(r) \cdot 0 = 0 \implies r \cdot a \in \ker(f).$$

Hence, $\ker(f)$ satisfies all axioms of an ideal.

Q.E.D.

III Ideals in a Field

Spoken Transcript [00:39:13 - 00:40:05]

What are its ideals? What are the ideals of a field?

Because a field is a ring, it has the notion of ideals. Every abstract concept of a ring we can apply to a field, because a field is a particular type of ring: it's a ring where we have multiplicative inverses for non-zero elements.

So, what could be an ideal here? What are the ideals?

I'll start with one simple thing: if I just take $I = \{0\}$. *[writes $I = \{0\}$]*

Then zero is in I . It is an abelian subgroup just by itself, because $0 + 0 = 0$ and everything... zero is sufficient unto itself, let us say, and it has this multiplicative property, because if I multiply by anybody it stays zero. So, zero is always an ideal.

So, what else? Yeah?

Student Answer

The whole field.

Spoken Transcript [00:40:06 - 00:40:21]

The whole field is always an ideal: $I = F$. *[writes $I = F$]*

And of course, there can be no doubt about that, because if you take the whole thing, it's an abelian group and you're not going to get out of it by multiplying.

So, can there be any more? Why not?

Student Answer

Because as soon as you have another element, then you have the multiplicative inverse, so you have 1, and with 1 you get...

Spoken Transcript [00:40:32 - 00:42:20]

Okay, so this argument is correct. The claim is: for a field, these are the only ideals. *[writes: "Claim: these are only ideals!"]*

Which is an important thing to always keep in mind. And the reason is the argument he said: suppose that I have an ideal, and suppose this ideal is not this silly zero ideal. *[writes $I \neq \{0\}$]* That means there exists somebody in the ideal that's not zero. *[writes $\exists a \in I, a \neq 0$]*

That somebody, because it's a field, has a multiplicative inverse: there exists an a^{-1} , so that $a^{-1} \cdot a$ is also in the ideal, because you're allowed to take anybody and multiply... this is anybody, and this is somebody in the ideal. So, if we start with a in the ideal, I can multiply by anybody.

In particular, I can multiply by its inverse. But then this is 1! [*writes* $\exists a^{-1} \implies a^{-1}a \in I \implies 1 \in I$] Once I have 1 in the ideal, then I can easily get anybody in the ideal, because I can just take any $r \cdot 1$. [*writes* $\implies r \cdot 1 \in I \implies I = F$]

Okay? So, it's an elementary argument; it's very good to understand. Fields have very few ideals — extremely few —, but rings can have quite a number of ideals. As I said here, this is the simplest example to see: ideals in the integers come from multiples of any integer, and ideals in...

Yeah, we will see some examples of it. But was this argument clear? Sometimes I do these things very quickly, it's just, you know...

So, the standard thing to say here is that you can view fields as sellouts, because they've lost all their ideals. [*laughter*]

Classification of Ideals in a Field

 **Theorem 1.27 (Ideals in a Field):** Let F be a field. The only ideals of F are the trivial zero ideal $\{0\}$ and the unit ideal F :

$$I \trianglelefteq F \implies I = \{0\} \quad \text{or} \quad I = F. \quad (1.18)$$

Proof

Let $I \subseteq F$ be an ideal of F . Clearly, $I = \{0\}$ and $I = F$ are ideals.

Now, suppose $I \neq \{0\}$. Then there exists a non-zero element:

$$a \in I \quad \text{with} \quad a \neq 0.$$

Since F is a field, every non-zero element has a multiplicative inverse $a^{-1} \in F$. By the multiplicative absorption property of ideals ([Definition 2.21](#)), choosing $r = a^{-1} \in F$:

$$a^{-1} \cdot a \in I \implies 1_F \in I.$$

Once the unit 1_F lies in I , applying absorption once more with any arbitrary element $r \in F$:

$$r = r \cdot 1_F \in I \quad \forall r \in F.$$

Therefore, $F \subseteq I$, which forces $I = F$.

Q.E.D.

 **Implications for Field Homomorphisms:** A direct consequence of this theorem is that every ring homomorphism $f: F \rightarrow R$ from a field F to a non-zero ring R is automatically **injective**:

$$\ker(f) \trianglelefteq F \implies \ker(f) = \{0\} \quad (\text{since } f(1_F) = 1_R \neq 0_R \implies \ker(f) \neq F).$$

Because $\ker(f) = \{0\}$, field homomorphisms into non-zero rings are always embeddings.

Insight: Fields as “Sellouts”

The lecturer’s humorous pedagogical quip — that “*fields are sellouts because they have lost all their ideals*” — highlights an essential contrast between field theory and ring theory:

- In a field, the abundance of units ($F^\times = F \setminus \{0\}$) destroys all non-trivial ideals, as any non-zero element immediately pulls the entire structure into the ideal ($1 \in I \implies I = F$).
- In general rings, the scarcity of units allows rich, complex lattices of proper non-zero ideals to flourish (e.g., $n\mathbb{Z} \trianglelefteq \mathbb{Z}$), which form the foundational building blocks for quotient rings R/I and algebraic geometry.

III Outlook: Zorn’s Lemma and Vector Space Bases

III Existence of Bases in Infinite-Dimensional Vector Spaces

Spoken Transcript [00:42:20 - 00:43:54]

For this week, I did some material that was billed for the second week, and I didn’t cover Zorn. So, I’ll say two words about Zorn, and I’ll say a little bit more next week. *[erases the lower board]*

The first question is about linear algebra, which is this statement which probably you’ve seen in the first year — which is a strange statement, but it’s about...

So, first of all, I will spend some time also next week on Zorn’s lemma, so you can read about that in the book. *[writes on the lower board: “Zorn’s Lemma”, “Let V be a vector space over a field k ”, “Does V have a basis?”]*

And it’s related to the statement: let V be a vector space over a field k . Does V have a basis? Did you consider this question in your first year?

If it’s finite-dimensional, of course it has a basis. And this issue is about whether if you have some massively infinite-dimensional vector space, it has a basis. And the traditional way to do this is an application of Zorn’s lemma, which is related to the Axiom of Choice and logic, and... not my favorite direction to think about, but we’ll say a little bit about it.

Okay, and I think that’s it. *[students applaud, lecturer gathers notes and lecture concludes]*

Zorn’s Lemma and the Existence of Vector Space Bases

Chalkboard Inscription:

- **Zorn’s Lemma**
- Let V be a vector space over a field k .
- **Question:** Does V have a basis?

 **Remark 1.28 (Infinite-Dimensional Vector Spaces and the Axiom of Choice):** For finite-dimensional vector spaces, the existence of a basis is constructive via Gaussian elimination or the Steinitz exchange lemma. For arbitrary, infinite-dimensional vector spaces (such as function spaces $C(\mathbb{R}, \mathbb{R})$ or sequence spaces $\mathbb{R}^{\mathbb{N}}$), the assertion that **every vec-**

tor space possesses a basis (known as a “*Hamel basis*”) cannot be proven constructively in Zermelo–Fraenkel set theory (ZF). It is logically equivalent to the “*Axiom of Choice*” (AC) and is standardly established by applying “*Zorn’s Lemma*” to the poset of linearly independent subsets ordered by inclusion.

Summary of Key Concepts

- **Gaussian Integers $\mathbb{Z}[i]$ and Primes:** The ring $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ possesses four units $\{\pm 1, \pm i\}$. An ordinary prime $p \in \mathbb{Z}$ factors in $\mathbb{Z}[i]$ if and only if $p = a^2 + b^2$, which occurs precisely when $p = 2$ or $p \equiv 1 \pmod{4}$ (Fermat’s two-square theorem). Integer primes $p \equiv 3 \pmod{4}$ remain prime in $\mathbb{Z}[i]$.
- **Rings and Elementary Arithmetic:** A ring $(R, +, \cdot, 0, 1)$ consists of an additive abelian group $(R, +, 0)$ and a multiplicative monoid $(R, \cdot, 1)$ coupled by the distributive laws. Distributivity yields the universal identities $0 \cdot a = 0$ and $(-1) \cdot a = -a$. When $1 = 0$, the ring collapses to the degenerate zero ring $R = \{0\}$.
- **Canonical Evaluation Map $\mathbb{Z} \rightarrow R$:** Defining integer multiples $n \cdot a$ yields the unique canonical ring homomorphism $\phi: \mathbb{Z} \rightarrow R$ with $\phi(n) = n \cdot 1_R$, establishing $\mathbb{Z}$ as the initial object in the category of unital rings.
- **Ring Homomorphisms:** A ring homomorphism preserves addition, multiplication, and the multiplicative identity ($f(1) = 1$). While $f(0) = 0$ is automatic via additive group cancellation, $f(1) = 1$ must be explicitly axiomatized because elements in arbitrary rings generally lack multiplicative inverses.
- **Ideals and Kernels:** An ideal $I \trianglelefteq R$ is an additive subgroup that multiplicatively absorbs the entire ambient ring ($r \cdot a \in I$ for all $a \in I, r \in R$). The kernel $\ker(f) = f^{-1}(\{0\})$ of every ring homomorphism is an ideal.
- **Ideals in Fields:** Because every non-zero element in a field F is invertible, any non-zero ideal contains 1_F and equals F . Thus, fields have only the trivial ideals $\{0\}$ and F , ensuring that every homomorphism from a field into a non-zero ring is injective.
- **Outlook on Zorn’s Lemma:** The existence of a Hamel basis for arbitrary infinite-dimensional vector spaces is non-constructive, logically equivalent to the Axiom of Choice, and proven using Zorn’s Lemma.

Tuesday 22 September 2026

Groups, Rings, Ideals, Subrings, and Introduction to Categories

Course Content Overview

This lecture reviews the fundamental algebraic structures introduced in Week 1—groups, rings, and fields—and explores their intrinsic properties, including the uniqueness of neutral elements and inverses. We formalize structure-preserving mappings (“*group and ring homomorphisms/isomorphisms*”), examine classical isomorphism classes over $\mathbb{R}$, and investigate substructures: “*subgroups*”, “*ideals*”, and “*subrings*” (including the classification of subrings and ideals of $\mathbb{Z}$, and ideals in smooth function spaces). Finally, we introduce the foundational language of “*category theory*”, defining categories, objects, morphisms, categorical isomorphisms, and “*functors*” with concrete algebraic examples.

Organizational Remarks

Spoken Transcript [00:00:00 - 00:01:48]

Okay, so we start, and there has been some progress on the website. There should be instructions there. Also, first of all, you should be able to register for the problem sessions now.

The second thing is that there should be instructions on the website about how to upload your homework.

And the third thing is about the quiz, and I think there’s a link there, too. We’ll see if it works on Friday. So, the rule for the quiz will be that you can take it anytime from 6 in the morning on Friday to midnight. And if this is not enough freedom, we can also discuss it. It’s not so...

So, another course hack is that if you look at this Metaphor page [*writes “Metaphor 2018” on the blackboard*] in 2018 — it was the last time I taught this course —, that Metaphor page is there. This course is going to follow roughly the same path. In fact, the topics and homework — I mean, there’s maybe some slight switching order of lectures —, but you can find all the information there. In particular, all the problem sets that Lieke will eventually upload, you can already look at them now if you want to get ahead.

And also, they're all the answers to all the problem sets. You can also look at them if you want, but in the modern world it's easy to find answers to these questions, so I don't see there's any reason to hide them from you. So, you can find everything there.

So, the other thing is that if you want to ask me a question, I'm happy to talk about math with anybody. You have to find me. Normally, I have an office, but I was kicked out because of this "*Fenstersanierung*". And it's going to last forever. So, I have a secret office on this floor. If you find me, you can talk to me. You just have to find it. It's actually pretty hard to find. But if you do, you can knock on the door. Sometimes I'm busy, sometimes I'm not, but I'm happy to talk to anyone. But you have to find the office yourself.

Organizational Information: Course Website and Metaphor Archive

The lecturer outlines administrative details for the semester:

- **Problem Sessions & Homework:** Registration for exercise sessions is open on the course website, where instructions for online homework submission are posted.
- **Weekly Quizzes:** Friday quizzes can be completed asynchronously between 06:00 and 24:00 (midnight).
- **Course Archive (Metaphor 2018):** Lecture topics, problem sets, and solution sets from the 2018 iteration of the course are publicly accessible for advance study.
- **Office Hours:** Due to ongoing building renovations ("*Fenstersanierung*"), the lecturer occupies a temporary office on the floor.

Review of Groups, Rings, and Fields

Group Axioms and Uniqueness of Identities

Spoken Transcript [00:01:48 - 00:03:22]

Okay, so last week we discussed kind of carefully these three objects which are the centerpiece for this course.

G is a group [*writes $(G, \cdot, e)$ on the blackboard*], and last week I used the star ($\star$) for the product. No one does that. That was purely for didactic purposes, and probably pointless didactic purposes. The standard for a group is dot ($\cdot$), but as I said, you can name this anything you want.

So, this is the standard data of the group, and the group — I don't repeat everything, but has the three axioms: associativity, existence of the neutral element, and existence of inverses.

So, now this is written now in slightly more abstract language than the explicit equations. And there's many things I didn't say, because they are all, in some sense, part of the definition. Like I didn't say explicitly... you could ask yourself: this neutral element, could there be some other element that also has the neutrality property? So, these are things you should think about yourself. I can't teach you everything; it's not possible.

Could there be another $\hat{e} \in G$ that has the neutrality property? And the answer is

no, because then you could look at $e \cdot \hat{e}$. If I use the fact that e is the neutral element, this equals $\hat{e}$, and if $\hat{e}$ is a pretend neutral element, then this equals e . Okay? Is that clear?

Definition of a Group and Uniqueness of the Identity

 **Definition 2.1 (Group):** A “group” is a triple $(G, \cdot, e)$, consisting of a non-empty set G , a binary operation $\cdot : G \times G \rightarrow G$, and a distinguished element $e \in G$ (called the “identity” or “neutral element”), satisfying the following three group axioms:

(1) **Associativity:** For all $a, b, c \in G$,

$$(a \cdot b) \cdot c = a \cdot (b \cdot c).$$

(2) **Existence of a Neutral Element:** There exists an element $e \in G$ such that for all $a \in G$,

$$a \cdot e = e \cdot a = a.$$

(3) **Existence of Inverses:** For every $a \in G$, there exists an element $b \in G$ such that

$$a \cdot b = b \cdot a = e.$$

 **Proposition 2.2 (Uniqueness of the Neutral Element):** Let $(G, \cdot)$ be a group. The neutral element $e \in G$ is unique.

Proof

Suppose $e, \hat{e} \in G$ are both neutral elements in G . Consider the product $e \cdot \hat{e}$:

- Since e is a neutral element, $e \cdot \hat{e} = \hat{e}$.
- Since $\hat{e}$ is a neutral element, $e \cdot \hat{e} = e$.

Equating both expressions yields:

$$\hat{e} = e \cdot \hat{e} = e.$$

Thus, the neutral element is unique.

Q.E.D.

III Uniqueness of Inverses

Spoken Transcript [00:03:22 - 00:05:04]

And then other questions you could have: if x is an element of the group, then this existence of inverse says that there exists a y such that $x \cdot y$ equals the neutral element. And you could say: could there be another $\hat{y}$ where $x \cdot \hat{y}$ is equal to e ?

And the answer is no, again. No! And how do you prove this? Well, you look at this equation, and you multiply both sides by the actual — the first one:

$$y \cdot (x \cdot \hat{y}) = y \cdot e = y.$$

And then this thing is equal by the original property — so the neutral element property is both sides —, so this thing is equal to e , so this is equal to $\hat{y}$. Okay?

Is that clear? So, all of this is embedded in some sense in the definitions. You could define a group by saying that there exists a neutral element and it’s unique, there exist inverses and they’re unique. You don’t have to; you could do that. That’s making the

definition bigger.

You can also try to make the definition smaller. There's some kind of strange direction where people think that, well, there's three axioms. We'd like a group to only have two axioms. And some guy in the '60s found some crazy way to write it with just one axiom, but it's just logically equivalent. You can go look that up if you want. I don't remember; it's some complicated axioms, you can imagine.

Uniqueness of Inverses in a Group

 **Proposition 2.3 (Uniqueness of Inverses):** Let $(G, \cdot, e)$ be a group. For each element $x \in G$, the inverse element $y \in G$ satisfying $x \cdot y = y \cdot x = e$ is unique.

Proof

Let $x \in G$. Suppose $y, \hat{y} \in G$ are both inverses of x , so that $x \cdot y = y \cdot x = e$ and $x \cdot \hat{y} = \hat{y} \cdot x = e$.

Multiplying the equation $x \cdot \hat{y} = e$ on the left by y gives:

$$y \cdot (x \cdot \hat{y}) = y \cdot e = y.$$

Using the associativity of the group operation on the left-hand side:

$$(y \cdot x) \cdot \hat{y} = y.$$

Since y is an inverse of x , we have $y \cdot x = e$. Substituting this in:

$$e \cdot \hat{y} = y \implies \hat{y} = y.$$

Therefore, the inverse of x is unique. We denote this unique inverse by x^{-1} .

Q.E.D.

 **Why Redundant Axioms Are Omitted:** The standard definition of a group requires only the **existence** of an identity and of inverses. As demonstrated above, their **uniqueness** is not an independent axiom but an immediate logical consequence of associativity and the existence axioms.

III Rings and Fields

Spoken Transcript [00:05:04 - 00:06:55]

Okay, so that's the group, and the other structure we had was this ring, which I don't repeat everything again. But the way to think about this is that if I look at just the additive part, this is an abelian group, and then I have some axioms for multiplication, and then two times the distributive laws. So, those were the axioms for the ring, and we did some examples.

So, certainly one part of this class is really internalizing these axioms. And there's some kind of idea about mathematics which is some kind of asymptotic understanding. Which is not just that you understand it, but like the really asymptotic understanding that you try to hope for, which you never really achieve, is that not only do you under-

stand it, but you cannot imagine the state of not understanding it. That's the thing you should really aim for. And you know, you can only do that kind of asymptotically. You can never really achieve that, but with these... So, this is just at the axiomatic level.

Spoken Transcript [00:06:55 - 00:08:02]

And then there was a notion of a field, and the field is somehow... the group is by itself an algebraic object. The field is a particular case of a ring, meaning a field is a ring with some additional structure, and the additional structure is that non-zero elements have multiplicative inverses.

So, you can think about this: in the whole world of rings, some of them are fields. But rings and groups are separate, although they are related by the fact that the additive part of the ring is an abelian group.

Okay, so those were the structures, and in algebra, after you have structures, there's the notion of the maps that respect the structures, which are the homomorphisms. And you've seen this before, and we discussed a little bit last week, but I want to put them on equal footing here.

Review of Rings and Fields

Definition 2.4 (Ring): A “ring” is a 5-tuple $(R, +, \cdot, 0, 1)$ consisting of a set R , two binary operations $+$ (addition) and $\cdot$ (multiplication), and two distinguished elements $0, 1 \in R$ ($0 \neq 1$) such that:

- (1) $(R, +, 0)$ is an “abelian group” (i.e., addition is associative, commutative, has identity 0 , and every element has an additive inverse $-a$).
- (2) Multiplication $\cdot$ is associative with identity 1 :

$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{and} \quad a \cdot 1 = 1 \cdot a = a \quad \forall a, b, c \in R.$$

- (3) Multiplication distributes over addition from both the left and the right:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c),$$

$$(a + b) \cdot c = (a \cdot c) + (b \cdot c) \quad \forall a, b, c \in R.$$

Definition 2.5 (Field): A “field” is a commutative ring $(F, +, \cdot, 0, 1)$ in which every non-zero element possesses a multiplicative inverse:

$$\forall x \in F \setminus \{0\}, \exists x^{-1} \in F \text{ such that } x \cdot x^{-1} = 1.$$

Equivalently, $(F \setminus \{0\}, \cdot, 1)$ is an abelian group under multiplication.

Insight: The Hierarchy of Algebraic Structures

Algebra organizes mathematical systems by layering structural axioms:

- **Groups** $(G, \cdot, e)$: A single operation with associativity, identity, and inverses.
- **Rings** $(R, +, \cdot, 0, 1)$: Two operations where $(R, +, 0)$ is an abelian group, multiplication is associative with 1 , and the operations are tied together via the distributive laws.
- **Fields** $(F, +, \cdot, 0, 1)$: Commutative rings where $(F \setminus \{0\}, \cdot, 1)$ is also an abelian

group.

Standing Convention: Commutativity of Rings

As established in Week 1 ([Remark 1.13](#)), unless explicitly stated otherwise, the term **ring** in this course always refers to a **commutative ring with unity** ($a \cdot b = b \cdot a$ for all $a, b \in R$). Non-commutative rings (such as matrix algebras $M_n(K)$ or endomorphism rings) are treated as specialized exceptions.

III Group Homomorphisms and Isomorphisms

III Definition and Properties of Homomorphisms

Spoken Transcript [00:08:02 - 00:09:40]

So, if you have two groups, then we want to consider maps that respect the group structure. So, this is called f is a group homomorphism if:

$$f(a \cdot b) = f(a) \cdot f(b) \quad \text{for all } a, b \in G.$$

Okay, so last week I did something complicated where we have G is $(G, \star)$ and then $\hat{G}$ is $(\hat{G}, \hat{\star})$ to be completely clear about the products being different. I'm never going to do that again. No one ever does that.

So, here when you read this, this multiplication is the operation in the first one, and that's the operation in the second one. Is that clear? They're both just called dot $(\cdot)$. And in fact, many people just don't write it at all. It'd be kind of normal to look in a book and there'd just be nothing, because that's the standard way you deal with algebra. You don't write $x \cdot y$, you just write xy .

Okay, so that's the definition of a group homomorphism. It respects the structure. It says you can multiply here and then go over by f , or you can go over separately and multiply. That's what's respecting the structure.

Spoken Transcript [00:09:40 - 00:10:40]

Then you could say: what about the neutral element? Why am I not saying anything about the neutral element? I could. I could add $f(e) = \hat{e}$, or if I want to be clear, $\hat{e}$, the e on that side.

I could add that, but I don't have to, because if I put e here, if I just use this property, I put $f(e \cdot e)$ — there's only one thing you can do always — so I just stick e in here. You agree?

And then I use the neutral property on this side: we get $f(e) = f(e) \cdot f(e)$.

And then I can use the cancellation property of the group: $f(e)$ has some inverse. There exists some $y \in \hat{G}$ such that $y \cdot f(e) = \hat{e}$, right? This is the cancellation property. We've used it many times, and at some point I will stop explaining it.

So, we multiply by y on both sides: we get $y \cdot f(e) = y \cdot (f(e) \cdot f(e))$. So, then this is

$\hat{e} = (y \cdot f(e)) \cdot f(e) = \hat{e} \cdot f(e) = f(e)$. So, $f(e) = \hat{e}$. Okay?
So, I don't have to, but you could put it in there.

Group Homomorphisms and Preservation of Identities

 **Definition 2.6 (Group Homomorphism):** Let $(G, \cdot, e)$ and $(\hat{G}, \cdot, \hat{e})$ be groups. A function $f: G \rightarrow \hat{G}$ is called a “group homomorphism” (or simply “homomorphism”) if it respects the group operation, that is:

$$f(a \cdot b) = f(a) \cdot f(b) \quad \forall a, b \in G. \quad (2.1)$$

 **Proposition 2.7 (Preservation of the Neutral Element):** Let $f: G \rightarrow \hat{G}$ be a group homomorphism. Then f automatically maps the identity of G to the identity of $\hat{G}$:

$$f(e) = \hat{e}. \quad (2.2)$$

Proof

Since $e \cdot e = e$ in G , applying the homomorphism property yields:

$$f(e) = f(e \cdot e) = f(e) \cdot f(e).$$

Since $f(e) \in \hat{G}$ belongs to a group, it possesses an inverse $y \in \hat{G}$ such that $y \cdot f(e) = \hat{e}$. Multiplying both sides of the equation on the left by y :

$$\begin{aligned} y \cdot f(e) &= y \cdot (f(e) \cdot f(e)) \\ \hat{e} &= (y \cdot f(e)) \cdot f(e) && \text{(by Associativity)} \\ \hat{e} &= \hat{e} \cdot f(e) \\ \hat{e} &= f(e). \end{aligned} \quad (2.3)$$

Thus, $f(e) = \hat{e}$.

Q.E.D.

III Group Isomorphisms

Spoken Transcript [00:10:40 - 00:11:47]

So, that's for the group. The group is the algebraic structure, and if you have two groups, you compare them with a homomorphism. It's a map that respects the structure.

And this is following the pattern of this kind of analysis in linear algebra also. There you have two vector spaces, and the map is a linear transformation. That's the notion of the map that respects the vector space structure on both sides.

And then there's some standard things to say: we say f is a group isomorphism if f is a group homomorphism and f is bijective. Okay? So, this should not surprise you. You've seen these terms in linear algebra, right?

Which language did you learn linear algebra in?

Student Answer

English.

Spoken Transcript [00:11:49 - 00:13:00]

English, okay, well, that's no problem then.

And there's another way to say this: f is a group isomorphism if and only if there exists an inverse, there exists a group homomorphism in the other direction. There exists a group homomorphism $g: \hat{G} \rightarrow G$ such that if I do f first and then g , I get the identity on the first group, and if I do g first and then f , I get the identity on the second group. These are equivalent. Okay?

So, if you have the bottom, it's certainly bijective, because you have the forward and backwards map. And if you have the top, since it's bijective there exists an inverse function, and you have to show that that's a homomorphism, which I leave to you to show. Okay? So, I hope this is clear.

Definition and Characterization of Group Isomorphisms

 **Definition 2.8 (Group Isomorphism):** Let G and $\hat{G}$ be groups. A map $f: G \rightarrow \hat{G}$ is a “group isomorphism” if:

- (1) f is a group homomorphism, and
- (2) f is bijective.

If such an isomorphism exists, we say that G and $\hat{G}$ are “isomorphic” and write $G \cong \hat{G}$.

 **Proposition 2.9 (Alternative Invertibility Characterization):** A map $f: G \rightarrow \hat{G}$ is a group isomorphism if and only if there exists a group homomorphism $g: \hat{G} \rightarrow G$ such that:

$$g \circ f = \text{id}_G \quad \text{and} \quad f \circ g = \text{id}_{\hat{G}}. \quad (2.4)$$

Proof

Direction ($\Leftarrow$): If such a homomorphism g exists, the two-sided inverse equations $g \circ f = \text{id}_G$ and $f \circ g = \text{id}_{\hat{G}}$ imply by standard set theory that f is a bijection. Since f is also given as a homomorphism, f is an isomorphism.

Direction ($\Rightarrow$): If f is an isomorphism, it is a bijective homomorphism, so a set-theoretic inverse function $g = f^{-1}: \hat{G} \rightarrow G$ exists satisfying (2.4). It remains only to verify that g is a homomorphism. For any $\hat{a}, \hat{b} \in \hat{G}$, let $a = g(\hat{a})$ and $b = g(\hat{b})$, so that $f(a) = \hat{a}$ and $f(b) = \hat{b}$. Then:

$$f(a \cdot b) = f(a) \cdot f(b) = \hat{a} \cdot \hat{b}.$$

Applying $g = f^{-1}$ to both sides gives:

$$g(\hat{a} \cdot \hat{b}) = a \cdot b = g(\hat{a}) \cdot g(\hat{b}).$$

Thus, g is a group homomorphism.

Q.E.D.

Examples of Groups and Isomorphisms over $\mathbb{R}$

Three Classical Groups of Real Numbers

Spoken Transcript [00:13:00 - 00:15:26]

All right, so maybe an example.

If I take the real numbers, one group is the real numbers with addition and zero: $(\mathbb{R}, +, 0)$. This is a rather basic group: real numbers, zero, you can add them.

Another group which you maybe don't think about as much is you could take the non-zero real numbers, multiplication, and one: $(\mathbb{R}^*, \cdot, 1)$. So, here I just remind you — this is pretty standard —, when you have a field and you put a star (*), this usually means you take the field and you throw out the zero:

$$\mathbb{R}^* := \mathbb{R} \setminus \{0\}.$$

Is that clear? So, this is the non-zero real numbers, and I put multiplication, one. That is also a group. Are you happy with that? Because you can invert non-zero elements.

And there's another one here: there's kind of two ways to be non-zero: you can be positive, or you can be negative. So, I could take the positive ones: $(\mathbb{R}_{>0}^*, \cdot, 1)$. That's still a group. Is that clear?

If I try to then take the negative ones, I kind of get in trouble because if I multiply negative ones I get positive, and the one is not there. So, anyway, there's these three groups.

So, which of these are isomorphic? Or which ones are not isomorphic? Do you know this? Let me write the question: Which are isomorphic? Which means: for which ones can I find an isomorphism?

Three Standard Groups over $\mathbb{R}$

We consider the following three algebraic groups defined on subsets of $\mathbb{R}$:

(1) The Additive Group of Real Numbers:

$$(\mathbb{R}, +, 0)$$

where $+$ is standard addition, 0 is the neutral element, and the inverse of x is $-x$.

(2) The Multiplicative Group of Non-Zero Real Numbers:

$$(\mathbb{R}^*, \cdot, 1) \quad \text{where } \mathbb{R}^* := \mathbb{R} \setminus \{0\},$$

equipped with standard multiplication, identity 1 , and inverse $x^{-1} = \frac{1}{x}$.

(3) The Multiplicative Group of Positive Real Numbers:

$$(\mathbb{R}_{>0}^*, \cdot, 1) \quad \text{where } \mathbb{R}_{>0}^* := \{x \in \mathbb{R} \mid x > 0\},$$

which forms a group under multiplication since the product and inverse of positive numbers remain strictly positive.

Non-Isomorphism between $(\mathbb{R}^*, \cdot, 1)$ and $(\mathbb{R}_{>0}^*, \cdot, 1)$

 Spoken Transcript [00:15:26 - 00:16:21]

So, let's start with these — this is maybe the more confusing one. If I take the non-zero real numbers, multiplication, 1, and I take the positive ones, $(\mathbb{R}_{>0}^*, \cdot, 1)$, could these be isomorphic?

It's kind of strange, because if you draw the picture, one of them is kind of both sides, and the other one is just one side. So, it seems kind of unlikely, although, you know, there's still bijections that do this. There are set-theoretic bijections, because both sets are just uncountable sets; they have the same cardinality.

What do you think?

Yeah?

 Student Answer

No, because we run into a problem for -1 , because if we had if $f(-1)$...

 Spoken Transcript [00:16:30 - 00:16:33]

Which way is f going?

 Student Answer continued

So, f goes from $\mathbb{R}^*$ to $\mathbb{R}_{>0}^*$.

 Spoken Transcript [00:16:37 - 00:16:39]

Yeah.

 Student Answer continued

And then $f(1)$ has to be 1.

 Spoken Transcript [00:16:42 - 00:16:45]

Has to be 1, that's the...

 Student Answer continued

Yeah, that's 1, but that's also $f(-1)$ squared.

 Spoken Transcript [00:16:48 - 00:16:50]

Yeah, so $f(-1)$ equals what? That's a good...

Student Answer continued

$f(-1) \cdot f(-1)$ will be 1, so the only positive number such that times itself is 1 will be 1, so $f(-1)$ has to be 1. So, $f(1) = f(-1)$, and then it's not bijective.

Spoken Transcript [00:17:09 - 00:18:00]

Yeah, this is all correct. So, the simplest way to say it is that $f(-1)$ has to be a square root of 1, okay, by this law. And there's only one square root of 1 in the positive, and that's already taken by 1. So, it violates the bijectivity.

Do you understand this statement? The point is that 1 has two square roots in here. There are two elements when you multiply them by themselves you get 1: +1 and -1. And those two elements have to go to two different elements here, because it's a bijection. If there were a bijection, they would have to go to two different elements. But here there's only one element whose square is 1. So, it's impossible that there is a bijection. Okay?

So, this is a solid argument. This is the standard argument.

Non-Isomorphism of $(\mathbb{R}^*, \cdot, 1)$ and $(\mathbb{R}_{>0}^*, \cdot, 1)$

 **Proposition 2.10:** The groups $(\mathbb{R}^*, \cdot, 1)$ and $(\mathbb{R}_{>0}^*, \cdot, 1)$ are not isomorphic:

$$(\mathbb{R}^*, \cdot, 1) \not\cong (\mathbb{R}_{>0}^*, \cdot, 1).$$

Proof

Suppose for contradiction that there exists an isomorphism $f: \mathbb{R}^* \rightarrow \mathbb{R}_{>0}^*$.

By [Proposition 2.7](#), any homomorphism must preserve the identity:

$$f(1) = 1. \quad (2.5)$$

Now consider the element $-1 \in \mathbb{R}^*$. Since $(-1) \cdot (-1) = 1$, the homomorphism property implies:

$$f(-1)^2 = f(-1) \cdot f(-1) = f((-1) \cdot (-1)) = f(1) = 1.$$

Thus, $f(-1)$ must be a real root of the equation $t^2 = 1$ in the codomain $\mathbb{R}_{>0}^*$. In the set of positive real numbers $\mathbb{R}_{>0}^*$, the only solution to $t^2 = 1$ is $t = 1$. Therefore:

$$f(-1) = 1. \quad (2.6)$$

Comparing (2.5) and (2.6), we obtain $f(1) = f(-1) = 1$. Since $1 \neq -1$ in $\mathbb{R}^*$, this contradicts the injectivity of f . Consequently, no such isomorphism can exist.

Q.E.D.

 **Algebraic Invariants: Torsion Elements:** This proof illustrates the power of “*algebraic invariants*”: an algebraic property preserved by all isomorphisms. In $(\mathbb{R}^*, \cdot, 1)$, the element -1 is an element of order 2 (an element $x \neq e$ such that $x^2 = e$, called a “*torsion element*”). However, $(\mathbb{R}_{>0}^*, \cdot, 1)$ contains no elements of order 2. Since an isomorphism must map elements of order k to elements of order k , the two groups cannot be isomorphic despite having the same uncountable cardinality ($2^{\aleph_0}$) as sets.

III Isomorphism between $(\mathbb{R}, +, 0)$ and $(\mathbb{R}_{>0}^*, \cdot, 1)$

Spoken Transcript [00:18:00 - 00:18:22]

And what about this one? $(\mathbb{R}, +, 0)$ and $(\mathbb{R}_{>0}^*, \cdot, 1)$?
 If I make any mistakes on the board, you have to tell me, all right?
 So, what about this? Could these be isomorphic?
 Yeah?

Student Answer

I think it could, we can take f from $\mathbb{R}$ to $\mathbb{R}_{>0}^*$ by the function e to the power of x .

Spoken Transcript [00:18:32 - 00:19:30]

Yeah, yeah. This is like the most famous function in mathematics, and this is its purpose, actually.

This is by $\exp$, $\exp(x)$, so $f(x) = \exp(x)$. So, you take a number, and you get a positive number. If you take 0, you get 1. And the main property of $\exp$ is that it intertwines the addition here with the multiplication there.

So, this is actually the most famous function in mathematics, basically, and this is its role. It gives you an isomorphism, a group isomorphism, between the additive structure and the multiplicative structure.

So, that's a kind of weird thing. It's saying that actually from the point of view of the group, the group doesn't see the difference between the additive structure and the multiplicative structure of the reals.

The Exponential Isomorphism

 **Proposition 2.11:** The additive group of real numbers $(\mathbb{R}, +, 0)$ is isomorphic to the multiplicative group of positive real numbers $(\mathbb{R}_{>0}^*, \cdot, 1)$:

$$(\mathbb{R}, +, 0) \cong (\mathbb{R}_{>0}^*, \cdot, 1).$$

Proof

Define the map $f: \mathbb{R} \rightarrow \mathbb{R}_{>0}^*$ by the exponential function:

$$f(x) := \exp(x) = e^x. \quad (2.7)$$

We verify the two isomorphism conditions:

(1) Homomorphism Property: For all $x, y \in \mathbb{R}$, by the functional equation of the exponential function:

$$f(x + y) = e^{x+y} = e^x \cdot e^y = f(x) \cdot f(y).$$

Thus, f transforms addition in $(\mathbb{R}, +)$ into multiplication in $(\mathbb{R}_{>0}^*, \cdot)$.

(2) Bijectivity: The exponential function $\exp: \mathbb{R} \rightarrow \mathbb{R}_{>0}^*$ is strictly monotonically increasing and continuous, with an explicit two-sided inverse given by the natural

logarithm:

$$g: \mathbb{R}_{>0}^* \rightarrow \mathbb{R}, \quad g(y) := \ln(y).$$

For all $x \in \mathbb{R}$, $g(f(x)) = \ln(e^x) = x = \text{id}_{\mathbb{R}}(x)$, and for all $y \in \mathbb{R}_{>0}^*$, $f(g(y)) = e^{\ln(y)} = y = \text{id}_{\mathbb{R}_{>0}^*}(y)$.

Therefore, f is a group isomorphism.

Q.E.D.

Insight: The Fundamental Role of the Exponential Function

From an algebraic perspective, the exponential function $x \mapsto e^x$ is not merely an analytic power series or a solution to $y' = y$; its primary algebraic purpose is to establish an exact equivalence (“*group isomorphism*”) between the continuous additive world $(\mathbb{R}, +)$ and the continuous positive multiplicative world $(\mathbb{R}_{>0}^*, \cdot)$.

The Kernel and Subgroups

Definition and Properties of the Kernel

Spoken Transcript [00:19:30 - 00:21:10]

There’s a notion of a subgroup. We did this for the ring; I’m going to repeat it all for the ring in one second. So, if you have this map, $f: G \rightarrow \hat{G}$ — and I’m not going to keep saying group homomorphism. The standard is to say homomorphism, which is abbreviated hom, and you’re supposed to understand, since I’m talking about groups, it’s a group homomorphism.

All right, so if f is a group homomorphism, then from the spirit of linear algebra, which is the algebra you’re supposed to be most familiar with, if you have a linear transformation between two vector spaces, the thing you immediately study is the kernel and the image of that linear transformation. So, we can do that also with groups.

So, then we can consider the kernel and the image, but let’s start with the kernel. So, what properties does this kernel enjoy?

The kernel of a group homomorphism is defined to be those elements in G such that when I apply f , I get the neutral element on the second side. In the linear algebra language, it’s things that went to zero, so now it’s the elements that go to the neutral element. Is that clear?

Well, the neutral element itself is in here. That’s one property, because the homomorphism takes the neutral element here to the neutral element there. So, that’s there, right?

Spoken Transcript [00:21:10 - 00:22:15]

Second property: suppose a and b are in the kernel of f , then what can we say? We can consider what happens to $f(ab)$.

There’s only one thing we can do here. These proofs at this level are strong proofs,

meaning that there's only one thing you can do at every stage, and you just have to do it.

So, this is $f(a) \cdot f(b)$, and now the assumption is they're both in the kernel, so these are both $\hat{e} \cdot \hat{e}$, and then we use the identity property, so that's $\hat{e}$.

Okay, so what have we learned here? That says that if a and b are in the kernel, we have proven that $a \cdot b$ is in the kernel. Okay?

Definition and Closure Properties of the Kernel

 **Definition 2.12 (Kernel of a Group Homomorphism):** Let $f: G \rightarrow \hat{G}$ be a group homomorphism between groups $(G, \cdot, e)$ and $(\hat{G}, \cdot, \hat{e})$. The “kernel” of f , denoted $\ker(f)$, is the pre-image of the identity element $\hat{e} \in \hat{G}$:

$$\ker(f) := \{x \in G \mid f(x) = \hat{e}\} \subseteq G. \quad (2.8)$$

 **Proposition 2.13 (Properties of the Kernel: Identity and Product Closure):** Let $f: G \rightarrow \hat{G}$ be a group homomorphism. Then $\ker(f)$ satisfies:

- (1) **Contains the Identity:** $e \in \ker(f)$.
- (2) **Closed under the Group Product:** If $a, b \in \ker(f)$, then $a \cdot b \in \ker(f)$.

Proof

Property (1): By Proposition 2.7, $f(e) = \hat{e}$. Thus, $e \in \ker(f)$ by definition.

Property (2): Let $a, b \in \ker(f)$, which means $f(a) = \hat{e}$ and $f(b) = \hat{e}$. Using the homomorphism property of f :

$$f(a \cdot b) = f(a) \cdot f(b) = \hat{e} \cdot \hat{e} = \hat{e}.$$

Therefore, $a \cdot b \in \ker(f)$.

Q.E.D.

III Closure under Inverses and the Definition of Subgroups

Spoken Transcript [00:22:15 - 00:24:05]

So, we're looking at properties of the kernel, and we have discovered that the neutral element's in the kernel, and the kernel is closed under the group law, the product.

We might be curious now whether the inverses are there also.

So, that's the third property of the kernel: if x is in the kernel of f , then the inverse...

I should make this comment that very often the inverse for the group law is denoted x^{-1} . That's just the name of it. The axioms say there exists an inverse. This is very often the name that's given to it. This kind of presupposes that the product you're calling dot $(\cdot)$. If the product you're calling is plus $(+)$, then you usually use the minus sign $(-x)$ here. But this is just notation. You could call it anything you want. You could call it Fred, or an elephant, anything. We're calling it this, okay?

Then that's the inverse is also in the kernel. Okay?

Spoken Transcript [00:24:05 - 00:26:00]

And how are you going to prove this? You have to start out by taking $f(x \cdot x^{-1})$. That's the one thing you know.

On the one hand, this is $f(e)$, and that's $\hat{e}$. On the other hand, this is $f(x) \cdot f(x^{-1})$, and this thing is assumed to be in the kernel, so this is equal to $\hat{e} \cdot f(x^{-1})$.

So, we get that the neutral element is equal to neutral element times that ($\hat{e} = \hat{e} \cdot f(x^{-1})$), so that must be the neutral element. Okay? We could go one more if we want to be really explicit:

$$f(x^{-1}) = \hat{e} \cdot f(x^{-1}) = \hat{e}.$$

So, now we have the full chain: $f(x^{-1}) = \hat{e}$.

So, we have these three properties: that the kernel enjoys these three properties. It contains the neutral element of G , the source group, it's closed under the group product, and it's closed under inverses.

Definition: if H is a subset of G , then H is a subgroup if $e \in H$, H is closed under the product, and H is closed under inverses. Meaning if anybody's in H , then its multiplicative inverse is also in H , okay?

Closure of the Kernel under Inverses and the Subgroup Axioms

 **Proposition 2.14 (Closure of the Kernel under Inverses):** Let $f: G \rightarrow \hat{G}$ be a group homomorphism. If $x \in \ker(f)$, then $x^{-1} \in \ker(f)$.

Proof

Let $x \in \ker(f)$, so that $f(x) = \hat{e}$. Consider the image of the identity element $e = x \cdot x^{-1}$:

$$\begin{aligned} \hat{e} &= f(e) && \text{(by Proposition 2.7)} \\ &= f(x \cdot x^{-1}) \\ &= f(x) \cdot f(x^{-1}) && \text{(homomorphism property)} \\ &= \hat{e} \cdot f(x^{-1}) && \text{(since } x \in \ker(f)) \\ &= f(x^{-1}) && \text{(neutral element property in } \hat{G}). \end{aligned}$$

Thus, $f(x^{-1}) = \hat{e}$, which demonstrates that $x^{-1} \in \ker(f)$.

Q.E.D.

 **Definition 2.15 (Subgroup):** Let $(G, \cdot, e)$ be a group, and let $H \subseteq G$ be a non-empty subset. H is called a "*subgroup*" of G (denoted $H \leq G$) if it satisfies the following three conditions:

- (1) **Identity Element:** $e \in H$.
- (2) **Closure under Multiplication:** For all $a, b \in H$, $a \cdot b \in H$.
- (3) **Closure under Inverses:** For all $x \in H$, $x^{-1} \in H$.

 **Inherited Group Structure:** If $H \subseteq G$ satisfies these three axioms, the restriction of the group operation $\cdot|_{H \times H}$ defines a binary operation on H , associativity is inherited from G , and $e \in H$ acts as the identity with inverses in H . Consequently, $(H, \cdot, e)$ is itself a group in its own right.

III The Kernel and the Image as Subgroups

Spoken Transcript [00:26:00 - 00:26:50]

So, in particular, this implies that if I take H and I take the group law, which makes sense on H because H is closed under it, and I take the zero, which also makes sense because zero's in H , that this is a group.

Which is a very nice thing. A subgroup is a group. It'd be terrible if it weren't, right?

So, that's the definition for a subgroup. And now we've proven a little proposition: The kernel of f is a subgroup of G .

Is that clear?

Subgroup Property of the Kernel

 **Proposition 2.16 (Kernel as a Subgroup):** Let $f: G \rightarrow \hat{G}$ be a group homomorphism. Then the kernel $\ker(f)$ is a subgroup of the domain group G :

$$\ker(f) \leq G. \quad (2.9)$$

Proof

By [Proposition 2.7](#), $e \in \ker(f)$. By [Proposition 2.13](#), $\ker(f)$ is closed under multiplication. By [Proposition 2.14](#), $\ker(f)$ is closed under taking inverses. Hence, by [Definition 2.15](#), $\ker(f) \leq G$.

Q.E.D.

III The Image of a Group Homomorphism

Spoken Transcript [00:26:50 - 00:28:30]

Okay, so that's the investigation of the kernel and the notion of a subgroup. But there's also the image. And so proposition — and I leave this for you, you can prove it —: The image of f is a subgroup of $\hat{G}$.

So, how do you prove this? You just check that the image contains the neutral element... it has to because the neutral element from the first group goes to the neutral element of the second group, so the image has to contain the neutral element. And then you have to check that if we have two elements in the image, then if I apply the group law for $\hat{G}$, they stay in the image, and the inverse. The proofs are all just the same. You just follow your nose. There's only one thing you can do, and you can do it.

All right. So, this is a nice situation for groups, that the understanding of the group is by these very simple axioms. There's lots of examples; in fact, part of this class is the examples for the groups, which will come later. And whenever you have a homomorphism, there's the kernel and the image.

So, this one I leave to you. You can look in the books too, but it's not worth looking in the books. You should read the books, I take it back. But the problem with reading

such a simple thing in the book is that you have to learn what notation the author is using and all that. It's much simpler just to write the proof of this yourself.

The Image of a Group Homomorphism

Definition 2.17 (Image of a Homomorphism): Let $f: G \rightarrow \hat{G}$ be a group homomorphism. The “*image*” of f , denoted $\text{Im}(f)$ (or $f(G)$), is the set:

$$\text{Im}(f) := \{\hat{y} \in \hat{G} \mid \exists x \in G \text{ such that } f(x) = \hat{y}\} \subseteq \hat{G}. \quad (2.10)$$

Proposition 2.18 (Image as a Subgroup): Let $f: G \rightarrow \hat{G}$ be a group homomorphism. Then the image $\text{Im}(f)$ is a subgroup of the codomain group $\hat{G}$:

$$\text{Im}(f) \leq \hat{G}. \quad (2.11)$$

Proof

We verify the three subgroup conditions for $\text{Im}(f) \subseteq \hat{G}$:

- (1) **Identity Element:** Since $f(e) = \hat{e}$ by [Proposition 2.7](#), $\hat{e} \in \text{Im}(f)$.
- (2) **Closure under Multiplication:** Let $\hat{y}_1, \hat{y}_2 \in \text{Im}(f)$. Then there exist $x_1, x_2 \in G$ such that $f(x_1) = \hat{y}_1$ and $f(x_2) = \hat{y}_2$. Using the homomorphism property:

$$\hat{y}_1 \cdot \hat{y}_2 = f(x_1) \cdot f(x_2) = f(x_1 \cdot x_2).$$

Since $x_1 \cdot x_2 \in G$, we have $\hat{y}_1 \cdot \hat{y}_2 \in \text{Im}(f)$.

- (3) **Closure under Inverses:** Let $\hat{y} \in \text{Im}(f)$ with $\hat{y} = f(x)$ for some $x \in G$. Since $x^{-1} \in G$, we compute:

$$\hat{y} \cdot f(x^{-1}) = f(x) \cdot f(x^{-1}) = f(x \cdot x^{-1}) = f(e) = \hat{e}.$$

By uniqueness of inverses in $\hat{G}$, $\hat{y}^{-1} = f(x^{-1}) \in \text{Im}(f)$.

Thus, $\text{Im}(f) \leq \hat{G}$.

Q.E.D.

Spoken Transcript [00:28:30 - 00:29:39]

But actually, I'm a big fan of books. It's one way to understand this kind of classical history of mathematics: read a lot of math books. And I know that somehow in the modern world, this is less popular, but in my generation we read a lot of math books.

And on the topic of books, there's the book for the course, which was picked a while ago, not by me even, and it has the advantage that it's freely available on the ETH website, so you can just click it and... Then there's some other references I gave which are, I think, a little bit better books. They're more advanced in perspective, they're written better.

But I think they're not freely available on the webpage, so it wasn't chosen for this class. But it's a good idea to read books. You'll learn something.

Pedagogical Aside: Recommended Literature

The lecturer emphasizes the pedagogical value of reading classical mathematical texts and consulting textbook references, noting the freely available course textbook on the

ETH website alongside advanced supplementary reading.

Ring Homomorphisms, Ideals, and Subrings

Ring Homomorphisms and Isomorphisms

Spoken Transcript [00:29:39 - 00:31:49]

Okay, so for the ring, we had also defined what a homomorphism is: it's a ring homomorphism. If I have two rings, then we had defined f is a ring hom, which, as I said, I will just say hom and you're supposed to understand they're rings. And then the axioms for a ring hom, if you remember, the first one is:

$$f(a + b) = f(a) + f(b).$$

And the interpretation of this: it says that the additive part of this is a group and the additive part of that is a group, and the ring hom should be a group hom for those parts. Is that clear? So, additively f is a group hom for the plus part, okay?

Of course, we also have this, which respects the multiplication: $f(ab) = f(a)f(b)$.

And then we also have that the identity has to be respected: $f(1) = 1$.

Okay? So, those are the notions of the ring homomorphism. And again, a ring isomorphism is a bijective ring homomorphism.

And another way to think about it is if there's an isomorphism, then that means that the inverse is a ring homomorphism in the other direction, just as I said with groups.

Review of Ring Homomorphisms and Isomorphisms

Definition 2.19 (Ring Homomorphism): Let $(R, +, \cdot, 0, 1)$ and $(\hat{R}, +, \cdot, \hat{0}, \hat{1})$ be rings. A map $f: R \rightarrow \hat{R}$ is a "*ring homomorphism*" if for all $a, b \in R$:

- (1) **Additive Homomorphism:** $f(a + b) = f(a) + f(b)$ (i.e., $f: (R, +, 0) \rightarrow (\hat{R}, +, \hat{0})$ is a group homomorphism on the additive group).
- (2) **Multiplicative Homomorphism:** $f(a \cdot b) = f(a) \cdot f(b)$.
- (3) **Preservation of the Multiplicative Identity:** $f(1) = \hat{1}$.

Definition 2.20 (Ring Isomorphism): A ring homomorphism $f: R \rightarrow \hat{R}$ is called a "*ring isomorphism*" if f is bijective. Equivalently, f is an isomorphism if and only if there exists a ring homomorphism $g: \hat{R} \rightarrow R$ such that $g \circ f = \text{id}_R$ and $f \circ g = \text{id}_{\hat{R}}$.

The Kernel of a Ring Homomorphism and Ideals

Spoken Transcript [00:31:49 - 00:34:03]

Okay, so then we can do the parallel study: if $f: R \rightarrow \hat{R}$ is a ring homomorphism, there are two things I can study as before: the kernel of f and the image of f . And we had studied the kernel last week: for the group, both the kernel and image were

subgroups: the former was the subgroup of the source group and the latter was the subgroup of the target group. But for rings, the symmetry is broken, so one has to concentrate a bit here.

And that's also true for vector spaces. For vector spaces, you have a linear map, the kernel is the subspace of the first one, and the image is the subspace of the second one. For the ring, something different happens.

And for the kernel of f , we had proven last week that the kernel of f is something called an ideal. An ideal I of a ring satisfies the following condition: one condition on the additive structure is that $(I, +, 0)$ inside $(R, +, 0)$ is a subgroup. That's the additive structure.

Spoken Transcript [00:34:03 - 00:35:23]

So for an ideal, there's kind of two pieces. One is how it works additively, and it must be a subgroup, and it's not a surprise because we already said that f on the additive structures is a group homomorphism, and we already know that the kernel from the point of view of the group homomorphism is an additive subgroup.

And then the second has to do with the multiplicative part. The kernel satisfies this property that if I take any element of the ring $r \in R$ and any element of the ideal $x \in I$, then $r \cdot x$ is in the ideal. So, these are the two conditions that define an ideal. And the proposition last week was: the kernel is an ideal. All right?

So, what about the image? Is the image an ideal? What is the image? The answer is no. Ideal is the notion for the kernel. The image is something else: the image is a subring.

Ideals in Rings and the Kernel as an Ideal

 **Definition 2.21 (Ideal):** Let $(R, +, \cdot, 0, 1)$ be a commutative ring. A subset $I \subseteq R$ is called an “*ideal*” (denoted $I \trianglelefteq R$) if it satisfies:

(1) **Additive Subgroup:** $(I, +, 0)$ is a subgroup of $(R, +, 0)$, i.e.,

$$0 \in I, \quad \forall x, y \in I \implies x + y \in I, \quad \text{and} \quad \forall x \in I \implies -x \in I.$$

(2) **Multiplicative Absorption (Ideal Property):** For every element of the ring $r \in R$ and every element of the ideal $x \in I$, the product belongs to I :

$$\forall r \in R, \forall x \in I \implies r \cdot x \in I.$$

Ideals and Commutativity

Because R is assumed commutative, left ideals ($r \cdot x \in I$), right ideals ($x \cdot r \in I$), and two-sided ideals ($r \cdot x \cdot s \in I$) coincide. In non-commutative ring theory, these concepts diverge, and kernels of ring homomorphisms are always two-sided ideals.

 **Proposition 2.22 (The Kernel of a Ring Homomorphism is an Ideal):** Let $f: R \rightarrow \hat{R}$ be a ring homomorphism. Then the kernel $\ker(f) := \{x \in R \mid f(x) = \hat{0}\}$ is an ideal of R .

Proof

We verify the two ideal conditions for $\ker(f)$:

- (1) **Additive Subgroup:** Since $f: (R, +, 0) \rightarrow (\hat{R}, +, \hat{0})$ is a group homomorphism between abelian groups, $\ker(f)$ is an additive subgroup of $(R, +, 0)$ by Proposition 2.16.
- (2) **Multiplicative Absorption:** Let $r \in R$ and $x \in \ker(f)$, so $f(x) = \hat{0}$. Using the multiplicative homomorphism property:

$$f(r \cdot x) = f(r) \cdot f(x) = f(r) \cdot \hat{0} = \hat{0}.$$

Thus, $r \cdot x \in \ker(f)$.

Therefore, $\ker(f) \trianglelefteq R$.

Q.E.D.

Insight: Broken Symmetry: Subrings vs. Ideals

In group theory and linear algebra, substructures are uniform: the kernel and image of a homomorphism are both subgroups (or vector subspaces). In ring theory, this symmetry is broken:

- The **kernel** $\ker(f)$ is an **ideal** (it absorbs multiplication from the entire ring, but does not contain 1 unless $\hat{R} = \{0\}$).
- The **image** $\text{Im}(f)$ is a **subring** (it contains the identity 1, but does not absorb multiplication from outside the image).

III Subrings and the Image of a Ring Homomorphism

Spoken Transcript [00:36:04 - 00:37:50]

So, what is a subring? Let R be a ring. Whenever I say let R be a ring, you should be thinking that I have in mind all the data: $(R, +, \cdot, 0, 1)$.

A subring S of R satisfies the following properties — this is the definition. The first property is that 0 and 1 should be in the subring. The second is that S is closed under the two operations: whenever I have elements of the subring, I can either add them or multiply them, and they stay in the subring, okay? And the third is that $(S, +, \cdot, 0, 1)$ is a ring. A subring inside here has 0 and 1, it's closed under the operations, and together it's a ring. This means it has the additive inverses and things like that, okay?

Spoken Transcript [00:37:50 - 00:40:21]

And so the proposition here is: if I have a ring homomorphism, then the image of f in the second one is a subring.

The kernel is an ideal, the image is a subring. You have to check that the image satisfies these properties. Does the image have 0? The answer is yes, because on the additive structure, it's a homomorphism of groups, and 0 goes to 0. Does the image have 1? Well, that's part of the definition of a ring homomorphism, so it has 1. Then

you have to check whether it's closed under $+$ and $\cdot$, and you just check that by writing out the equations. And whether this is a ring, you have to check whether it has the additive inverses, but that's also clear.

Okay. So, it's good to have these ideas extremely clear in your head: you have the group, you have the ring, there's a notion of group hom, there's a notion of ring hom, and there's kernel and image in both cases. For the group, the kernel and the image are subgroups; for the ring, something different happens: the image is a subring, but the kernel is an ideal. And so now we should do some examples.

Definition of a Subring and the Image of a Ring Homomorphism

 **Definition 2.23 (Subring):** Let $(R, +, \cdot, 0, 1)$ be a commutative ring with unity. A subset $S \subseteq R$ is called a “*subring*” of R (denoted $S \leq R$) if:

- (i) **Contains Neutral Elements:** $0 \in S$ and $1 \in S$.
- (ii) **Closure under Addition and Multiplication:** For all $a, b \in S$,

$$a + b \in S \quad \text{and} \quad a \cdot b \in S.$$

- (iii) **Ring Axioms (Closure under Additive Inverses):** $(S, +, \cdot, 0, 1)$ is itself a commutative ring with the induced operations (in particular, $-a \in S$ for all $a \in S$).

 **Proposition 2.24 (The Image of a Ring Homomorphism is a Subring):** Let $f: R \rightarrow \hat{R}$ be a ring homomorphism. Then the image $\text{Im}(f) \subseteq \hat{R}$ is a subring of $\hat{R}$.

Proof

We check the subring conditions for $\text{Im}(f)$:

- (1) **Neutral Elements:** Since $f: (R, +, 0) \rightarrow (\hat{R}, +, \hat{0})$ is a group homomorphism, $f(0) = \hat{0} \in \text{Im}(f)$. By axiom (3) of ring homomorphisms, $f(1) = \hat{1} \in \text{Im}(f)$.
- (2) **Closure under Addition and Multiplication:** Let $\hat{a}, \hat{b} \in \text{Im}(f)$ with $\hat{a} = f(a)$ and $\hat{b} = f(b)$ for $a, b \in R$. Then:

$$\hat{a} + \hat{b} = f(a) + f(b) = f(a + b) \in \text{Im}(f),$$

$$\hat{a} \cdot \hat{b} = f(a) \cdot f(b) = f(a \cdot b) \in \text{Im}(f).$$

- (3) **Additive Inverses:** For $\hat{a} = f(a) \in \text{Im}(f)$, $-\hat{a} = -f(a) = f(-a) \in \text{Im}(f)$.

Thus, $\text{Im}(f)$ is a subring of $\hat{R}$.

Q.E.D.

Examples of Ideals and Subrings

The Even Integers $2\mathbb{Z}$ in $\mathbb{Z}$

Spoken Transcript [00:40:21 - 00:40:57]

Okay, let's try here: $2\mathbb{Z}$ inside $\mathbb{Z}$. So, this is a ring, $\mathbb{Z}$. I can take the even integers.

So, what is this inside that? Is it an ideal? Is it a subring?

What's the answer?

Student Answer

I think it is an ideal.

Spoken Transcript [00:41:00 - 00:41:35]

It's an ideal. So, claim: this is an ideal.

Why is it an ideal? We have to check that additively it's a subgroup: if you add two even numbers you get an even number, 0 is an even number, if you have an even number the minus of it is an even number, so you're totally solid on the subgroup. Then if you take any other element of the integers and multiply it by an even number, you stay even. So, that looks pretty good. Answer is yes.

Okay, what about a subring? Is it a subring?

Student Answer

No, because we don't have the 1.

Spoken Transcript [00:41:38 - 00:41:48]

Yeah, so it's not a subring. Ideal: yes. Subring: no.

Okay, so let's try something else.

The Even Integers: Ideal vs. Subring

 **Example 2.25 (The Even Integers $2\mathbb{Z} \subset \mathbb{Z}$):** Let $R = \mathbb{Z}$ and consider the subset of even integers $2\mathbb{Z} := \{2k \mid k \in \mathbb{Z}\}$.

- **$2\mathbb{Z}$ is an Ideal of $\mathbb{Z}$ (YES):**

(1) $(2\mathbb{Z}, +, 0)$ is an additive subgroup of $\mathbb{Z}$: $0 = 2 \cdot 0 \in 2\mathbb{Z}$, the sum of two even integers $2k + 2\ell = 2(k + \ell)$ is even, and $-(2k) = 2(-k) \in 2\mathbb{Z}$.

(2) Multiplicative absorption holds: For any $r \in \mathbb{Z}$ and any $x = 2k \in 2\mathbb{Z}$,

$$r \cdot x = r \cdot (2k) = 2(r \cdot k) \in 2\mathbb{Z}.$$

- **$2\mathbb{Z}$ is NOT a Subring of $\mathbb{Z}$ (NO):** The multiplicative identity $1 \notin 2\mathbb{Z}$ since 1 is odd.

III Subrings between $\mathbb{Z}$ and $\mathbb{Q}$

Spoken Transcript [00:41:48 - 00:42:37]

Okay, this is maybe a trickier question.

Certainly $\mathbb{Z}$ is a subring of $\mathbb{Q}$. I hope that's not controversial. You accept that $\mathbb{Z}$ is a subring of $\mathbb{Q}$?

Can you find a subring that's in between $\mathbb{Z}$ and $\mathbb{Q}$? Does there exist a subring in between $\mathbb{Z} \subsetneq S \subsetneq \mathbb{Q}$?

How about that? Can you do that? You have 3 minutes.

Student Answer

So we look at pairs of integers and denominator is just powers of two?

Spoken Transcript [00:42:44 - 00:44:10]

Okay, that's good. So, he is saying that we let S be the set of rational numbers of this form $\frac{a}{2^n}$, where a is an integer and n is also an integer. Is this exactly what you're saying?

Okay, so that's a proposal. One thing that's good about this proposal is this set S is certainly more than $\mathbb{Z}$ because it has like $3/2$ in it, and it's certainly less than $\mathbb{Q}$ because it doesn't have $2/3$ in it. So, that part is solid.

Then the claim is you have to check that this satisfies all these properties. What are the properties about having 0 and 1? Basically the answer is yes, it's going to satisfy, because you need to know how to add fractions, which I don't teach you. I've never taught anyone how to add fractions; I'm not going to teach you. But when you think about the rule about adding fractions, you keep the 2^n in the bottom, that's the main thing here. When you add fractions, the denominator doesn't give you things that you didn't put in.

And so the answer is yes. This is true.

So, that's an example. That's a very important one. Of course, 2 is not important here; you could have put here 17, it'd be fine. That's another example. And in fact, being prime is not that important either.

Subrings of the Rational Numbers

Example 2.26 (Dyadic Rationals $\mathbb{Z}[\frac{1}{2}] \subset \mathbb{Q}$): Consider the problem of finding an intermediate subring strictly between $\mathbb{Z}$ and $\mathbb{Q}$:

$$\mathbb{Z} \subsetneq S \subsetneq \mathbb{Q}.$$

Define the set of “*dyadic rationals*” $S = \mathbb{Z}[\frac{1}{2}]$ by:

$$S := \left\{ \frac{a}{2^n} \mid a \in \mathbb{Z}, n \in \mathbb{N}_{\geq 0} \right\} \subset \mathbb{Q}. \quad (2.12)$$

- **Proper Inclusions:** $S \supsetneq \mathbb{Z}$ since $\frac{1}{2} = \frac{1}{2^1} \in S \setminus \mathbb{Z}$, and $S \subsetneq \mathbb{Q}$ since $\frac{1}{3} \notin S$ (powers of 2 cannot produce a denominator divisible by 3).
- **Subring Verification:**
 - (i) $0 = \frac{0}{2^0} \in S$ and $1 = \frac{1}{2^0} \in S$.
 - (ii) Closed under addition: For $\frac{a}{2^n}, \frac{b}{2^m} \in S$ with $n \geq m$,

$$\frac{a}{2^n} + \frac{b}{2^m} = \frac{a + b \cdot 2^{n-m}}{2^n} \in S.$$

(iii) Closed under multiplication: $\frac{a}{2^n} \cdot \frac{b}{2^m} = \frac{a \cdot b}{2^{n+m}} \in S$.

(iv) Closed under additive inverses: $-\left(\frac{a}{2^n}\right) = \frac{-a}{2^n} \in S$.

Thus, $\mathbb{Z}[\frac{1}{2}]$ is a subring of $\mathbb{Q}$. More generally, for any $m \in \mathbb{Z}_{\geq 2}$ (e.g., $m = 17$), the localized ring $\mathbb{Z}[\frac{1}{m}] := \{\frac{a}{m^n} \mid a \in \mathbb{Z}, n \geq 0\}$ forms a proper intermediate subring $\mathbb{Z} \subsetneq \mathbb{Z}[\frac{1}{m}] \subsetneq \mathbb{Q}$.

III Polynomial Subrings: $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$

Spoken Transcript [00:44:10 - 00:45:05]

Okay, so another example.

Do we get a bell? I don't remember. Do I get to go to the bell, or do I have to stop?

Okay, let's try one more example before the bell, hopefully.

If I look at polynomials here, these are polynomials in the variable x , that's a ring: $\mathbb{Z}[x]$. Inside it are polynomials in the variable x^2 : $\mathbb{Z}[x^2]$. Every polynomial in the variable x^2 is a polynomial in x , so this is a subset: $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$. I hope that's clear.

So, is this an ideal, or a subring, or both, or neither, or what?

You ran out of time. Okay, we take a break, you can think about it.

Mid-Lecture Break

The lecturer announces a short pause for the students to consider whether $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$ is an ideal or a subring. The lecture resumes at timestamp 00:45:06.

Spoken Transcript [00:45:06 - 00:45:48]

Okay, so maybe we start, or is it a minute?

I don't know whether we get a bell, I don't remember, but anyway. So, what's the answer to this question?

Yeah?

Student Answer

I'd say it's a subring but not an ideal.

Spoken Transcript [00:45:48 - 00:46:22]

Yeah, so subring: yes. Ideal: no. So, for the subring, again, you just have to check that everything's there.

[The bell rings] Oh, we do get a bell.

So, why is it not an ideal? Yeah?

Student Answer

If you take x^2 , and multiply by x , then you get x^3 , which is not in the set.

Spoken Transcript [00:46:22 - 00:46:37]

So, x^2 is in this, and if it were an ideal, then I should be able to take any r in the ring and multiply it by this x^2 . So, I take any r in the ring, multiply by x^2 , it should stay in here. That's if it were an ideal.

So, I could take r to be x , and then this would be odd, and it wouldn't be in there, okay? So, you can check it yourself.

These are not supposed to be hard questions; it's just about whether you have absorbed the definitions. Which, as I said, is not an easy task.

Polynomial Subrings with Even Powers

Example 2.27 (Even Degree Polynomials $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$): Let $R = \mathbb{Z}[x]$ be the polynomial ring over $\mathbb{Z}$, and consider the subset $\mathbb{Z}[x^2] \subset \mathbb{Z}[x]$ of polynomials containing only even powers of x :

$$\mathbb{Z}[x^2] := \left\{ \sum_{k=0}^m a_k (x^2)^k = a_0 + a_1 x^2 + a_2 x^4 + \cdots + a_m x^{2m} \mid a_k \in \mathbb{Z} \right\}. \quad (2.13)$$

- **$\mathbb{Z}[x^2]$ is a Subring of $\mathbb{Z}[x]$ (YES):** $0, 1 \in \mathbb{Z}[x^2]$, the sum and difference of even polynomials are even, and the product of even powers yields even powers ($x^{2k} \cdot x^{2\ell} = x^{2(k+\ell)}$). Thus, $\mathbb{Z}[x^2] \leq \mathbb{Z}[x]$.
- **$\mathbb{Z}[x^2]$ is NOT an Ideal of $\mathbb{Z}[x]$ (NO):** Taking $r = x \in \mathbb{Z}[x]$ and $p(x) = x^2 \in \mathbb{Z}[x^2]$, their product is:

$$r \cdot p(x) = x \cdot x^2 = x^3 \notin \mathbb{Z}[x^2]$$

since x^3 has an odd degree. Thus, $\mathbb{Z}[x^2]$ fails the multiplicative absorption property.

III Subrings of the Integers $\mathbb{Z}$

Spoken Transcript [00:46:37 - 00:47:30]

Okay, so let's go to the integers. What are the subrings?

We've seen the rational numbers have lots of interesting subrings, because we found some: there's the integers, and you could invert 2, invert 3, 17. But what are the subrings of $\mathbb{Z}$?

Yeah?

Student Answer

Only $\mathbb{Z}$.

Spoken Transcript [00:46:54 - 00:47:30]

Yeah. There's no real interesting subrings: only $\mathbb{Z}$ in $\mathbb{Z}$. That's the only subring.

The whole ring is always a subring, just by definition. And this is the only one.

And why is that? Because the subring would have to contain 1. And once it contains 1, it contains every positive number. But the subring also is a subgroup, so it has to contain the negative ones too.

Okay? So, there's nothing you can do.

Classification of the Subrings of the Integers

 **Proposition 2.28 (Subrings of $\mathbb{Z}$):** The only subring of $(\mathbb{Z}, +, \cdot, 0, 1)$ is $\mathbb{Z}$ itself:

$$S \leq \mathbb{Z} \implies S = \mathbb{Z}. \quad (2.14)$$

Proof

Let $S \leq \mathbb{Z}$ be a subring.

(1) By definition of a subring, $1 \in S$.

(2) Since S is closed under addition, by induction $n = \underbrace{1 + 1 + \cdots + 1}_{n \text{ times}} \in S$ for all positive integers $n \in \mathbb{N}_{>0}$.

(3) By definition, $0 \in S$.

(4) Since $(S, +, 0)$ is an additive subgroup, S is closed under additive inverses: $-n \in S$ for all $n \in \mathbb{N}_{>0}$.

Combining these, S contains all integers $\mathbb{Z} \subseteq S$. Since $S \subseteq \mathbb{Z}$ by assumption, $S = \mathbb{Z}$.

Q.E.D.

 **Contrast: Ideals vs. Subrings in $\mathbb{Z}$:** This result highlights the sharp contrast between subrings and ideals in the ring of integers $\mathbb{Z}$:

- **Subrings of $\mathbb{Z}$:** There is only **one** subring, namely $\mathbb{Z}$ itself (because a subring must contain 1, which generates all of $\mathbb{Z}$ additively).
- **Ideals of $\mathbb{Z}$:** There are **infinitely many** distinct ideals of $\mathbb{Z}$, each of the form $n\mathbb{Z} = \langle n \rangle$ for $n \in \mathbb{N}_{\geq 0}$ (because an ideal does not need to contain 1).

III Classification of Ideals of $\mathbb{Z}$

Spoken Transcript [00:47:30 - 00:48:25]

So, what are the ideals? What are the ideals of $\mathbb{Z}$?

What would you guess the answer to this question is?
Yeah?

Student Answer

The set of multiples of an integer.

Spoken Transcript [00:48:25 - 00:49:35]

Yeah, so you pick some integer n , and you look at the multiples of n .

That's the standard notation: $n \cdot \mathbb{Z}$. So, if n is 2, it's the even numbers; if n is 3, it's the numbers divisible by 3, and so on.

Comment 1: this is certainly an ideal. It's just because the way we checked carefully that 2, the even ones, were an ideal, it's the same argument, okay?

Comment 2: it's not sensitive to whether you put $+n$ or $-n$ here, because inside $\mathbb{Z}$ is a -1 . So, if you put $+7$ or -7 , it's the same set. Is that also clear?

So, typically when you write $n\mathbb{Z}$, you should take $n \geq 0$. You don't have to, but that's the convention. In the literature, you'll never see minus... it's just a weird thing to do. There's no reason...

Principal Ideals of the Integers $\mathbb{Z}$

 **Definition 2.29** (*Principal Ideals of $\mathbb{Z}$*): For each integer $n \in \mathbb{Z}$, the set of integer multiples of n :

$$n\mathbb{Z} := \{n \cdot k \mid k \in \mathbb{Z}\} \quad (2.15)$$

is an ideal of $\mathbb{Z}$.

Remark 2.30 (*Conventions on $n\mathbb{Z}$*):

- **Ideals of $\mathbb{Z}$** : For every $n \in \mathbb{Z}$, $(n\mathbb{Z}, +, 0)$ is an additive subgroup of $\mathbb{Z}$, and for any $r \in \mathbb{Z}$ and $x = nk \in n\mathbb{Z}$, $r \cdot x = n(rk) \in n\mathbb{Z}$.
- **Sign Invariance**: Since $-1 \in \mathbb{Z}$ is a unit, $(-n)\mathbb{Z} = n\mathbb{Z}$. By convention, we restrict the generator index to $n \in \mathbb{N}_{\geq 0} = \{0, 1, 2, \dots\}$.

III Historical Origin of Ideals and David Hilbert

Spoken Transcript [00:49:35 - 00:50:40]

So, we're going to prove these are all the ideals. But this is an important example because the word ideal, where does it come from? The idea is that it's an idealized number, idealized integer.

For the integers, the ideals are the integers, but when you think about other rings, the ideals are some kind of idea of a number. This was in some sense the origin of this word ideal here: ideal number. This terminology goes back to about 1900, and there

were lots of people around, but certainly one person who played a role was the great David Hilbert, who was a mathematician in Göttingen. And a piece of math culture: does anyone know what the epitaph on the grave of David Hilbert is?

Not one person?

Yeah?

Student Question

What is an epitaph?

Spoken Transcript [00:50:40 - 00:51:45]

Ah, epitaph! So, you have a gravestone *[draws a sketch of an arched gravestone on the blackboard]*, and then what's written here? There's the person's name, and then there's like a little motto or slogan. That's the epitaph.

Okay, so this is a real piece of math culture, what's the epitaph of David Hilbert's grave. This is the kind of thing you learn to be a mathematician; it's part of being in the culture. Yeah, do you know it?

Student Answer

Wir müssen wissen, wir werden wissen.

Spoken Transcript [00:51:45 - 00:52:30]

Yeah: "*Wir müssen wissen, wir werden wissen.*" Two lines. It's in Göttingen.

[The lecturer writes "Motto / Slogan" inside the drawn gravestone on the board, and then erases it]

I have to teach you some culture, too, you know.

Okay, so we have to prove something.

Historical Context: Ideal Numbers and David Hilbert



≡ Origin of the Term "*Ideal*": The concept of an "*ideal*" originated in algebraic

number theory with Ernst Kummer (1840s) as “*ideal complex numbers*” to restore unique prime factorization in rings of algebraic integers such as $\mathbb{Z}[\zeta_p]$, and was later formalized by Richard Dedekind and David Hilbert around 1900. In $\mathbb{Z}$, every ideal is represented by a concrete integer ($I = n\mathbb{Z}$); in more general rings, ideals generalize the behavior of numbers.

III Proof: Every Ideal in $\mathbb{Z}$ is Principal

 **Theorem 2.31 (Ideals of $\mathbb{Z}$ are Principal):** Every ideal I of the ring of integers $\mathbb{Z}$ is of the form:

$$I = n\mathbb{Z} = \langle n \rangle \quad (2.16)$$

for a uniquely determined non-negative integer $n \in \mathbb{N}_{\geq 0}$.

Proof that Every Ideal in $\mathbb{Z}$ is of the Form $n\mathbb{Z}$

Spoken Transcript [00:52:30 - 00:54:10]

What do we have to prove?

Let $I \subseteq \mathbb{Z}$ be an ideal.

I want to prove it's $n\mathbb{Z}$. What about math what I like is truth. Proof is good, but truth is better. And this stuff is true, but anyway, we should prove it; it's a math course.

So, if you want to prove it: **Case (1)** is that I could be just the zero set ($\{0\}$). And this means it's $0 \cdot \mathbb{Z}$. It's a degenerate case. That's possible. And that's an ideal: 0 is an ideal.

So, **Case (2)** is that I has some non-zero element.

Proof: Case 1 and Identification of Positive Elements

Case 1 ($I = \{0\}$):

If $I = \{0\}$, then $I = 0\mathbb{Z}$ with $n = 0$. This settles the degenerate case.

Case 2 ($I \neq \{0\}$):

If I contains a non-zero element $x \in I$ with $x \neq 0$:

- If $x > 0$, then I contains a positive integer.
- If $x < 0$, since I is an ideal and $-1 \in \mathbb{Z}$, $-x = (-1) \cdot x \in I$, where $-x > 0$.

In either case, the set of positive elements $I \cap \mathbb{N}_{>0}$ is non-empty.

Spoken Transcript [00:54:10 - 00:55:40]

Because I has to have 0 because I is an additive subgroup. If you're not in this case, you have to have some non-zero element.

Let's call that element x . x is in I .

Now, x is an integer.

So, we can multiply by ± 1 and get a positive integer. One of these is a positive integer.

So, if it has a non-zero element, then I has a positive integer.

And if it has a positive integer, what am I going to say now?

Yeah?

Student Answer

It has a minimum.

Spoken Transcript [00:55:40 - 00:57:00]

Yeah, has a least positive integer. That's the next line.

So, I has a least positive integer.

Basically every property of integers at some point you have to use this principle. So, once it has a least positive integer, let's call that one n . So, n is equal to the least positive integer of I .

So, now we have to show that I is actually $n\mathbb{Z}$. That's what we want to show, all right?

You look at $y \in I$. There's only one thing you can do now, which is the division algorithm. The division algorithm says y is equal to some multiple of n plus a remainder term:

$$y = q \cdot n + r.$$

That is the division, long division.

Application of the Well-Ordering Principle and Division Algorithm

Selection of the Minimal Generator n :

By the “*Well-Ordering Principle*” of $\mathbb{N}$, the non-empty subset $I \cap \mathbb{N}_{>0} \subseteq \mathbb{N}$ contains a unique smallest element:

$$n := \min(I \cap \mathbb{N}_{>0}). \quad (2.17)$$

Claim: $I = n\mathbb{Z}$.

- **Inclusion $n\mathbb{Z} \subseteq I$:** Since $n \in I$ and I is an ideal, for any $k \in \mathbb{Z}$, $k \cdot n \in I$. Thus, $n\mathbb{Z} \subseteq I$.
- **Inclusion $I \subseteq n\mathbb{Z}$:** Let $y \in I$ be an arbitrary element. By the “*Division Algorithm*” (division with remainder in $\mathbb{Z}$), there exist unique integers $q, r \in \mathbb{Z}$ such that:

$$y = q \cdot n + r \quad \text{with} \quad 0 \leq r < n. \quad (2.18)$$

Spoken Transcript [00:57:00 - 00:58:30]

And the important part of this remainder term is $0 \leq r < n$.

Suppose y is not in $n\mathbb{Z}$, that means that it's not divisible, so there is a remainder term.

But this remainder term is equal to $y - qn$, and so the remainder term is also in I , because y is in I , n is in I so qn is in I , and since I is an additive subgroup, this is also in I .

It can't be zero because otherwise it would divide, so the long division says r is between 0 and n .

But then r is a lesser positive integer, which is a contradiction. Okay, that's the proof.

Spoken Transcript [00:58:30 - 00:59:10]

So, we use what you have to use for integers: you use this fact that non-empty subsets have a least one. Once you have that least one, you know what you're trying to prove, and then you use the long division algorithm and use the remainder. This is what you always do with these kind of elementary properties of integers.

So, that proves this claim.

And that's the most basic properties of the ideals in the integers.

Conclusion of the Proof: Minimality Contradiction

Continuation of Proof of Theorem 2.31

Rearranging equation (2.18) for the remainder r :

$$r = y - q \cdot n.$$

Since $y \in I$, $n \in I$, and I is an ideal, $q \cdot n \in I$. Since $(I, +, 0)$ is an additive subgroup:

$$r = y - q \cdot n \in I.$$

We now analyze the remainder r :

- If $r > 0$, then $r \in I \cap \mathbb{N}_{>0}$ with $r < n$. But this directly contradicts the defining minimality of $n = \min(I \cap \mathbb{N}_{>0})$ from (2.17).
- Therefore, we must have $r = 0$.

Setting $r = 0$ in (2.18) gives $y = q \cdot n \in n\mathbb{Z}$. Since $y \in I$ was arbitrary, $I \subseteq n\mathbb{Z}$. Combining both inclusions yields $I = n\mathbb{Z}$. Q.E.D.

Q.E.D.

Insight: Principal Ideal Domains (PIDs)

A ring in which every ideal is generated by a single element (i.e., $I = \langle a \rangle = aR$) is called a "*Principal Ideal Domain*" ("*PID*"). Theorem 2.31 proves that the ring of integers $\mathbb{Z}$ is a PID.

III Rings of Smooth Functions and Compactly Supported Ideals

Spoken Transcript [00:59:10 - 01:00:50]

For any ring R , we've seen before that $\{0\}$ is an ideal, and the whole ring R is an ideal.

And we haven't discussed fields at all today, but we discussed last time that if you have a field, these are the only two ideals.

But the integers have a lot of ideals: all these $n\mathbb{Z}$ are ideals.

Okay, so let's throw in another example. One object you certainly know in mathematics: if I take the real numbers, you can take C^∞ functions.

You don't have to take C^∞ , you could take 17 times differentiable functions, but we'll just go ∞ . These are infinitely differentiable functions.

So an element here is some function $f: \mathbb{R} \rightarrow \mathbb{R}$, which I can differentiate as many times as I want, so it has a nice graph; I can graph it like this *[sketches a wavy function curve on the board]*.

And this beautiful object is a ring.

Because why? You can add functions, you can multiply functions, 0 is just the constant function 0, and 1 is the constant function 1.

Is that clear? It's a ring.

The Ring of Smooth Functions $C^\infty(\mathbb{R})$

 **Definition 2.32 (Ring of Smooth Functions):** The set of infinitely differentiable functions on the real line:

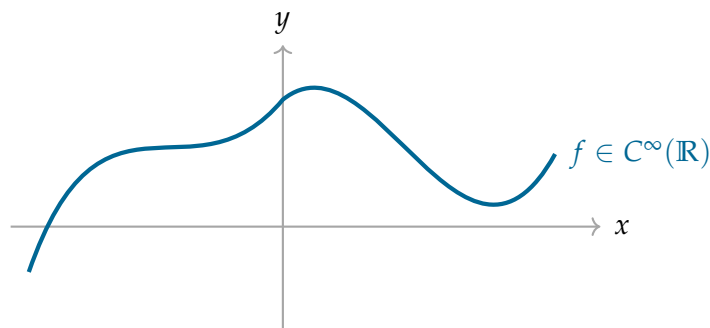
$$C^\infty(\mathbb{R}) := \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is smooth}\} \quad (2.19)$$

forms a commutative ring $(C^\infty(\mathbb{R}), +, \cdot, 0, 1)$ under pointwise operations:

$$(f + g)(x) := f(x) + g(x),$$

$$(f \cdot g)(x) := f(x) \cdot g(x),$$

with additive identity $0(x) \equiv 0$ and multiplicative identity $1(x) \equiv 1$.



Spoken Transcript [01:00:50 - 01:02:05]

[A student asks a question about the board text] Here? Which board? Infinitely differentiable function.

Yeah, my handwriting's not good: function. And it's not because I have this wrist injury [laughs]; I just have bad handwriting.

And I don't even know how I got this wrist injury. It's not like I'm an MMA fighter or anything like that. It just started hurting at one point. I thought it was going to be a career-ending injury, but it turns out that if I wear this thing, it's fine.

Spoken Transcript [01:02:05 - 01:03:45]

In this class, it's not primarily about such rings of differentiable functions — that's some other class —, but I can show you something interesting in it.

There's an interesting ideal inside $C^\infty(\mathbb{R})$, and these are functions with compact support: $C_c^\infty(\mathbb{R})$.

It's the set of functions with compact support.

A general function can do whatever it wants, totally unconstrained. But a function with compact support means that its entire life happens somewhere compact. So it has to be zero here, and then suddenly it comes to life, and then it goes to zero. So it's like this kind of hat function that looks like that, okay? That's a function of compact support.

Spoken Transcript [01:03:45 - 01:05:10]

So, is this an ideal?

To be an ideal, it has to be the case that zero's there, and very fortunately zero has compact support. What does not have compact support is the constant one function; that does not have compact support.

So, this whatever it is, it's surely not a subring, because it does not have the constant one function.

If you add compact support things, they have compact support, that's true. And the interesting thing is if I take a function of compact support and I multiply it by any function, it still has compact support, because once this function is zero, it just cannot be brought back to life, no matter how much you multiply.

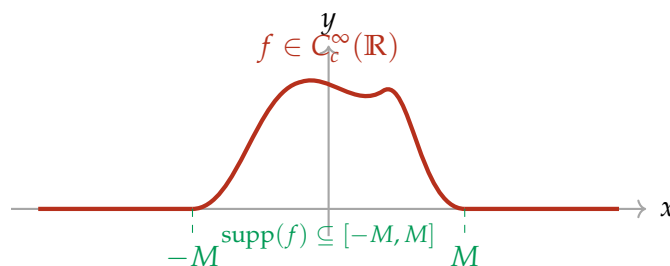
So, this is an example.

Compactly Supported Smooth Functions as an Ideal

 **Definition 2.33 (Functions with Compact Support):** The set of smooth functions with “compact support” on $\mathbb{R}$, denoted $C_c^\infty(\mathbb{R})$, is:

$$C_c^\infty(\mathbb{R}) := \{f \in C^\infty(\mathbb{R}) \mid \text{supp}(f) \text{ is compact}\}, \quad (2.20)$$

where $\text{supp}(f) := \overline{\{x \in \mathbb{R} \mid f(x) \neq 0\}}$ is bounded (i.e., $f(x) = 0$ outside some finite interval $[-M, M]$).



Proposition 2.34: $C_c^\infty(\mathbb{R})$ is an ideal of $C^\infty(\mathbb{R})$, but it is not a subring.

Proof

- (1) **Additive Subgroup:** The zero function $0 \in C_c^\infty(\mathbb{R})$ has $\text{supp}(0) = \emptyset$. For $f, g \in C_c^\infty(\mathbb{R})$, $\text{supp}(f + g) \subseteq \text{supp}(f) \cup \text{supp}(g)$, which is compact as the union of two compact sets.
- (2) **Multiplicative Absorption (Ideal Property):** For any arbitrary $g \in C^\infty(\mathbb{R})$ and $f \in C_c^\infty(\mathbb{R})$,

$$\text{supp}(g \cdot f) \subseteq \text{supp}(f).$$

Since $\text{supp}(f)$ is compact and closed subsets of compact sets in $\mathbb{R}$ are compact, $g \cdot f \in C_c^\infty(\mathbb{R})$.

- (3) **Not a Subring:** The constant unit function $1(x) \equiv 1$ has $\text{supp}(1) = \mathbb{R}$, which is not compact. Since $1 \notin C_c^\infty(\mathbb{R})$, $C_c^\infty(\mathbb{R})$ is not a subring.

Q.E.D.

Spoken Transcript [01:05:10 - 01:06:40]

Maybe I ask you a question: is there some ideal J that's bigger?

Can you find an ideal that's bigger than this, and not the whole ring, of course, because that's always an ideal that's bigger?

The traditional vision is there's an algebra direction, an analysis direction, a geometry direction. But actually, it doesn't really make sense to separate them all. It's all just one way of looking at numbers.

Can you think of a way to make this ideal bigger?

Spoken Transcript [01:06:40 - 01:07:55]

Well, we used the fact that it's zero on both sides here, but it's not necessary, right? Can't we just say functions that eventually go to zero on the right? Those are more. And it satisfies the same thing.

So here you could say J is functions that become zero for all sufficiently large x .

Those are functions that look like: they do whatever they want, and then eventually they become zero. These functions are certainly inside there, but there are more of them.

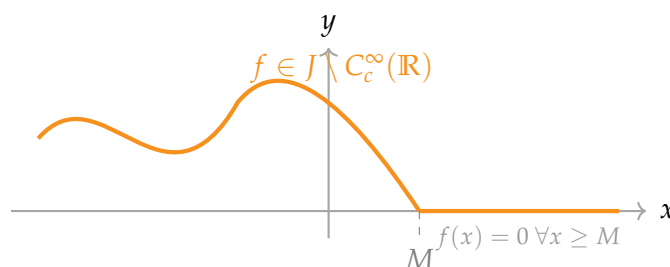
And these things are also an ideal, because if you multiply by something that's zero here, it's going to stay zero.

📖 A Strictly Larger Ideal: Functions Vanishing at Positive Infinity

💡 **Example 2.35 (Functions Vanishing for Large x):** Consider the set of smooth functions that vanish identically for all sufficiently large positive x :

$$J := \{f \in C^\infty(\mathbb{R}) \mid \exists M \in \mathbb{R} \text{ such that } f(x) = 0 \text{ for all } x \geq M\}. \quad (2.21)$$

- **Proper Inclusions:** $C_c^\infty(\mathbb{R}) \subsetneq J \subsetneq C^\infty(\mathbb{R})$. Any compactly supported function vanishes for large x , so $C_c^\infty(\mathbb{R}) \subseteq J$. However, a function that is non-zero on all of $(-\infty, 0)$ and vanishes on $[1, \infty)$ belongs to $J \setminus C_c^\infty(\mathbb{R})$. Furthermore, $1 \notin J$, so $J \neq C^\infty(\mathbb{R})$.
- **Ideal Verification:** If $f \in J$ vanishes on $[M, \infty)$ and $g \in C^\infty(\mathbb{R})$, then $(g \cdot f)(x) = g(x) \cdot 0 = 0$ for all $x \geq M$, so $g \cdot f \in J$. Thus, J is an ideal.



III The Restriction Homomorphism

🎧 Spoken Transcript [01:07:55 - 01:09:12]

If I look at C^∞ functions of $\mathbb{R}$, there's a ring homomorphism to C^∞ functions of positive reals ($\mathbb{R}_{>0}$):

$$C^\infty(\mathbb{R}) \xrightarrow{\text{res}} C^\infty(\mathbb{R}_{>0}).$$

This is a good example of a homomorphism.

And this is called restriction, res .

Meaning if I have a C^∞ function here, it's a function from all of $\mathbb{R}$ to $\mathbb{R}$. If I have one, I can just restrict it:

$$f: \mathbb{R} \rightarrow \mathbb{R} \implies f|_{\mathbb{R}_{>0}}: \mathbb{R}_{>0} \rightarrow \mathbb{R}.$$

Is that clear? Every element here, I can just restrict, and that gives me a map. First, it's a map on sets, and then you have to show it respects the zero, it respects the function that's constant 1, and addition and multiplication behave well under restriction. So, this is a legitimate ring homomorphism.

So for function spaces, restriction is a very useful way of making all these ring homomorphisms.

Now your analysis question: is this surjective?

That's properly a question in your analysis class. You can think about that.

The Restriction Homomorphism on Function Spaces

 **Definition 2.36 (Restriction Homomorphism):** Let $U \subset \mathbb{R}$ be an open subset (e.g., $U = \mathbb{R}_{>0} = (0, \infty)$). The “restriction mapping”:

$$\text{res}: C^\infty(\mathbb{R}) \rightarrow C^\infty(U), \quad f \mapsto f|_U \quad (2.22)$$

is a ring homomorphism satisfying:

- $\text{res}(f + g) = (f + g)|_U = f|_U + g|_U = \text{res}(f) + \text{res}(g),$
- $\text{res}(f \cdot g) = (f \cdot g)|_U = f|_U \cdot g|_U = \text{res}(f) \cdot \text{res}(g),$
- $\text{res}(1_{\mathbb{R}}) = 1_U.$

III Introduction to Category Theory

III Categories, Objects, and Morphisms

Spoken Transcript [01:09:12 - 01:10:48]

There are two foundational topics which I’ve avoided discussing, but I think I’ll discuss now. And it has to do with category theory and Zorn’s lemma.

So, you had a foundations, this Grundstrukturen class. Did you discuss categories there?

You can say no.

Okay, so category. We discuss category.

A category consists of objects and homs.

Homs are sometimes called morphisms, or arrows. The languages differ across different authors.

So, the first thing to know about a category: it has objects and it has homs.

More specifically, given two objects A, B in the category, there is a set of homs, and this set has a name:

$$\text{Hom}(A, B).$$

Foundational Data of a Category

 **Definition 2.37 (Category: Objects and Hom-Sets):** A “category” $\mathcal{C}$ consists of:

- (1) A collection of “objects”, denoted $\text{Obj}(\mathcal{C})$. We write $A \in \text{Obj}(\mathcal{C})$ to indicate that A is an object of $\mathcal{C}$.
- (2) For every pair of objects $A, B \in \text{Obj}(\mathcal{C})$, a set of “morphisms” (also called “homs” or “arrows”), denoted:

$$\text{Hom}_{\mathcal{C}}(A, B) \quad (\text{or } \text{Mor}_{\mathcal{C}}(A, B)). \quad (2.23)$$

An element $f \in \text{Hom}_{\mathcal{C}}(A, B)$ is represented diagrammatically as an arrow $f: A \rightarrow B$ or $A \xrightarrow{f} B$, where A is the “domain” (source) and B is the “codomain” (target).

III Examples of Categories: Groups, Rings, and Vector Spaces

Spoken Transcript [01:10:48 - 01:12:06]

For example, the category of groups.

The category of groups has objects, and the objects are groups.

An object is a group $(G, \cdot, e)$. That's the objects. In the world of all groups, every possible group is in here.

And what are the homs going to be? The homs between two particular groups are the set of group homomorphisms from G to $\hat{G}$.

Another example is the category of rings. There the objects are rings with the full data, and the homs are ring homomorphisms.

From last year you could consider the category of vector spaces over the complex numbers: the objects are vector spaces over $\mathbb{C}$, and the homs are linear maps.

To be a category, I have to tell you what the objects are, given two objects I have to tell you what the homs are, and the resulting structure has to have some properties.

Spoken Transcript [01:12:06 - 01:13:00]

Did you figure out this question about the restriction homomorphism?

[A student in the lecture hall answers]

Student Answer

I think $\sin(1/x)$...

Spoken Transcript [01:13:00 - 01:14:00]

Yeah, so the answer is no, it's not surjective.

A picture is worth a thousand words.

The picture is an element here that's not there: $\sin(1/x)$ oscillating near $x = 0$.

[draws a sketch of $\sin(1/x)$ oscillating near $x = 0$ on the blackboard]

You should recognize that from your analysis class. You know this function, right?

Non-Surjectivity of Restriction and the Topologist's Sine Curve

 **Proposition 2.38 (Non-Surjectivity of Restriction):** The restriction ring homomorphism:

$$\text{res}: C^\infty(\mathbb{R}) \rightarrow C^\infty(\mathbb{R}_{>0}), \quad f \mapsto f|_{\mathbb{R}_{>0}}$$

is **not** surjective.

Proof

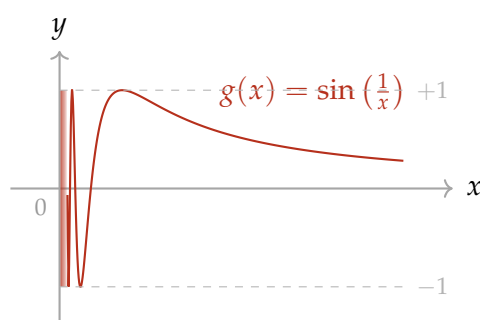
To show that res is not surjective, it suffices to exhibit a smooth function $g \in C^\infty(\mathbb{R}_{>0})$ that cannot be extended to any continuous (and hence no smooth) function $\tilde{g}: \mathbb{R} \rightarrow \mathbb{R}$.

Consider the oscillating function:

$$g(x) := \sin\left(\frac{1}{x}\right) \quad \text{for } x > 0. \quad (2.24)$$

Since the composition of smooth functions on $(0, \infty)$ is smooth, $g \in C^\infty(\mathbb{R}_{>0})$. However, as $x \rightarrow 0^+$, the argument $\frac{1}{x} \rightarrow \infty$, and $g(x)$ oscillates infinitely often between -1 and $+1$ on every neighborhood $(0, \varepsilon)$. Consequently, the limit $\lim_{x \rightarrow 0^+} \sin(1/x)$ does not exist. Thus, g cannot be extended continuously to $x = 0$, proving that $g \notin \text{Im}(\text{res})$.

Q.E.D.



Standard Algebraic Categories

Example 2.39 (The Category of Groups $\mathbf{Grp}$): The “category of groups”, denoted $\mathbf{Grp}$, is defined by:

- **Objects:** $\text{Obj}(\mathbf{Grp}) = \{(G, \cdot, e) \mid G \text{ is a group}\}$.
- **Morphisms:** For any two groups $G, \hat{G} \in \text{Obj}(\mathbf{Grp})$,

$$\text{Hom}_{\mathbf{Grp}}(G, \hat{G}) := \{f: G \rightarrow \hat{G} \mid f \text{ is a group homomorphism}\}. \quad (2.25)$$

Example 2.40 (The Category of Rings $\mathbf{Ring}$): The “category of rings”, denoted $\mathbf{Ring}$, is defined by:

- **Objects:** $\text{Obj}(\mathbf{Ring}) = \{(R, +, \cdot, 0, 1) \mid R \text{ is a ring}\}$.
- **Morphisms:** For any two rings $R, \hat{R} \in \text{Obj}(\mathbf{Ring})$,

$$\text{Hom}_{\mathbf{Ring}}(R, \hat{R}) := \{f: R \rightarrow \hat{R} \mid f \text{ is a ring homomorphism (with } f(1) = \hat{1})\}. \quad (2.26)$$

Example 2.41 (The Category of Vector Spaces $\mathbf{Vect}_{\mathbb{C}}$): The “category of complex vector spaces” $\mathbf{Vect}_{\mathbb{C}}$ is defined by:

- **Objects:** Vector spaces V over $\mathbb{C}$.
- **Morphisms:** Linear maps $T: V \rightarrow W$, so $\text{Hom}_{\mathbf{Vect}_{\mathbb{C}}}(V, W) = \mathcal{L}(V, W)$.

III Axioms of a Category: Composition and Identity

Spoken Transcript [01:14:00 - 01:15:45]

Okay, so the properties that the homs have to satisfy for the category are very simple properties.

The first property is that $\text{Hom}(A, B)$ is a set.

The second is that there's composition.

Composition means that if I have three objects, A, B , and C , then here I have the set of homs from A to B , and homs from B to C , and composition means there has to be a rule that tells you how, if you have a hom from A to B and a hom from B to C , this gives me a hom from A to C .

If I'm telling you a category, it's up to me to tell you how to compose homs. And in all examples, you just compose them as functions.

And inside $\text{Hom}(A, A)$ there's a distinguished element, called the identity (id_A). That's part of the data.

Spoken Transcript [01:15:45 - 01:17:26]

And there's one more condition, which is how the identity relates to composition.

If you take the identity in A , and you take any element $f \in \text{Hom}(A, B)$, then if I do the identity first and then I do f , what should this be equal to?

Yeah, f ! That's the identity axiom:

$$f \circ \text{id}_A = f.$$

And also the other way: if you do f first and then id_B , then this should just be f :

$$\text{id}_B \circ f = f.$$

The data of this category is to capture what happens when we think about these algebraic objects abstractly. We have objects, we have homs, the homs compose, there's an identity hom, and they satisfy these identity properties and associativity.

The Axioms of a Category

 **Definition 2.42 (Category (Full Formal Definition)):** A “category” $\mathcal{C}$ consists of the following data:

- (1) A collection $\text{Obj}(\mathcal{C})$ of “objects”.
- (2) For any two objects $A, B \in \text{Obj}(\mathcal{C})$, a set $\text{Hom}_{\mathcal{C}}(A, B)$ of “morphisms”.
- (3) For any three objects $A, B, C \in \text{Obj}(\mathcal{C})$, a “composition law”:

$$\circ: \text{Hom}_{\mathcal{C}}(B, C) \times \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{C}}(A, C), \quad (g, f) \mapsto g \circ f. \quad (2.27)$$

- (4) For every object $A \in \text{Obj}(\mathcal{C})$, a distinguished “identity morphism”:

$$\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A). \quad (2.28)$$

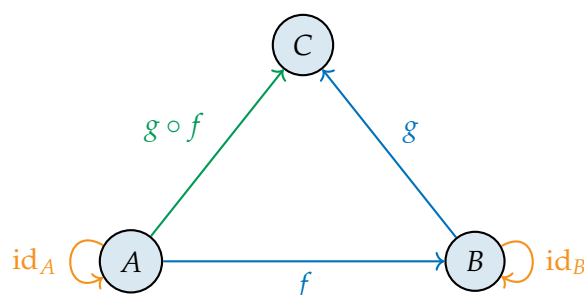
These data are required to satisfy the following category axioms:

- **Associativity of Composition:** For all $f \in \text{Hom}_{\mathcal{C}}(A, B)$, $g \in \text{Hom}_{\mathcal{C}}(B, C)$, and $h \in \text{Hom}_{\mathcal{C}}(C, D)$,

$$(h \circ g) \circ f = h \circ (g \circ f). \quad (2.29)$$

- **Identity Axioms (Unit Laws):** For any morphism $f \in \text{Hom}_C(A, B)$, composing with the identity morphism from either the left or the right acts as the identity:

$$f \circ \text{id}_A = f \quad \text{and} \quad \text{id}_B \circ f = f. \quad (2.30)$$



III Category-Theoretic Definition of Isomorphism

Spoken Transcript [01:17:26 - 01:19:58]

If you have a category, built into the category is a notion of isomorphic objects.
 Suppose **Cat** is a category. I want to define the notion of isomorphism purely in categorical terms.
 Is there a purely arrow-theoretic notion of isomorphism? Category-theoretic?
 And the answer is yes.

Spoken Transcript [01:19:58 - 01:22:08]

So, I have this category **Cat**, and I have two objects: A and B . If I have $f \in \text{Hom}(A, B)$, then f is an isomorphism if..
 What do you normally need for an isomorphism? You need a map the other way.
 If there exists a $g \in \text{Hom}(B, A)$ going in the other direction such that what?

Student Answer

The two compositions give the identity.

Spoken Transcript [01:21:22 - 01:22:08]

Yeah, there exists a g in the other direction such that:

$$f \circ g = \text{id}_B \quad \text{and} \quad g \circ f = \text{id}_A.$$

 That's the definition. That's a purely category-theoretic definition of isomorphism.
 Once I tell you the category of groups or rings, I tell you the homs, the identity, and this composition property, that already tells you what the notion of isomorphism is.

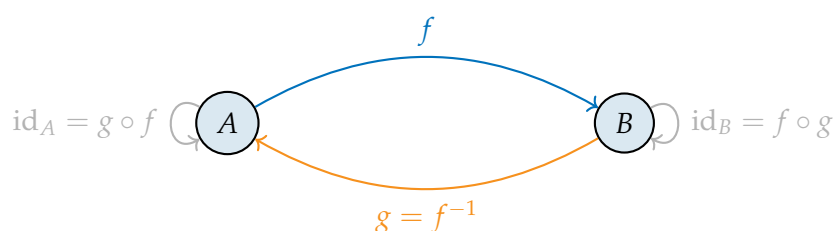
📖 Categorical Isomorphisms

Definition 2.43 (Isomorphism in a Category): Let $\mathcal{C}$ be a category, and let $A, B \in \text{Obj}(\mathcal{C})$. A morphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$ is called an “*isomorphism*” if there exists a morphism $g \in \text{Hom}_{\mathcal{C}}(B, A)$ such that:

$$g \circ f = \text{id}_A \quad \text{and} \quad f \circ g = \text{id}_B. \quad (2.31)$$

The morphism g is uniquely determined and is called the “*inverse morphism*” of f , denoted $g = f^{-1}$.

Two objects A and B are said to be “*isomorphic*” in $\mathcal{C}$ (written $A \cong B$) if there exists an isomorphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$.



≡ Arrow-Theoretic Universality of Isomorphisms: Notice that Definition 2.49 makes **no reference to internal set elements, injectivity, or surjectivity**. Bijectivity is replaced purely by the existence of a two-sided structural inverse arrow. In concrete categories:

- In **Grp**, categorical isomorphisms are precisely bijective group homomorphisms.
- In **Ring**, categorical isomorphisms are bijective ring homomorphisms.
- In **Vect_K**, categorical isomorphisms are bijective linear transformations (invertible matrices).

💡 Insight: The Power of Category-Theoretic Abstraction

Category theory unifies mathematics by defining concepts purely in terms of external interactions (“*morphisms*”) between systems rather than internal set elements. The definition of an isomorphism $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$ is universally identical across group theory, ring theory, linear algebra, topology, and algebraic geometry.

📖 Functors Between Categories

📖 Definition of a Covariant Functor

🎧 Spoken Transcript [01:22:08 - 01:23:55]

There’s a notion of functor.

A functor is a notion of a map from one category to another. Functors compare two different categories:

$$F: \mathbf{Cat} \rightarrow \widehat{\mathbf{Cat}}.$$

And now what's the data of the functor?

For every object A in the first category, I have to give you an object $F(A)$ in the second category. That's part of the data of the functor.

Spoken Transcript [01:23:55 - 01:25:28]

That's not enough: whenever I have an arrow $A \xrightarrow{f} B$ on the **Cat** side, I also have to tell you how this hom goes to a hom between these on the other side:

$$F(f): F(A) \rightarrow F(B).$$

And it has to respect all of the structure, which means this rule must take the identity to the identities:

$$F(\text{id}_A) = \text{id}_{F(A)},$$

and it must respect composition:

$$F(g \circ f) = F(g) \circ F(f).$$

That's the notion of a functor.

Definition of a Covariant Functor

 **Definition 2.44 (Covariant Functor):** Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A “(covariant) functor” $F: \mathcal{C} \rightarrow \mathcal{D}$ consists of:

- (1) An **Object Mapping**: Assigning to each object $A \in \text{Obj}(\mathcal{C})$ an object $F(A) \in \text{Obj}(\mathcal{D})$.
- (2) A **Morphism Mapping**: Assigning to each morphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$ a morphism:

$$F(f) \in \text{Hom}_{\mathcal{D}}(F(A), F(B)). \quad (2.32)$$

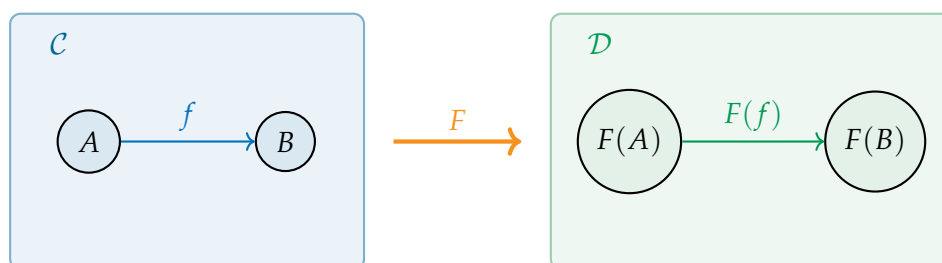
Such that the following two functoriality axioms are satisfied:

- **Preservation of Identities**: For every object $A \in \text{Obj}(\mathcal{C})$,

$$F(\text{id}_A) = \text{id}_{F(A)}. \quad (2.33)$$

- **Preservation of Composition**: For all composable morphisms $A \xrightarrow{f} B \xrightarrow{g} C$ in $\mathcal{C}$,

$$F(g \circ f) = F(g) \circ F(f). \quad (2.34)$$



III Examples of Functors: The Forgetful Functor and the Double Dual Functor

Spoken Transcript [01:25:28 - 01:28:46]

What's an example of a functor? There's one that's been staring us in the face the whole time: if I take the category of rings, there's a way to get to the category of groups. It's just taking the additive structure of the ring. That's a rule to go from rings to groups, and that is a functor.

The first example of a functor is: take the category of rings **Ring** and the category of groups **Grp**. There's a functor $F: \mathbf{Ring} \rightarrow \mathbf{Grp}$ defined by: if I take a ring $(R, +, \cdot, 0, 1)$, I associate to it the additive group $(R, +, 0)$.

And then if I have a ring homomorphism on the ring side, it also gives you an additive group homomorphism on the group side. That's the forgetful functor.

The Additive Forgetful Functor from Rings to Groups

 **Example 2.45 (The Additive Functor $\mathbf{Ring} \rightarrow \mathbf{Grp}$):** There is a canonical “forgetful functor” $F: \mathbf{Ring} \rightarrow \mathbf{Grp}$ mapping rings to their underlying additive abelian groups:

- **On Objects:** For each ring $(R, +, \cdot, 0, 1) \in \text{Obj}(\mathbf{Ring})$,

$$F((R, +, \cdot, 0, 1)) := (R, +, 0) \in \text{Obj}(\mathbf{Grp}). \quad (2.35)$$

- **On Morphisms:** For any ring homomorphism $f: R \rightarrow \hat{R}$,

$$F(f) := f: (R, +, 0) \rightarrow (\hat{R}, +, \hat{0}). \quad (2.36)$$

Since a ring homomorphism satisfies $f(a + b) = f(a) + f(b)$ and $f(0) = \hat{0}$, $F(f)$ is automatically a group homomorphism.

- **Functoriality:** $F(\text{id}_R) = \text{id}_{(R,+)}$ and $F(g \circ f) = g \circ f = F(g) \circ F(f)$ hold trivially.

Spoken Transcript [01:28:46 - 01:30:08]

A more interesting example that you learned: if I take the category of complex vector spaces $\mathbf{Vect}_{\mathbb{C}}$, there's a very nice functor here to itself.

This functor takes V to its double dual $(V^*)^*$.

If you take the dual V^* , it reverses the arrows (contravariant). But the double dual $(V^*)^*$ doesn't switch the arrows; it goes forward (covariant). Every vector space, you can associate its double dual.

And if you have a linear map $V \xrightarrow{T} W$, then the dual gives $W^* \xrightarrow{T^*} V^*$, and doing it again gives $(V^*)^* \xrightarrow{(T^*)^*} (W^*)^*$.

So this functor associates $T \mapsto (T^*)^*$.

Spoken Transcript [01:30:08 - 01:30:33]

[A student asks a question about vector space dimensions]

👤 Student Question

Is the dimension of vector space only finite or can it also be infinite?

🗣️ Spoken Transcript [01:30:08 - 01:30:33]

I didn't say anything. You could, if you want, say you want the category of finite-dimensional vector spaces. But the way I said it, it is anything.

For finite-dimensional vector spaces it happens the double dual is isomorphic. I'm just saying there's a functor giving a vector space its double dual.

Usually in linear algebra you spend all your time dualing and double dualing, right?

Okay, good. That's it. *[students applaud / lecturer concludes the lecture]*

📖 The Double Dual Endofunctor on Vector Spaces

💡 **Example 2.46 (The Double Dual Functor on $\mathbf{Vect}_{\mathbb{C}}$):** Let $\mathbf{Vect}_{\mathbb{C}}$ be the category of complex vector spaces and $\mathbb{C}$ -linear maps. The “double dual operation” defines an “endofunctor” (a functor from a category to itself):

$$(-)^{**} : \mathbf{Vect}_{\mathbb{C}} \rightarrow \mathbf{Vect}_{\mathbb{C}}$$

defined by:

(1) **On Objects:** For each vector space V ,

$$V \mapsto V^{**} := (V^*)^* = \mathrm{Hom}_{\mathbb{C}}(V^*, \mathbb{C}). \quad (2.37)$$

(2) **On Morphisms:** For any linear map $T: V \rightarrow W$, the single dual map $T^*: W^* \rightarrow V^*$ reverses arrows (acting “contravariantly”: $T^*(\psi) = \psi \circ T$). Taking the dual a second time restores the original arrow direction, yielding the “double dual map”:

$$T^{**} := (T^*)^*: V^{**} \rightarrow W^{**}, \quad T^{**}(\xi) := \xi \circ T^*. \quad (2.38)$$

$$\begin{array}{ccc} V & \xrightarrow{\quad T \quad} & W \\ & \vdots & \\ V^* & \xleftarrow{\quad T^* \text{ (Dual map / Contravariant)} \quad} & W^* \\ & \vdots & \\ V^{**} & \xrightarrow{\quad T^{**} = (T^*)^* \text{ (Double dual / Covariant)} \quad} & W^{**} \end{array}$$

≡ **Functoriality of Double Duality:** The composition of two arrow-reversing (contravariant) operations produces an arrow-preserving (covariant) functor:

- **Preservation of Identities:** $(\mathrm{id}_V)^{**} = (\mathrm{id}_{V^*})^* = \mathrm{id}_{V^{**}}$.
- **Preservation of Composition:** For $U \xrightarrow{S} V \xrightarrow{T} W$, the dual satisfies $(T \circ S)^* = S^* \circ T^*$. Taking the dual again yields:

$$(T \circ S)^{**} = (S^* \circ T^*)^* = (T^*)^* \circ (S^*)^* = T^{**} \circ S^{**}.$$

This holds for all vector spaces regardless of dimension. When $\dim(V) < \infty$, the canonical evaluation map $\text{ev}: V \rightarrow V^{**}$, $v \mapsto (\psi \mapsto \psi(v))$ is an isomorphism, making $V \cong V^{**}$ a natural isomorphism of functors.

■ Summary of Key Concepts

This lecture established foundational concepts across modern abstract algebra and category theory:

- **Groups, Rings, and Fields:** Reviewed core definitions, proving the uniqueness of identity elements and inverses. Explored the exponential isomorphism $(\mathbb{R}, +) \cong (\mathbb{R}_{>0}^*, \cdot)$ and proved $(\mathbb{R}^*, \cdot) \not\cong (\mathbb{R}_{>0}^*, \cdot)$ via torsion invariants.
- **Structure-Preserving Maps:** Formalized group and ring homomorphisms/isomorphisms. Analyzed how symmetry breaks between rings and groups: for group homomorphisms, kernels and images are subgroups; for ring homomorphisms, the kernel is an **ideal** while the image is a **subring**.
- **Substructures and Ideals:** Investigated subrings and ideals of $\mathbb{Z}$ (proving that $\mathbb{Z}$ is a Principal Ideal Domain where every ideal is $n\mathbb{Z}$) and smooth function rings $C^\infty(\mathbb{R})$ (where compactly supported functions $C_c^\infty(\mathbb{R})$ form an ideal).
- **Category Theory:** Introduced categories $\mathcal{C}$ (objects, hom-sets, composition, identity laws) and arrows-only definitions of categorical isomorphisms. Defined co-variant functors $F: \mathcal{C} \rightarrow \mathcal{D}$ with canonical examples: the additive forgetful functor $\mathbf{Ring} \rightarrow \mathbf{Grp}$ and the double dual endofunctor $(-)^{**}: \mathbf{Vect}_{\mathcal{C}} \rightarrow \mathbf{Vect}_{\mathcal{C}}$.

Wednesday 23 September 2026

Category Theory Basics, Integral Domains, and Groups of Units

Lecture Overview & Context

This lecture concludes the foundational discussion of category theory before returning to core ring theory and modular arithmetic:

- **Category Theory Basics:** Abstract definitions of objects, morphism sets (hom-sets), composition laws, identity morphisms, isomorphisms, monomorphisms (left-cancellation / injectivity), and epimorphisms (right-cancellation / surjectivity) along with categorical duality.
- **Integral Domains:** Zero divisors, formal definition of an integral domain (including the mandatory exclusion of the zero ring), and the proof that $\mathbb{Z}/m\mathbb{Z}$ is an integral domain if and only if m is prime.
- **Divisibility and Units:** Divisibility in arbitrary commutative rings, units, and precise linguistic distinctions (“the units”, “the unit”, “a unit”).
- **Groups of Units & Modular Isomorphisms:** Proof that the set of units $U(R)$ forms a multiplicative group, characterization of $U(\mathbb{Z}/m\mathbb{Z})$ via coprimality, and explicit proof that $U(\mathbb{Z}/7\mathbb{Z}) \cong (\mathbb{Z}/6\mathbb{Z}, +)$ as a cyclic group generated by $\bar{3}$.

Administrative Remarks

Spoken Transcript [00:00:00 - 00:01:26]

Okay, so we start. So, I wanted to make this comment: there were several questions about the course. The first one is if you want to watch the lectures, you have to be enrolled. So, if you're not enrolled, it will stop you. And there's not much I can do about that, or if there is, I don't know what to do, but the idea is you should enroll in the course if you want to watch the lectures.

Other people had some difficulty uploading the problem sets. Maybe you have to be enrolled in the course for that, too — I don't know —, but many people were successful, so if you have difficulty, find someone who's successful, because I don't know how to solve the problem in another way.

Okay, a question that's more about the actual course was that every week there are these suggested pages to read in the book, and the question is: how do they relate to the

course? That's a good question.

And the answer is that, roughly speaking, I do that so you can find the parallel material on which I'm lecturing in the book. And I don't always follow exactly the order there, I don't follow exactly the notation, and I don't follow exactly the text of the definitions. But I do something that's logically equivalent, so it can be helpful. As far as matters like the exam, I will only test you on material that has appeared in the lecture.

Okay? So, this is my best statement about this so far. Okay.

Category Theory: Abstract Definitions

Definition and Axioms of a Category

Spoken Transcript [00:01:26 - 00:02:50]

So, the last topic yesterday was categories, and I wanted to say a few things.

When I had this discussion about categories yesterday, it was mixed in with all the examples, so I want to just very quickly give the pure abstract definition and two small points. But this course is not about category theory. It's a topic here because it's natural to introduce when you talk about these algebraic objects, so, anyway, the course is not about category theory.

So, a category has two types of data: it has objects, as we said, and it has homs. So, A can be an object, and for homs, if you pick any two objects, you get a hom space, sometimes called morphisms.

And the abstract properties of this hom space — I said them, but I will repeat —: first, that it's a set, that's a technical property. And that's to avoid some contradictions in foundations, which I'm not going to discuss now.

The second is that there's a composition law.

And the third is that there are identity elements — existence of identity.

So... *[writes on board]* Okay, that's the data. So, is there any question about this data?

Constituents of a Category: Data

 **Definition 2.47 (Category Data):** A “category” $\mathcal{C}$ consists of the following mathematical data:

- **Objects:** A class of objects, denoted $\text{Obj}(\mathcal{C})$. We write $A \in \text{Obj}(\mathcal{C})$ (or simply $A \in \mathcal{C}$) to denote that A is an object of $\mathcal{C}$.
- **Morphism Sets (Homs):** For any pair of objects $A, B \in \text{Obj}(\mathcal{C})$, a set of morphisms from A to B , denoted:

$$\text{Hom}(A, B) \quad \text{or} \quad \text{Hom}_{\mathcal{C}}(A, B).$$

An element $f \in \text{Hom}(A, B)$ is represented diagrammatically as an arrow $f: A \rightarrow B$.

- **Composition Law:** For any three objects $A, B, C \in \text{Obj}(\mathcal{C})$, a binary composition

mapping:

$$\begin{aligned} \circ: \operatorname{Hom}(B, C) \times \operatorname{Hom}(A, B) &\rightarrow \operatorname{Hom}(A, C), \\ (g, f) &\mapsto g \circ f. \end{aligned}$$

- **Identities:** For each object $A \in \operatorname{Obj}(\mathcal{C})$, a designated morphism called the identity morphism on A :

$$\operatorname{Id}_A \in \operatorname{Hom}(A, A) \quad (\text{or } 1_A \in \operatorname{Hom}(A, A)).$$

≡ **Foundational Remark on Hom-Sets:** The requirement that $\operatorname{Hom}(A, B)$ forms a legitimate set (rather than a proper class) is a standard set-theoretic convention defining a **locally small category**. This prevents Russell-style self-referential paradoxes when indexing collections of arrows.

Spoken Transcript [00:02:50 - 00:03:45]

And then there's the axioms. The first one is that the composition law is associative, which means, you know, if you have three things, you can compose them in the two different ways. I hope I don't have to say more than that.

And then the second is the axioms about the identity, which is roughly: if I do this with f , this is f , and the other way around also. The identity doesn't change f . So, we discussed this last time.

Associativity and Identity Axioms

Definition 2.48 (Category Axioms): The data of a category $\mathcal{C}$ must satisfy the following two fundamental axioms:

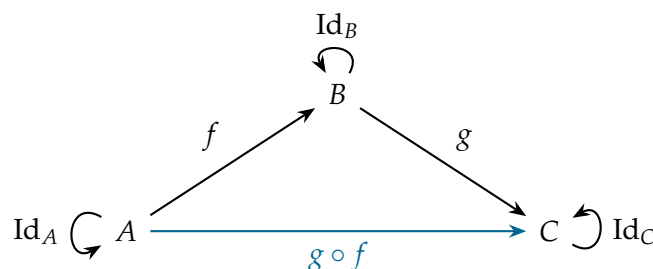
- (1) **Associativity of Composition:** For all objects $A, B, C, D \in \operatorname{Obj}(\mathcal{C})$ and all morphisms $f \in \operatorname{Hom}(A, B)$, $g \in \operatorname{Hom}(B, C)$, and $h \in \operatorname{Hom}(C, D)$, composition is associative:

$$(h \circ g) \circ f = h \circ (g \circ f). \quad (2.39)$$

- (2) **Identity Axioms (Neutrality):** For any morphism $f \in \operatorname{Hom}(A, B)$, composition with the respective identity morphisms leaves f unchanged:

$$f \circ \operatorname{Id}_A = f, \quad (2.40)$$

$$\operatorname{Id}_B \circ f = f. \quad (2.41)$$



III Categorical Morphisms: Isomorphisms, Monomorphisms, and Epimorphisms

Spoken Transcript [00:03:45 - 00:05:35]

So, that's the abstract definition, so there's really not that much here. And we discussed this last time, and we had lots of examples.

And so you could say: well, why do we want to do this? And as I said, there's a setting where many concepts that we talk about when we talk about groups, rings, fields, vector spaces — they're categorical concepts, meaning that they only depend on this structure. So, it simplifies thinking to consider them all together.

So, I already told you about isomorphism. The categorical isomorphism is that if I have this arrow from A to B — this picture means this: $f \in \text{Hom}(A, B)$, okay? That's why in category theory, you can say this, or you can say that, and you mean the same thing — that this f is an isomorphism — that's a definition — if and only if there exists g the other way, $g: B \rightarrow A$, such that the two compositions are identities.

And if you want to say which identity, you have to be careful. So, g starts with B , so if you do g first, then f back, this should be the identity in B , and the other is the identity in A .

So, that's the abstract definition of isomorphism. And in all of the categories we discussed, that coincides with the notion of isomorphism in the category that you already know, like, for example, isomorphism of vector spaces.

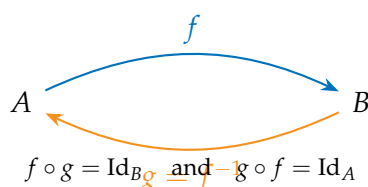
Categorical Isomorphism

 **Definition 2.49 (Isomorphism):** Let $\mathcal{C}$ be a category and let $f \in \text{Hom}(A, B)$ (written $f: A \rightarrow B$). The morphism f is an “*isomorphism*” if and only if there exists a morphism $g \in \text{Hom}(B, A)$ (written $g: B \rightarrow A$) such that:

$$f \circ g = \text{Id}_B, \quad (2.42)$$

$$g \circ f = \text{Id}_A. \quad (2.43)$$

If such a morphism g exists, it is uniquely determined, called the “*inverse*” of f , and denoted $f^{-1} := g$. Two objects A and B are said to be “*isomorphic*” (written $A \cong B$) if there exists an isomorphism between them.



 **Consistency across Concrete Categories:** In concrete categories where objects are sets endowed with structure and morphisms are structure-preserving maps (homomorphisms):

- In **Set**, isomorphisms are precisely bijective functions.
- In **Grp**, **Ring**, and **Vect_K**, isomorphisms are bijective homomorphisms.

The power of the categorical definition is that it never inspects elements $x \in A$; it char-

acterizes invertibility purely through diagrammatic composition.

Spoken Transcript [00:05:35 - 00:07:20]

There are two other notions, and that's the last thing I want to say. And this is the notion of monomorphism.

So, f from A to B is a monomorphism — that's a long word, that's a definition — if for all g_1 arrows from another object, X , to A , and another one, g_2 — so we take any two —, then the composition $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$. I hope I wrote that clearly here, okay?

So, it's a little bit complicated to say. If I have an arrow in the category, it may or may not be a monomorphism. I'm defining a monomorphism. And the condition of being a monomorphism means that whenever I take an object X and map it two ways to A , the composition with f being equal implies they were equal to start with. Of course, if they're equal to start with, the composition is equal, but it's the other implication. That's the definition of a monomorphism.

And, you know, you have to think about this and unwind it. But once you have this definition, you can imagine: what is a monomorphism in all of those categories you know? And what do you think it is?

Spoken Transcript [00:07:20 - 00:08:15]

So, I leave this for you to think about, but roughly speaking, monomorphism is being injective.

Just imagine they're sets. If this is injective, and you take any two maps, you can compare them after you do f . So, you have to think about it, but this is the definition.

And once you take this definition, you can say: okay, I can take the category of sets and functions, and what is a monomorphism there? And the answer is: injective. That's the “*mono*” in monomorphism.

So, this is a categorical formulation of being injective.

I don't know if you find this appealing or not, but that's what this discussion is about. Because the nice thing about this formulation is that it applies to all examples of categories. Every single one we discussed, and all the other ones you can think about, there's a notion of monomorphism.

Left-Cancellable Morphisms: Monomorphisms

 **Definition 2.50 (Monomorphism):** Let $\mathcal{C}$ be a category. A morphism $f \in \text{Hom}(A, B)$ (or $f: A \rightarrow B$) is called a “*monomorphism*” (often abbreviated as a **monic morphism** or **mono**) if it is left-cancellable: for every object $X \in \text{Obj}(\mathcal{C})$ and all pairs of parallel morphisms $g_1, g_2 \in \text{Hom}(X, A)$,

$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2. \quad (2.44)$$

$$\begin{array}{ccccc}
 & & g_1 & & \\
 & \nearrow & & \searrow & \\
 X & & & & A \xrightarrow{f} B \\
 & \searrow & & \nearrow & \\
 & & g_2 & &
 \end{array}$$

$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2 \quad (\text{Categorical Injective})$$

≡ **Why Monomorphisms Characterize Injectivity:** In the category of sets (**Set**), suppose $f: A \rightarrow B$ is a monomorphism. To test whether f is injective on points $a_1, a_2 \in A$, let $X = \{*\}$ be a one-element set and define two maps $g_1, g_2: \{*\} \rightarrow A$ by $g_1(*) = a_1$ and $g_2(*) = a_2$. Then $(f \circ g_1)(*) = f(a_1)$ and $(f \circ g_2)(*) = f(a_2)$. Thus:

$$f(a_1) = f(a_2) \implies f \circ g_1 = f \circ g_2 \implies g_1 = g_2 \implies a_1 = a_2.$$

Hence, left-cancellation with respect to arbitrary test objects X precisely recovers element-wise injectivity without mentioning elements.

Spoken Transcript [00:08:15 - 00:09:20]

And this one has a sibling, or a partner — I don't know their relationship —, which is... whenever you talk about injective, you can also talk about surjective, and the categorical term there is epimorphism.

So, you can define epimorphism. So, if you have $f: A \rightarrow B$, it is an epimorphism if — it's a definition — for all...

And this is now about this kind of feeling of mathematics: how do you think I'm going to define the opposite notion of injective? Like here we had injective by X mapping to A . So, what do you think I should do here?

You should guess. Guessing is like one of the most important tools of a mathematician.

Yeah?

Student Answer

Well, we do the same, except we also switch like we compose...

Spoken Transcript [00:09:25 - 00:10:35]

This is a long answer. Here we had X mapping to A , so the opposite of that is mapping from B . It's like the other side.

So, for every $B \xrightarrow[g_2]{g_1} X$, then... So, this is the opposite of that, by just flipping everything to the other side. So, for every one of these, then we can consider $f...$

In that one it was the g 's first, then f ; now it's f first, then the g 's: $g_1 \circ f = g_2 \circ f \implies g_1 = g_2$.

So, that's the categorical notion of epimorphism, and it's somehow dual to monomorphism. If you like this world, it almost makes epimorphism and monomorphism the same definition up to this flipping. And, you know, thinking about things

categorically is good for such understanding.

If you look at this definition, and you look at it for the examples we know, this is surjectivity.

Okay, so that's the end of this discussion about categories. If you like it, you can go read more. As I said, this course is not a course on categories. There's a bunch of other things that I would say here, but I'm not going to, because, yeah, maybe I will at the end if I have time.

Right-Cancellable Morphisms: Epimorphisms

Definition 2.51 (Epimorphism): Let $\mathcal{C}$ be a category. A morphism $f \in \text{Hom}(A, B)$ (or $f: A \rightarrow B$) is called an "*epimorphism*" (often abbreviated as an **epic morphism** or **epi**) if it is right-cancellable: for every object $X \in \text{Obj}(\mathcal{C})$ and all pairs of parallel morphisms $g_1, g_2 \in \text{Hom}(B, X)$,

$$g_1 \circ f = g_2 \circ f \implies g_1 = g_2. \quad (2.45)$$

$$\begin{array}{ccccc} A & \xrightarrow{\quad f \quad} & B & \begin{array}{c} \xrightarrow{\quad g_1 \quad} \\ \xleftarrow{\quad g_2 \quad} \end{array} & X \\ & & & \text{(Categorical Surjective)} & \end{array}$$

Insight: Categorical Duality: Reversing Arrows

Monomorphisms and epimorphisms represent our first concrete encounter with **categorical duality**. If we construct the opposite category $\mathcal{C}^{\text{op}}$ by reversing all morphism arrows ($f^{\text{op}}: B \rightarrow A$ for every $f: A \rightarrow B$), then:

$$f \text{ is a monomorphism in } \mathcal{C} \iff f^{\text{op}} \text{ is an epimorphism in } \mathcal{C}^{\text{op}}.$$

Every theorem or statement about monomorphisms immediately yields an identical, dual statement about epimorphisms simply by reversing the order of all composition arrows.

Ring Theory: Zero Divisors and Integral Domains

Zero Divisors and the Definition of an Integral Domain

Spoken Transcript [00:10:35 - 00:12:35]

So, let us return to rings now. This was kind of a foundational side note.

So, we have this ring $(R, +, \cdot, 0, 1)$ (assumed commutative with unity $1 \neq 0$).

And the rings here, there's kind of a different flavor of rings: there are the rings that are like $\mathbb{Z}$, where when you have things that are non-zero and you multiply them, they stay non-zero. That's one of the principles of $\mathbb{Z}$. But it's not true for rings like $\mathbb{Z}/6\mathbb{Z}$, this ring.

I can take two elements here, like the residue class of 2 times the residue class of 3, neither of which is zero. So, this is not zero, and that is not zero. So, I take two elements that are legitimately not zero, but when I multiply these residue classes, this is 6, and that is 0.

So, that's a different type of multiplication. It's a multiplication where when you multiply non-zero things, you get zero, which is unexpected from the point of view of the integers.

And this is also related to the fact that there's no hope of having a multiplicative inverse if you're mapping things to zero... Okay, so there are these two differences, and there's a definition for this:

Zero Divisors in Modular Arithmetic

Let $(R, +, \cdot, 0, 1)$ be a commutative ring with unity $1 \neq 0$.

 **Example 2.52 (Multiplication in $\mathbb{Z}/6\mathbb{Z}$):** In the ring of residue classes modulo 6, denoted $\mathbb{Z}/6\mathbb{Z}$ (or $\mathbb{Z}_6$):

$$\bar{2} \neq \bar{0},$$

$$\bar{3} \neq \bar{0}.$$

However, their product in the ring is:

$$\bar{2} \cdot \bar{3} = \overline{2 \cdot 3} = \bar{6} = \bar{0} \in \mathbb{Z}/6\mathbb{Z}.$$

 **Zero Divisors vs. Invertibility:** Non-zero elements whose product is zero are called “*zero divisors*”. If an element $a \in R$ is a zero divisor (with $a \cdot b = 0$ for some $b \neq 0$), it can never possess a multiplicative inverse a^{-1} :

$$b = 1 \cdot b = (a^{-1} \cdot a) \cdot b = a^{-1} \cdot (a \cdot b) = a^{-1} \cdot 0 = 0,$$

which contradicts $b \neq 0$. Hence, rings containing zero divisors fail to exhibit regular arithmetic cancellation.

Spoken Transcript [00:12:35 - 00:14:05]

A ring R is an integral domain... So, it's a definition of this term “*integral domain*”. And some authors would just say it's a domain, and also if you speak about rings, very often mathematicians will say it's a domain... if whenever I have a and b , neither of which is zero — so maybe I'll just be really explicit — whenever I have $a \neq 0$ and $b \neq 0$, this implies the product is not equal to zero ($a \cdot b \neq 0$).

So, that's the definition. It's a really basic definition in ring theory, and it separates this type of ring from that type of ring. Is that clear?

Fields, for example, are always domains, because you have the multiplicative inverse, if you think about it, okay?

So, we can ask...

Definition of an Integral Domain (Initial Board State)

 **Definition 2.53** (*Integral Domain / Domain*): A commutative ring R is called an “*integral domain*” (or simply a “*domain*”) if for all $a, b \in R$:

$$a \neq 0 \quad \text{and} \quad b \neq 0 \implies a \cdot b \neq 0. \quad (2.46)$$

Equivalently, taking the contrapositive:

$$a \cdot b = 0 \implies a = 0 \quad \text{or} \quad b = 0.$$

 **Example 2.54** (*Fields are Domains*): Every field F is an integral domain: if $a \neq 0$ and $a \cdot b = 0$, then multiplying by the inverse $a^{-1} \in F$ immediately yields:

$$b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0.$$

Spoken Transcript [00:14:05 - 00:14:42]

So, see, I did that already there: I called it a domain, not an integral domain, because it takes a lot of syllables to say “*integral domain*”. Formally, this is defined as an integral domain; colloquially, almost everyone says a domain.

So, these terms, monomorphism and epimorphism, have you ever heard these terms before? In which context?

Student Question/Answer

Linear algebra.

Spoken Transcript [00:14:42 - 00:14:52]

Ah, okay. And they were just used synonymously with injection and surjection?

Student Answer

We were introduced to them, but then we went back to using injection and surjection.

Spoken Transcript [00:14:52 - 00:14:58]

Okay, yeah.

Student Answer

So, they were like injective and surjective homomorphisms.

Spoken Transcript [00:14:58 - 00:15:20]

Yes, exactly, yes. Injective and surjective homomorphisms. That’s precisely what

this categorical thing says, right? Because in the category of groups, the arrows are homomorphisms, and when a homomorphism is a monomorphism, it's an injective homomorphism.

In my experience with undergraduates, some really like the idea of categories and really like to think in that language, and others like the examples, so either way is okay.

III Characterization of $\mathbb{Z}/m\mathbb{Z}$ as an Integral Domain

Spoken Transcript [00:15:20 - 00:16:24]

All right, so here we come to the first proposition, which is something my guess is you are already familiar with, but we do it again. So, we take this ring $\mathbb{Z}/m\mathbb{Z}$, and the claim is that this $\mathbb{Z}/m\mathbb{Z}$ is a domain if and only if — and what am I going to write here? Yeah?

Student Answer

m is prime.

Spoken Transcript [00:16:04 - 00:16:24]

m is prime. m is prime.

Oh, yeah, I should say one thing about domains which is important, I didn't say, and it's kind of annoying: part of the definition... I actually should put that in.

Spoken Transcript [00:16:24 - 00:17:15]

It's a super annoying thing to talk about, but I talk about any ring here, so my definition has to deal with the zero ring, the one that, you know, is unpleasant to think about. And you could think about: what would this definition be for the zero ring? Because for all a and b , there are no a 's and b 's that are not equal to zero. So, formally this statement in the formal deductive logical sense is true for the zero ring.

So, it is part of the definition: a ring R which is not the zero ring ($R \neq \{0\}$) is a domain. Zero rings are not allowed to be domains, by fiat, in the definition.

Correction: Exclusion of the Zero Ring

 **Definition 2.55** (*Integral Domain (Formal Standard Definition)*): A commutative ring R is an “*integral domain*” (or “*domain*”) if:

- (i) $R \neq \{0\}$ (that is, $1_R \neq 0_R$, excluding the zero ring).
- (ii) For all $a, b \in R$:

$$a \neq 0 \quad \text{and} \quad b \neq 0 \implies a \cdot b \neq 0.$$

≡ **Why the Zero Ring Satisfies the Conditional Vacuously:** In the zero ring $R = \{0\}$, the only element is 0. Therefore, there exist no non-zero elements $a, b \in R$. The premise $(a \neq 0 \wedge b \neq 0)$ is vacuously false for all elements, making the logical implication $(a \neq 0 \wedge b \neq 0 \implies ab \neq 0)$ vacuously true! To avoid having the degenerate zero ring classified as a domain, mathematicians add $R \neq 0$ explicitly by fiat.

Commutativity in Integral Domains

By definition, an “*integral domain*” is strictly required to be **commutative**. A non-commutative ring without zero divisors $(a \neq 0 \wedge b \neq 0 \implies ab \neq 0)$ is commonly referred to in advanced algebra as a **domain** or an **entire ring** (e.g., division rings such as the real quaternions $\mathbb{H}$ or free associative algebras). In this course, following standard convention, the terms “*domain*” and “*integral domain*” are interchangeable and presuppose commutativity.

Proposition: When Residue Class Rings are Domains

 **Proposition 2.56:** Let $m \in \mathbb{Z}_{>0}$ be a positive integer. The residue class ring $(\mathbb{Z}/m\mathbb{Z}, +, \cdot, \bar{0}, \bar{1})$ is an integral domain if and only if m is a prime number:

$$\mathbb{Z}/m\mathbb{Z} \text{ is an integral domain} \iff m \text{ is prime.} \quad (2.47)$$

Proof of Proposition 2.56

Spoken Transcript [00:17:15 - 00:19:24]

Okay, $\mathbb{Z}/m\mathbb{Z}$ is a domain if and only if m is prime. Okay, so let's try to prove this. So, proof.

This is like one of the issues I said: this issue about the zero ring is a little bit problematic everywhere in the statements, because for any statement you write, “*let R be a ring*”, you have to think — if you want that statement to be literally true in the mathematical sense from the definitions in this class —, you have to think about what that statement could mean for the zero ring and whether it's true. And as I said, I don't view it as a deep or really important thing to do, but in terms of bookkeeping and saying things correctly, you have to say it...

Okay, so let's try to prove this: $\mathbb{Z}/m\mathbb{Z}$ is a domain if and only if m is prime.

So, **(1)** suppose m is not prime.

Here we have m as some positive integer ($m \in \mathbb{Z}_{>0}$). Suppose m is not prime. Then this means that m admits a factorization into $a \cdot b$, where a is strictly greater than 1 and strictly less than m , and b is strictly like this ($1 < b < m$). That's what it means. If it's not prime, I can factor it, and if I can factor it, I break it into two smaller pieces.

And then if I look at this equation in residue classes, this equals that, which is equal to zero in $\mathbb{Z}/m\mathbb{Z}$; and this shows that that's not zero ($\bar{a} \neq \bar{0}$), and this shows that that's not zero ($\bar{b} \neq \bar{0}$).

Okay? So, this says that if m is not prime, there is no way for it to be a domain, because if it's not prime, it factors, and those factors precisely give the violation of that property. All right?

Proof of Proposition: Direction 1 (Non-Prime Case)

Direction 1: m Not Prime $\implies \mathbb{Z}/m\mathbb{Z}$ Not a Domain.

Suppose m is not prime (composite with $m > 1$). Then m admits a proper non-trivial factorization in $\mathbb{Z}$:

$$\begin{aligned} m &= a \cdot b, \\ 1 &< a < m, \\ 1 &< b < m. \end{aligned} \tag{2.48}$$

Passing to the residue classes in the quotient ring $\mathbb{Z}/m\mathbb{Z}$:

$$\bar{a} \cdot \bar{b} = \overline{a \cdot b} = \bar{m} = \bar{0} \in \mathbb{Z}/m\mathbb{Z}. \tag{2.49}$$

However, because $1 < a < m$ and $1 < b < m$, neither a nor b is divisible by m . Therefore:

$$\bar{a} \neq \bar{0} \quad \text{and} \quad \bar{b} \neq \bar{0}. \tag{2.50}$$

The existence of non-zero elements $\bar{a}, \bar{b} \in \mathbb{Z}/m\mathbb{Z}$ satisfying $\bar{a} \cdot \bar{b} = \bar{0}$ demonstrates that $\mathbb{Z}/m\mathbb{Z}$ possesses zero divisors, so $\mathbb{Z}/m\mathbb{Z}$ is not an integral domain.

Spoken Transcript [00:19:24 - 00:21:50]

So, then the other thing is suppose it is prime. Suppose $m = p$ is prime.

Then, we let a and b be integers such that $\bar{a} \cdot \bar{b} = \bar{0}$ in $\mathbb{Z}/p\mathbb{Z}$. All right?

What are we going to do here? This means that if I translate this equation on residue classes to integers, this means that $a \cdot b$ is equal to some integer $d \cdot p$.

So, I am using this notation and arithmetic in $\mathbb{Z}/m\mathbb{Z}$ freely here, because I assume you've seen this before. Is this true? If it's not, you should tell me. All right. If you haven't seen it, you should study it.

So, if I take these two residue classes and their product is zero, that exactly means that the product is divisible by p . Is that clear?

And then this means that p divides $a \cdot b$, because $a \cdot b = d \cdot p$.

And then there's the principle of Euclid we discussed in the first week that says either p divides a or p divides b . It wasn't such a long time ago, it was just last week.

And this means if $p \mid a$, then that means $\bar{a} = \bar{0}$; and if $p \mid b$, that means $\bar{b} = \bar{0}$.

So, we've proven that if the product is zero in $\mathbb{Z}/p\mathbb{Z}$, the only way that happens is one of them is zero, or the other is zero. And that's logically equivalent to this thing: if they were both not zero, then the product is not zero. Or the logical equivalent, which is the contrapositive, is that if the product is zero, then one of the factors must be zero, which we've established. So, this proposition is proven.

Proof of Proposition: Direction 2 (Prime Case via Euclid's Lemma)

Direction 2: $m = p$ **Prime** $\implies \mathbb{Z}/p\mathbb{Z}$ **Is a Domain.**

Suppose $m = p$ is a prime number. Let $\bar{a}, \bar{b} \in \mathbb{Z}/p\mathbb{Z}$ and assume that their product vanishes:

$$\bar{a} \cdot \bar{b} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}. \quad (2.51)$$

By definition of modular multiplication, this implies $\overline{a \cdot b} = \bar{0}$, meaning that $a \cdot b$ is congruent to 0 (mod p). Hence, there exists some integer $d \in \mathbb{Z}$ such that:

$$a \cdot b = d \cdot p \implies p \mid (a \cdot b). \quad (2.52)$$

By Euclid's Lemma (which holds whenever p is prime):

$$p \mid (a \cdot b) \implies p \mid a \text{ or } p \mid b. \quad (2.53)$$

Translating each case back into modular equivalence in $\mathbb{Z}/p\mathbb{Z}$:

- If $p \mid a$, then $\bar{a} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}$.
- If $p \mid b$, then $\bar{b} = \bar{0} \in \mathbb{Z}/p\mathbb{Z}$.

Thus, $\bar{a} \cdot \bar{b} = \bar{0}$ implies $\bar{a} = \bar{0}$ or $\bar{b} = \bar{0}$. This establishes that $\mathbb{Z}/p\mathbb{Z}$ has no zero divisors. Because $p \geq 2$, $\mathbb{Z}/p\mathbb{Z}$ contains at least two distinct elements ($\bar{0} \neq \bar{1}$), so $\mathbb{Z}/p\mathbb{Z}$ is not the zero ring. Therefore, $\mathbb{Z}/p\mathbb{Z}$ is an integral domain.

Spoken Transcript [00:21:50 - 00:23:19]

So, you could say again in this proposition, when someone writes this, I here put m to be a positive number ($m \in \mathbb{Z}_{>0}$), and now it matches carefully, but these boundary cases... you know, there's some aspect of this foundational part where you have to be careful yourself about boundary cases.

So, if I put $m = 1$, then I get the zero ring, but 1 is not considered a prime number, so it matches. You could say: what happens if I put zero? It's not here; we don't think about primality of zero, so zero is left out of this, but if m is zero, this is $\mathbb{Z}$, and it is a domain. Okay? All right.

[erases the blackboard]

Sometimes being too clear is actually more confusing in lectures. So, there's this kind of principle of lecturing where you lie about all the little details, and the lectures come out cleaner if you do that. And it's not clear didactically that that isn't a better way to do it.

There's a certain mathematician that takes pleasure in these edge cases, and you can tell that I'm not one of them.

Q.E.D.

Pedagogical Analysis of Boundary Cases

Remark 2.57 (Boundary Cases: $m = 1$ and $m = 0$): The proposition specifies $m \in \mathbb{Z}_{>0}$. Let us examine the boundary cases:

- **Case $m = 1$:** The ideal is $1\mathbb{Z} = \mathbb{Z}$. The quotient ring is the zero ring:

$$\mathbb{Z}/1\mathbb{Z} \cong \{0\},$$

where $\bar{1} = \bar{0}$. Since the definition of an integral domain explicitly requires $R \neq \{0\}$, $\mathbb{Z}/1\mathbb{Z}$ is **not** an integral domain. This perfectly matches the number-theoretic convention that 1 is not a prime number.

- **Case $m = 0$:** The zero ideal is $0\mathbb{Z} = \{0\}$. The quotient ring is canonically isomorphic to the integers:

$$\mathbb{Z}/0\mathbb{Z} \cong \mathbb{Z}.$$

The integers $\mathbb{Z}$ are the prototypical integral domain. In modern commutative algebra, $\{0\} \subset \mathbb{Z}$ is a prime ideal precisely because $\mathbb{Z}/\{0\} \cong \mathbb{Z}$ is an integral domain!

Divisibility and Units in Commutative Rings

Divisibility in Arbitrary Rings

Spoken Transcript [00:23:19 - 00:24:29]

Okay, so something more fun is we get to talk about units, and this is actually pretty fun.

This has to do with every ring, so let's take a ring R .

And this is now related to last week; in fact, it was the point of the discussion last week. So, there we discussed the ring $\mathbb{Z}$, but now you can take any ring.

So, the definition here is that if we have elements a and b in the ring ($a, b \in R$), then we say — this is the definition of the term —, we say “ a divides b ”. What this means is there exists some element of the ring, c ($c \in R$), such that $a \cdot c = b$ (in a commutative ring, the order does not matter: $a \cdot c = c \cdot a = b$).

Okay, that's just the actual definition of division, and it's the same definition as with the integers.

So, this is a notion of division for any ring. Okay?

Divisibility Relations

Let R be a ring (assumed commutative with unity $1 \neq 0$).

 **Definition 2.58 (Divisibility in a Ring):** Let $a, b \in R$. We say that “ a divides b ”, denoted:

$$a \mid b, \tag{2.54}$$

if there exists an element $c \in R$ such that:

$$a \cdot c = b. \tag{2.55}$$

In this case, we also say that b is a “multiple” of a , or that a is a “divisor” of b .

Divisibility and Commutativity

In non-commutative rings, divisibility splits into **left divisibility** ($b = a \cdot c$) and **right divisibility** ($b = c' \cdot a$). Under our standing convention that R is a commutative ring, left and right divisibility coincide ($a \cdot c = c \cdot a = b$), ensuring that divisibility is a well-defined, order-independent binary relation.

≡ **Connection with Principal Ideals:** In ideal-theoretic language, $a \mid b$ in a commutative ring R is equivalent to the statement that the principal ideal generated by b is contained in the principal ideal generated by a :

$$a \mid b \iff b \in aR \iff (b) \subseteq (a).$$

III The Definition of a Unit

Spoken Transcript [00:24:29 - 00:26:22]

Now, u in R ($u \in R$) is a unit — that's the definition of a unit.

The definition is if u divides 1 ($u \mid 1$). Oh, I should say that also: “ a divides b ”, that's how you say it in English, and the symbol for that is $a \mid b$. Okay?

So, here this is now a definition: an element is a unit if u divides 1 ($u \mid 1$). And that is equivalent — this is just equivalent — to there existing a w ($w \in R$) such that $u \cdot w = 1$. I never have to worry about the order here because I said that for the rings in this course, unless I tell you explicitly otherwise, we're always in commutative rings. Okay?

So, this is the definition, this is unwinding division, and if I want to unwind this using English — so this is the definition of division, and this is just English —, we say that if and only if u has a multiplicative inverse in R .

Okay? So, this is a really fundamental definition: a unit in our ring. So, are you happy with this? That's the definition, this is just translation, and that's what you say: u has a multiplicative inverse, w is the multiplicative inverse. So, is there any problem with this?

Invertible Elements

📖 **Definition 2.59 (Unit in a Ring):** Let R be a commutative ring with unity 1. An element $u \in R$ is called a “*unit*” if u divides 1:

$$u \mid 1. \tag{2.56}$$

By the definition of divisibility (Definition 2.58), this is equivalent to:

$$\exists w \in R \text{ such that } u \cdot w = 1. \tag{2.57}$$

In words, u is a unit if and only if u has a multiplicative inverse in R . The element w is uniquely determined and denoted $u^{-1} := w$.

≡ **Equivalence of Characterizations on the Blackboard:** The blackboard outlines

three equivalent perspectives on units:

- (1) **Divisibility Perspective:** $u \mid 1$ (the element divides the multiplicative identity).
- (2) **Algebraic Perspective (Unwinding Division):** $\exists w \in R$ such that $u \cdot w = 1$.
- (3) **Verbal Perspective (English):** u possesses a multiplicative inverse in R .

Since all rings considered here are commutative, left-inverses and right-inverses coincide.

III Linguistic Nuances: “The Units” vs. “The Unit” vs. “A Unit”

Spoken Transcript [00:26:22 - 00:28:16]

Actually, people asked me some questions about language, and you know, English is a language I know, which means that I speak it without thinking. It was pointed out to me that the use of “unit” is subtle, and I was doing it already in the first week; if you were sensitive to language, you’d have seen. So, for $\mathbb{Z}$ here, the units: what are the units for $\mathbb{Z}$?

In $\mathbb{Z}$, and it comes as no surprise, this is $+1$ and -1 . Those are the only invertibles in it.

So, the units: if I say “the units” here, that means all of them. The units.

If I say “the unit” — this is standard use of mathematical English —, if I say “the unit”, that’s just 1. If I have a ring and say “let’s consider the unit”, that means 1.

You could say “a unit”. If I say “let’s consider a ring, and let’s consider a unit”, that could be either one; we don’t know. Could be either. A unit — this could be either.

So, that’s actually standard usage. It’s a little confusing, but once you get used to it, you know, it’s just language. Is it clear? It was pointed out to me that I was doing that, and anyway, that’s right. But if someone comes to you with a ring and says “consider the unit”, that person means 1.

Units of the Integers and Linguistic Distinctions

 **Example 2.60** (*Units of the Ring of Integers $\mathbb{Z}$*): In the ring of integers $\mathbb{Z}$, the only elements possessing multiplicative inverses are 1 and -1 :

$$\text{The units of } \mathbb{Z} = \{1, -1\}. \quad (2.58)$$

 **Remark 2.61** (*Linguistic Distinction in Mathematical English*): In mathematical English, the word “unit” carries three distinct grammatical usages depending on the article:

- **“The units” (Plural):** Refers to the entire set of all invertible elements in the ring R , denoted $U(R)$ or $R^\times$. For $\mathbb{Z}$, the units are $\{1, -1\}$.
- **“The unit” (Singular, Definite Article):** Refers specifically to the multiplicative identity element $1_R \in R$.
- **“A unit” (Singular, Indefinite Article):** Refers to an arbitrary invertible element $u \in R$ (i.e., any element $u \in U(R)$). For $\mathbb{Z}$, “a unit” could be either 1 or -1 .

III The Group of Units and Modular Arithmetic

III The Group of Units of a Ring

Spoken Transcript [00:28:16 - 00:29:11]

Okay, so we discussed the units of the Gaussian integers, and now, since everything is defined, I recommend proving what the units of the Gaussian integers are. We wrote them last time.

But let's do something else now.

So, this is an exercise... *[writes on board]* Exercise: prove the units of the Gaussian integers are $\pm 1, \pm i$. You need an argument for that, and that argument comes from the norm in complex analysis, so prove it.

Okay, so observation...

Exercise: Units of the Gaussian Integers

 **Exercise 2.62 (Units of the Gaussian Integers):** Prove that the units of the ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ are precisely:

$$U(\mathbb{Z}[i]) = \{\pm 1, \pm i\}. \quad (2.59)$$

Proof Strategy via the Multiplicative Norm: Consider the field norm $N: \mathbb{Q}(i) \rightarrow \mathbb{Q}$ defined by $N(a + bi) = a^2 + b^2$. For all $z, w \in \mathbb{Z}[i]$, the norm is multiplicative:

$$N(z \cdot w) = N(z) \cdot N(w).$$

If $z \in \mathbb{Z}[i]$ is a unit, there exists $w \in \mathbb{Z}[i]$ with $z \cdot w = 1$. Taking norms yields:

$$N(z) \cdot N(w) = N(1) = 1.$$

Since $a, b \in \mathbb{Z}$, $N(z) = a^2 + b^2$ is a non-negative integer. The only positive integer divisor of 1 in $\mathbb{Z}$ is 1, so $a^2 + b^2 = 1$. The only integer solutions are:

$$(a, b) \in \{(\pm 1, 0), (0, \pm 1)\} \implies z \in \{\pm 1, \pm i\}.$$

Spoken Transcript [00:29:11 - 00:30:47]

...and it's a really important observation: if I take any ring R — let R be a ring — and let U — you could write $U(R)$ if you want — be the set of units of R .

Then this observation is: if I take the units, I take multiplication — that's multiplication in the ring —, and I take the unit 1, this is a group.

Okay? What does it say? If I take two units and multiply them, I get a unit.

And that's clear because, you know, if you have two elements here, if I multiply them, I have to show that's a unit. That means I have to show that if I multiply by some more stuff, I get 1.

But this is a unit, so there's already its multiplicative inverse, and that one is a unit, so there's already its multiplicative inverse; so if I multiply these together, I get 1, because

I can flip the orders around.

So, this says the units are closed under multiplication.

And there's the neutral element, and then you have to show the existence of inverses, but a unit is a unit because it has an inverse.

So, that's the proof of this. This is a good thing to think about: this is always a group.

The Group of Units

 **Proposition 2.63 (Group of Units):** Let R be a (commutative) ring with unity 1_R , and let $U(R)$ (also denoted U_R or $R^\times$) be the set of units of R :

$$U(R) := \{u \in R \mid \exists w \in R: u \cdot w = 1_R\}. \quad (2.60)$$

Then $(U(R), \cdot, 1_R)$ is a group under the multiplication of R .

Proof

We verify the group axioms for $(U(R), \cdot, 1_R)$:

- **Closure under Multiplication:** Let $u_1, u_2 \in U(R)$. By definition, there exist $w_1, w_2 \in R$ such that $u_1 w_1 = 1_R$ and $u_2 w_2 = 1_R$. Since R is commutative:

$$(u_1 u_2)(w_2 w_1) = (u_1 w_1)(u_2 w_2) = 1_R \cdot 1_R = 1_R.$$

Thus, $u_1 u_2$ has a two-sided multiplicative inverse in R , which shows $u_1 u_2 \in U(R)$.

- **Associativity:** Multiplicative associativity in $U(R)$ is directly inherited from the ring R .
- **Identity Element:** Since $1_R \cdot 1_R = 1_R$, the unity 1_R is its own inverse, so $1_R \in U(R)$. For all $u \in U(R)$, $1_R \cdot u = u$.
- **Existence of Inverses:** For any $u \in U(R)$, there exists $w \in R$ such that $uw = 1_R$. By symmetry of the relation $wu = 1_R$, w is also invertible with inverse u , so $w = u^{-1} \in U(R)$.

Hence, $(U(R), \cdot, 1_R)$ is a group. If R is commutative, $U(R)$ is an abelian group.

Q.E.D.

The Units Group in Commutative vs. Non-Commutative Rings

- **Commutative Rings (Course Setting):** Because all rings in this course are commutative by default, the group of units $(U(R), \cdot, 1_R)$ is always an **abelian** group.
- **General Non-Commutative Rings:** In general non-commutative ring theory, an element u is a unit if and only if it possesses a two-sided inverse ($uw = wu = 1_R$). In that broader setting, $(U(R), \cdot, 1_R)$ is still a group (with $(u_1 u_2)^{-1} = u_2^{-1} u_1^{-1}$), but it is generally non-abelian (e.g., $U(M_n(K)) = \text{GL}_n(K)$ for $n \geq 2$).

III Characterization of Units in $\mathbb{Z}/m\mathbb{Z}$

Spoken Transcript [00:30:47 - 00:31:44]

So, a very natural question, and one of the first things to think about regarding $\mathbb{Z}/m\mathbb{Z}$, if I take this ring R *[writes on board]*...

And it was implicit there: I will always consider $m > 0$ when I do this, because there's no reason to consider negative ones, as that's just counting everything twice. And then there's the confusing case when $m = 0$, which is usually a special case, so I exclude it.

Usually we allow $m = 1$, which is the zero ring, but anyway, this is kind of standard. When I write $\mathbb{Z}/m\mathbb{Z}$, I always mean this, and typically I also mean $m > 1$, but anyway.

So, the question is: what are the units of $\mathbb{Z}/m\mathbb{Z}$?

Spoken Transcript [00:31:44 - 00:33:33]

So, let's say: suppose we have $\bar{x} \in \mathbb{Z}/m\mathbb{Z}$, and $\bar{x}$ is a unit.

This means that there exists $\bar{y} \in \mathbb{Z}/m\mathbb{Z}$ such that $\bar{x} \cdot \bar{y} = \bar{1}$ in $\mathbb{Z}/m\mathbb{Z}$.

I can write this, or sometimes you write $\text{mod } 1$, but anyway, that's what it means to be a unit — I'm just repeating it, right?

And then I can translate: what does this mean in this ring? What does it mean for the integers x , y , and 1 ? This is equivalent to $x \cdot y = 1 + d \cdot m$.

Do you agree with that? That's what it means, for some d . Okay?

And this, if you remember from the first week, means that x and m have $\gcd = 1$.

The only thing that could divide both x and m , it would also have to divide 1 . So, that's why we say $\gcd = 1$.

In English, this condition is usually said as x and m are relatively prime.

Those are just words that express $\gcd(x, m) = 1$. Okay?

Spoken Transcript [00:33:33 - 00:35:02]

So, we have now discovered what the units of $\mathbb{Z}/m\mathbb{Z}$ are: they are all the elements which are relatively prime.

Because if it is a unit, it's the residue class of something relatively prime; if you have $\gcd = 1$, then you go backwards in this diagram.

So, I state that as a result.

So, if you write this as a result, then it says that the units of $\mathbb{Z}/m\mathbb{Z}$ consist of all elements $\bar{x}$ where $\gcd(x, m) = 1$. Did I write $\gcd$ here? Maybe $\gcd$. Okay? Is that clear?

And so the thing we've learned is that this units group, $U(\mathbb{Z}/m\mathbb{Z})$, is a group.

So, it says in particular, if you take $\mathbb{Z}/m\mathbb{Z}$, and you take the residue classes that are relatively prime, they form a group under multiplication.

So, let me do an example.

Units of the Quotient Ring $\mathbb{Z}/m\mathbb{Z}$

 **Theorem 2.64 (Units in $\mathbb{Z}/m\mathbb{Z}$):** Let $m \in \mathbb{Z}_{>0}$ be a positive integer. An element $\bar{x} \in \mathbb{Z}/m\mathbb{Z}$ is a unit if and only if x and m are coprime:

$$U(\mathbb{Z}/m\mathbb{Z}) = (\mathbb{Z}/m\mathbb{Z})^\times = \{\bar{x} \in \mathbb{Z}/m\mathbb{Z} \mid \gcd(x, m) = 1\}. \quad (2.61)$$

The order of this group is given by Euler's totient function $\varphi(m) := |U(\mathbb{Z}/m\mathbb{Z})|$.

Proof

We establish the equivalence by unwinding the definition of divisibility:

- **Forward Direction ($\implies$):** Suppose $\bar{x} \in U(\mathbb{Z}/m\mathbb{Z})$. Then there exists $\bar{y} \in \mathbb{Z}/m\mathbb{Z}$ such that:

$$\bar{x} \cdot \bar{y} = \bar{1} \in \mathbb{Z}/m\mathbb{Z}.$$

In terms of integer divisibility, this means $xy \equiv 1 \pmod{m}$. Hence, there exists an integer $d \in \mathbb{Z}$ such that:

$$xy = 1 + d \cdot m \iff xy - d \cdot m = 1.$$

Any common divisor $g \in \mathbb{Z}$ of x and m must divide the linear combination $xy - dm = 1$. Therefore, $g \mid 1$, which forces $\gcd(x, m) = 1$.

- **Reverse Direction ($\impliedby$):** Conversely, suppose $\gcd(x, m) = 1$. By Bézout's Identity, there exist integers $y, k \in \mathbb{Z}$ such that:

$$x \cdot y + m \cdot k = 1 \implies x \cdot y \equiv 1 \pmod{m}.$$

Projecting this relation into residue classes in $\mathbb{Z}/m\mathbb{Z}$, we obtain $\bar{x} \cdot \bar{y} = \bar{1}$. Thus, $\bar{x}$ is invertible, so $\bar{x} \in U(\mathbb{Z}/m\mathbb{Z})$.

By Proposition 2.63, $(U(\mathbb{Z}/m\mathbb{Z}), \cdot, \bar{1})$ forms a finite abelian group.

Q.E.D.

III Example: The Unit Group of $\mathbb{Z}/7\mathbb{Z}$

Spoken Transcript [00:35:02 - 00:36:09]

$\mathbb{Z}/7\mathbb{Z}$.

So, that's actually a prime one, but let me now do this example. So, the elements of this are $\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}$, right?

And which of these residue classes come here? The classes of x where x and 7 are relatively prime.

So, the question is: what are the units? Of course, it's some of these, but which of these residue classes correspond to x and 7 being relatively prime?

What would you say? As I said, if you don't know, guess.

Student Answer

All of them.

Spoken Transcript [00:36:09 - 00:36:14]

All of them? What about this one [*points at $\bar{0}$*]?

Student Answer

Only 1. 1 to 6 are the units.

Spoken Transcript [00:36:14 - 00:36:50]

Exactly. So, the answer to this question is these: $\{\bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}\}$.

And the reason is because, look, this was the answer: it's the residue classes of those numbers that are relatively prime. So, this is the residue class of those numbers that are relatively prime to 7. And these are those residue classes.

If you're $0 \pmod{7}$, then you're not relatively prime to 7. Like an example of $0 \pmod{7}$ would be 14, and that's not relatively prime to 7. So, it's these. Is that clear?

The Unit Group of $\mathbb{Z}/7\mathbb{Z}$

 **Example 2.65 (Units of $\mathbb{Z}/7\mathbb{Z}$):** Consider the quotient ring $\mathbb{Z}/7\mathbb{Z}$. The elements of the ambient ring are:

$$\mathbb{Z}/7\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}\}.$$

Since 7 is a prime number, every non-zero residue class $\bar{x} \in \{\bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}\}$ satisfies $\gcd(x, 7) = 1$. The zero class $\bar{0}$ is not coprime to 7 (since $\gcd(0, 7) = 7 \neq 1$, or for example $14 \equiv 0 \pmod{7}$ where $\gcd(14, 7) = 7$). Consequently, the group of units is:

$$U(\mathbb{Z}/7\mathbb{Z}) = (\mathbb{Z}/7\mathbb{Z})^\times = \{\bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}\}. \quad (2.62)$$

The order of the unit group is $|U(\mathbb{Z}/7\mathbb{Z})| = \varphi(7) = 7 - 1 = 6$.

II Powers of the Element $a = \bar{3}$ and Cyclic Structure

Spoken Transcript [00:36:50 - 00:37:09]

Okay, and the claim is this is a group under multiplication. Well, anyway, this is a group under addition, but if you take the relatively prime ones, this is a group under multiplication.

So, we should be able to figure out what group it is. So, what's the neutral element of this group?

Student Answer

One.

Spoken Transcript [00:37:09 - 00:37:56]

One, it's going to be $\bar{1}$.

What is this group? It's a commutative group of order 6. And so later we'll see that we know all commutative groups of order 6. We'd like to know which one this is. It's isomorphic to some group, we know.

And so it matters what you pick. I propose we pick 3. So, let me say we take this element in this group, $a = \bar{3}$.

And let's try to consider its powers. So, there's the neutral element, then I can take a, a^2, a^3, a^4 . You should know the powers of 3. Okay, let's see how many powers of 3 you know.

Spoken Transcript [00:37:56 - 00:39:44]

So, what is this one? So I'm just going to do this: this is $\bar{1}$. So, what is a ? Well, that's just that: $\bar{3}$.

What is this? That's the square of a , so that's the residue class of 3^2 , which is 9. So, I should put $\bar{2}$ here, right? Are you happy with this?

And this is a^3 : 3^3 is 27. For 27, the residue mod 7 is $\bar{6}$, right?

a^4 is 3^4 , which is 81, and the residue mod 7 of 81 is $\bar{4}$, I think.

3^5 is 243, and its residue is going to be $\bar{5}$.

What do you think this is going to be? Do you know what 3^6 is? 3^6 is 729. And if I look at its residue class mod 7, well, 700 is already here, and 28 is divisible, so its residue class is $\bar{1}$. This is back to 1.

So, this is understanding the group. You should know the powers of 3; they're very useful, you know.

So, this is what happens when you multiply in this group: you take a , a^2 , the next one a^3 , the next one a^4 , the next one a^5 , and then when you go to the sixth, you go back to the identity.

What group do you know this happens in? Does this remind you of anything? Yeah?

Student Answer

A cyclic group.

Spoken Transcript [00:39:44 - 00:39:54]

It's $\mathbb{Z}/6\mathbb{Z}$. So, the proposition that we've kind of proven here...

Powers of the Generator $a = \bar{3}$ in $U(\mathbb{Z}/7\mathbb{Z})$

To examine the internal structure of the abelian group $U(\mathbb{Z}/7\mathbb{Z})$ of order 6, we choose the candidate generator $a := \bar{3} \in U(\mathbb{Z}/7\mathbb{Z})$ and compute its successive powers:

Power	a^0	a^1	a^2	a^3	a^4	a^5	a^6
Symbolic	$\bar{1}$	a	a^2	a^3	a^4	a^5	a^6
Value in $\mathbb{Z}$	1	3	9	27	81	243	729
Residue modulo 7	$\bar{1}$	$\bar{3}$	$\bar{2}$	$\bar{6}$	$\bar{4}$	$\bar{5}$	$\bar{1}$

≡ Modular Arithmetic Derivations for Powers of 3:

$$\begin{aligned}
 a^0 &= \bar{1}, \\
 a^1 &= \bar{3}, \\
 a^2 &= \overline{3^2} = \bar{9} = \bar{2} & (9 = 7 \cdot 1 + 2), \\
 a^3 &= \overline{3^3} = \overline{27} = \bar{6} & (27 = 7 \cdot 3 + 6), \\
 a^4 &= \overline{3^4} = \overline{81} = \bar{4} & (81 = 7 \cdot 11 + 4), \\
 a^5 &= \overline{3^5} = \overline{243} = \bar{5} & (243 = 7 \cdot 34 + 5), \\
 a^6 &= \overline{3^6} = \overline{729} = \bar{1} & (729 = 7 \cdot 104 + 1).
 \end{aligned}$$

Because the powers of $a = \bar{3}$ exhaust all 6 elements of $U(\mathbb{Z}/7\mathbb{Z})$ before returning to the identity $\bar{1}$ at exponent 6, the element $\bar{3}$ is a “*generator*” (or “*primitive root*” modulo 7). Consequently, $U(\mathbb{Z}/7\mathbb{Z})$ is a cyclic group of order 6.

II The Group Isomorphism $(U(\mathbb{Z}/7\mathbb{Z}), \cdot) \cong (\mathbb{Z}/6\mathbb{Z}, +)$

Spoken Transcript [00:39:54 - 00:41:24]

...we’ve proven what that calculation of powers of 3... and I picked 3. If you picked another one, it would have been slightly different, but that calculation proves a group isomorphism, which maybe I’ll write here and we think about.

You should check my powers of 3, and then after you take the power of 3, you have to take the residue mod 7.

So, this great 20th century Hungarian mathematician (Paul Erdős) said that if numbers aren’t beautiful, I don’t know what is. That’s kind of the mathematical perspective on numbers.

So, we have here the units of $\mathbb{Z}/7\mathbb{Z}$, and then there’s another group, which is $\mathbb{Z}/6\mathbb{Z}$ with $+$ and $\bar{0}$.

And I claim these are isomorphic. So, this is the units with multiplication and $\bar{1}$. This group is isomorphic to that group. And what is the isomorphism? It’s written on the board.

The isomorphism — this is a slightly painful way to write the isomorphism, but let’s just be painful about it —: so, here $\bar{1}$ is going to go to $\bar{0}$. And bar means something

different: it means mod7 here, and mod6 here, right?

Spoken Transcript [00:41:24 - 00:42:33]

a , which is 3 here, a is $\bar{3}$. So, the nice way to do it is: a goes to $\bar{1}$, a^2 goes to $\bar{2}$. I guess I could write this in a slightly better notation, but anyway, you get the pattern. There are not so many more: a^3, a^4, a^5 . These go to $\bar{3}, \bar{4}, \bar{5}$.

So, this is the simple way to write it. And the reason this is true is because on this side when we multiply, we just add up the exponents, and on this side we add up those. And the important thing is the sixth one goes to zero, which is what we have here, okay?

And if you want to say this explicitly, you have to write these a 's in terms of the elements of the units, which I've done there. So, this is totally explicit.

Okay, that's all. *[students applaud and pack up; video ends]*

Group Isomorphism: $U(\mathbb{Z}/7\mathbb{Z}) \cong (\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$

 **Proposition 2.66 (Cyclic Group Isomorphism):** The multiplicative group of units $(U(\mathbb{Z}/7\mathbb{Z}), \cdot, \bar{1})$ is isomorphic to the additive cyclic group $(\mathbb{Z}/6\mathbb{Z}, +, \bar{0})$:

$$(U(\mathbb{Z}/7\mathbb{Z}), \cdot, \bar{1}) \cong (\mathbb{Z}/6\mathbb{Z}, +, \bar{0}). \quad (2.63)$$

Proof

Let $a = \bar{3} \in U(\mathbb{Z}/7\mathbb{Z})$. We define the mapping $\Phi: U(\mathbb{Z}/7\mathbb{Z}) \rightarrow \mathbb{Z}/6\mathbb{Z}$ by sending powers of the generator a to their integer exponents modulo 6:

$$\begin{aligned} \Phi: U(\mathbb{Z}/7\mathbb{Z}) &\rightarrow \mathbb{Z}/6\mathbb{Z}, \\ \bar{1} = a^0 &\mapsto \bar{0}, \\ \bar{3} = a^1 &\mapsto \bar{1}, \\ \bar{2} = a^2 &\mapsto \bar{2}, \\ \bar{6} = a^3 &\mapsto \bar{3}, \\ \bar{4} = a^4 &\mapsto \bar{4}, \\ \bar{5} = a^5 &\mapsto \bar{5}. \end{aligned}$$

Homomorphism Property: For any exponents $j, k \in \{0, 1, 2, 3, 4, 5\}$, multiplication in the unit group corresponds to addition of exponents:

$$a^j \cdot a^k = a^{j+k} = a^{(j+k) \bmod 6},$$

since $a^6 = \bar{1}$. Applying the map Φ :

$$\Phi(a^j \cdot a^k) = \overline{j+k} = \bar{j} + \bar{k} = \Phi(a^j) + \Phi(a^k) \in \mathbb{Z}/6\mathbb{Z}.$$

Hence, Φ is a group homomorphism. Because Φ maps the generator a of order 6 to the generator $\bar{1}$ of order 6, Φ is a bijective homomorphism, establishing the isomorphism.

Q.E.D.

 **Disambiguating Residue Class Notations Across Groups:** The bar notation $\bar{\cdot}$ operates over two distinct moduli in this isomorphism:

- In the domain $(U(\mathbb{Z}/7\mathbb{Z}), \cdot)$, bars denote residue classes modulo 7 under multiplication: $\bar{x} = x + 7\mathbb{Z}$.
- In the codomain $(\mathbb{Z}/6\mathbb{Z}, +)$, bars denote residue classes modulo 6 under addition: $\bar{k} = k + 6\mathbb{Z}$.

The isomorphism Φ is the “*discrete logarithm*” with respect to the base 3 modulo 7.

💡 *Insight: Paul Erdős on the Beauty of Numbers*

The lecturer references the celebrated quote by the Hungarian mathematician Paul Erdős (1913–1996): “*If numbers are not beautiful, I don't know what is.*” The unexpected isomorphism between the multiplicative units of a prime field $\mathbb{F}_7^\times$ and the simple additive clock arithmetic of $\mathbb{Z}/6\mathbb{Z}$ is a classic embodiment of this aesthetic: multiplication modulo 7 transforms into addition modulo 6 via the powers of a single primitive root.

📌 Summary of Key Concepts

- **Category Theory:** A category $\mathcal{C}$ comprises objects $\text{Obj}(\mathcal{C})$, hom-sets $\text{Hom}(A, B)$, associative composition $\circ$, and identity morphisms Id_A .
- **Categorical Morphisms:**
 - **Isomorphism:** Invertible morphism ($f \circ g = \text{Id}_B, g \circ f = \text{Id}_A$).
 - **Monomorphism:** Left-cancellable morphism ($f \circ g_1 = f \circ g_2 \implies g_1 = g_2$), generalizing injectivity.
 - **Epimorphism:** Right-cancellable morphism ($g_1 \circ f = g_2 \circ f \implies g_1 = g_2$), generalizing surjectivity.
- **Integral Domains:** A commutative ring with $1 \neq 0$ having no zero divisors ($a \neq 0 \wedge b \neq 0 \implies ab \neq 0$). The quotient ring $\mathbb{Z}/m\mathbb{Z}$ is an integral domain if and only if m is prime.
- **Units and Divisibility:** An element $u \in R$ is a unit if $u \mid 1$, i.e., u has a multiplicative inverse. The units form a multiplicative group $U(R)$.
- **Units of Residue Class Rings:** $U(\mathbb{Z}/m\mathbb{Z}) = \{\bar{x} \in \mathbb{Z}/m\mathbb{Z} \mid \gcd(x, m) = 1\}$ with order $\varphi(m)$. For prime $p = 7$, $U(\mathbb{Z}/7\mathbb{Z}) \cong (\mathbb{Z}/6\mathbb{Z}, +)$ generated by $\bar{3}$.

Tuesday 29. September 2026

Groups of Units, Fields of Fractions, and Polynomial Rings

Course Content Overview

This lecture investigates the transition from commutative rings to fields and polynomials, focusing on the invertible elements of rings, the localization process that produces the minimal ambient field of an integral domain, and the algebraic construction of polynomial rings:

- **The Group of Units** $\text{Unit}(R)$: Structure of multiplicatively invertible elements, characterization of units in $\mathbb{Z}$ and the Gaussian integers $\mathbb{Z}[i]$ via the norm, and the group of units of residue class rings $(\mathbb{Z}/m\mathbb{Z})^*$.
- **Euler's Totient Function** $\phi(m)$: Definition as the order of $(\mathbb{Z}/m\mathbb{Z})^*$, with explicit computations for prime powers $\phi(p) = p - 1$ and $\phi(p^2) = p(p - 1)$.
- **Universal Property of $\mathbb{Q}$** : Rigorous formulation of $\mathbb{Q}$ as the smallest field containing $\mathbb{Z}$, and the unique extension of ring embeddings into fields.
- **Construction of the Field of Fractions** $\text{Frac}(R)$: Generalization to arbitrary integral domains via equivalence classes of formal pairs $(x, y) \in R \times (R \setminus \{0\})$, verification of well-defined operations, and canonical embedding $i: R \hookrightarrow \text{Frac}(R)$.
- **Concrete Quotient Fields**: Field of fractions of $\mathbb{Z}$, the Gaussian rationals $\mathbb{Q}(i) = \text{Frac}(\mathbb{Z}[i])$, subrings of $\mathbb{Q}$ such as $\mathbb{Z}[\frac{1}{3}]$, and self-isomorphism for existing fields $\text{Frac}(K) \cong K$.
- **Polynomial Rings** $R[x]$: Intuitive formulation via formal linear combinations, rigorous foundation using almost-zero sequences in $R^{\mathbb{N}_0}$, componentwise addition, and Cauchy product (convolution) multiplication.

Review: Rings and the Group of Units

Spoken Transcript [00:00:00 - 00:00:32]

Okay, so we start.

About the quiz: as I said, I'm very much pro-guessing in mathematics; you're never penalized for getting the wrong answer. Lika said that there was some implementation problem and some people were penalized for getting the right answer, and she's going to fix that, so.

If you think the quiz was too easy, or too hard, or too medium, tell us. As I said, I've never gone through this quiz thing before. Okay, so then...

III The Group of Units of a Ring

Spoken Transcript [00:00:32 - 00:01:51]

Last week we discussed rings, and associated to a ring there's an additive group. This is easy, because it's part of the definition, but there's another group which was also discussed in the last lecture, which is the group of units.

And I remind you of this group... *[writes on the board]* Various people use different notation for describing it, but here I just say $\text{Unit } R$ so there's no question: it's the group of units of the ring R . It's those elements such that there exists somebody else where the product is 1, so it's those elements with a multiplicative inverse. And this is a group — we discussed this last week.

So, this group here is $\text{Unit } R$, and it has, well, the units by this definition, multiplication, and 1.

Whenever you see a ring, you immediately have these two groups. And this one is, you know, often the more complicated one to think about, because it's not part of the definition in some sense.

The Group of Units

Let $(R, +, \cdot, 0, 1)$ be a ring with unity. Associated with R are two natural canonical groups:

- (1) **The Additive Group:** $(R, +, 0)$, which forms an abelian group by the ring axioms.
- (2) **The Group of Units:** The set of multiplicatively invertible elements in R :

$$\text{Unit}(R) := \{x \in R \mid \exists y \in R \text{ such that } xy = 1\}. \quad (3.1)$$

Under ring multiplication, $(\text{Unit}(R), \cdot, 1)$ forms a group with identity element 1.

≡ Multiplicative Group Structure: The group of units is closed under multiplication: if $x, z \in \text{Unit}(R)$ with inverses $x^{-1}, z^{-1} \in R$, then $(xz)(z^{-1}x^{-1}) = 1$, so $xz \in \text{Unit}(R)$. Associativity is inherited from the ring, and the multiplicative identity 1 is invertible since $1 \cdot 1 = 1$.

III Units in the Integers and Gaussian Integers

Spoken Transcript [00:01:51 - 00:03:03]

We've seen this — I just review, but you know this — if you take $\mathbb{Z}$, the units are ± 1 . And I said this many times, but maybe since I didn't actually ever give a proof: in $\mathbb{Z}[i]$, the units are $\pm 1, \pm i$.

If you're taking complex analysis, you should be able to prove this, but let's actually

write a proof. I don't think I ever wrote the proof for this.

So, proof: this is a proof that these are the only invertible elements in this ring. Is it clear what I'm trying to prove?

Proof: let $a + bi \in \mathbb{Z}[i]$ be a unit. When I write this, of course that means that a and b are integers, okay? That's what being in there means.

All right, so then there exists somebody else, $c + di$, also here, such that $(a + bi)(c + di) = 1$. And this is multiplication of complex numbers.

Spoken Transcript [00:03:03 - 00:03:58]

The whole idea of the proof is understanding multiplication of complex numbers. And if you've taken a complex numbers class, or you've met complex numbers, however you know it: when you multiply two complex numbers, their angles add and their lengths multiply, right?

So, we just take the norm — the lengths — take the norms. The norm here is $\sqrt{a^2 + b^2}$; that's the length of this one. I hope that's not controversial, that's Euclid or something. This one is $\sqrt{c^2 + d^2}$, and their norms multiply, so this has to be just 1.

Okay? So, this is pretty restrictive: a, b, c , and d are all integers.

Spoken Transcript [00:03:58 - 00:05:13]

So, we can square that equation: $(a^2 + b^2)(c^2 + d^2) = 1^2$, which is 1. Now you have an equation where you have integers all multiplying to 1, and you know the only way this can happen is if this integer itself is ± 1 , because you can't factor 1 in the integers without having just $1 \cdot 1$ or $(-1) \cdot (-1)$. Is that clear?

How could this be 1? Well, a^2 for an integer is either 0, or 1, or something bigger. So, the only way this could be ± 1 is if a is equal to 0 or ± 1 , and b is equal to 0 or ± 1 — those are the only cases. If these are both positive integers, it's going to be bigger than 1. So, you only have to check these cases. Is that clear?

And if you look at these cases, well, these are the answers. Okay? That's a complete proof; that's as complete a proof as you get in this class. I think this proof was 100 percent complete.

That's a nice fact, and if you draw them, they look like this *[draws a sketch of the complex plane with four points on the axes]*. Those are the four units in the Gaussian integers.

Classification of Units in the Gaussian Integers

 **Example 3.1 (Units of the Integers):** For the ring of integers $\mathbb{Z}$:

$$\text{Unit}(\mathbb{Z}) = \mathbb{Z}^* = \{1, -1\}.$$

 **Proposition 3.2 (Units of the Gaussian Integers):** The group of units of the Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is:

$$\text{Unit}(\mathbb{Z}[i]) = \{1, -1, i, -i\}.$$

Proof

Let $z = a + bi \in \mathbb{Z}[i]$ be a unit with $a, b \in \mathbb{Z}$. By definition, there exists $w = c + di \in \mathbb{Z}[i]$ ($c, d \in \mathbb{Z}$) such that:

$$(a + bi)(c + di) = 1. \quad (3.2)$$

Taking the complex modulus (Euclidean length) on both sides of (3.2):

$$|a + bi| \cdot |c + di| = |1| = 1 \implies \sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2} = 1.$$

Squaring this relation yields an equation in non-negative integers:

$$(a^2 + b^2)(c^2 + d^2) = 1. \quad (3.3)$$

Since $a^2 + b^2 \in \mathbb{Z}_{\geq 0}$ and divides 1, we must have:

$$a^2 + b^2 = 1.$$

Because $a, b \in \mathbb{Z}$, the only integer solutions to $a^2 + b^2 = 1$ are:

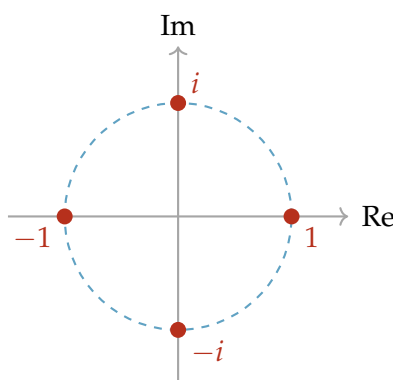
$$a = \pm 1, \quad b = 0 \implies z = \pm 1,$$

$$a = 0, \quad b = \pm 1 \implies z = \pm i.$$

Conversely, $1 \cdot 1 = 1$, $(-1)(-1) = 1$, and $i(-i) = 1$, so all four elements are invertible. Thus:

$$\text{Unit}(\mathbb{Z}[i]) = \{1, -1, i, -i\}.$$

Q.E.D.



≡ Algebraic Meaning of the Norm: The field norm $N(a + bi) = a^2 + b^2$ is multiplicative: $N(zw) = N(z)N(w)$. Since $N(z) \in \mathbb{Z}_{\geq 0}$ for any Gaussian integer, any invertible element must divide $N(1) = 1$, forcing $N(z) = 1$. The only lattice points on the circle of radius 1 are the four points shown above.

III Units in Residue Class Rings

Spoken Transcript [00:05:14 - 00:07:21]

We did a much more interesting example last time, which was also with proof, but since various people asked me about it afterwards, I want to do just another example. And that was this ring: $\mathbb{Z}/m\mathbb{Z}$.

That's a nice ring, and so it has a group of units, $\text{Unit}(\mathbb{Z}/m\mathbb{Z})$. This was a result I proved last time: the units are equal to those classes such that this is equal to 1. And as I said, it's standard here to say $m \geq 2$, so we're not dealing with the zero ring, okay?

So, that's the units, and these are the residue classes which are relatively prime. I did an example last time. This is a very interesting thing to study; it's just not a trivial thing, in the sense that one might say that if I take the additive structure here, this is very simple to think about. It's just adding up integers mod m . In some sense, most people learn about this in one name or the other in school.

But the units are more complicated, more interesting. And so I wanted to do another... there were various people who asked me some questions. So, let's try this: $\mathbb{Z}/10\mathbb{Z}$, just to say a little bit...

The standard notation for this is that: $(\mathbb{Z}/m\mathbb{Z})^*$. This is the usual notation for the units of $\mathbb{Z}/m\mathbb{Z}$. And also this notation is sometimes used in general: R^* sometimes denotes the units, depending on the author. I've just been writing Unit, so there's no question about what I mean, but in this case it's so common to use the star here that I'll also use the star. Is that clear?

Spoken Transcript [00:07:21 - 00:08:37]

How to think about this? Well, I'll just do an example, because as I said, there were various questions. How to think about this? This is all the residue classes that are relatively prime. So, you can just write them all first and cross them out: 0, 1, 2, 3, 4, 5... maybe I should have picked something less than 10...

These are all the residue classes, and then we have to cancel all the ones that are not relatively prime. So, which ones? Is 0 relatively prime? No, so we cross that out. 1 is relatively prime; 1 is always relatively prime. What about 2? No. 3? Yes, so we skip it. 4? That has a 2 in it, and 10 has a 2, so we cross that out. 5? 10 has a 5 in it, so we cross that one out, cross that one out, and what's left? 4, yeah. So, these are the four residue classes left.

The group of units here consists of these residue classes, okay? And the claim is that this is a group under multiplication. That's what that claim is. So, for example, 3 is in this group, so what is its inverse?

Student Answer

7.

Spoken Transcript [00:08:42 - 00:08:52]

Yeah, 7, because 7×3 is 21, and so on. So, anyway, you can play with this.

Units in $\mathbb{Z}/m\mathbb{Z}$ and the Example $\mathbb{Z}/10\mathbb{Z}$

 **Proposition 3.3 (Units in the Residue Class Ring):** Let $m \in \mathbb{Z}_{\geq 2}$. The group of units

of the residue class ring $\mathbb{Z}/m\mathbb{Z}$ (frequently denoted by $(\mathbb{Z}/m\mathbb{Z})^*$ or R^* for general rings) consists of those residue classes whose representatives are coprime to m :

$$(\mathbb{Z}/m\mathbb{Z})^* = \text{Unit}(\mathbb{Z}/m\mathbb{Z}) = \{\bar{x} \in \mathbb{Z}/m\mathbb{Z} \mid \gcd(x, m) = 1\}. \quad (3.4)$$

Example 3.4 (Computing the Units of $\mathbb{Z}/10\mathbb{Z}$): Consider the ring $\mathbb{Z}/10\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}, \bar{7}, \bar{8}, \bar{9}\}$. To find the units, we eliminate all classes that share a common divisor with 10 greater than 1:

- **Class $\bar{0}$:** $\gcd(0, 10) = 10 \neq 1$ (eliminate $\bar{0}$),
- **Even classes:** $\gcd(2, 10) = 2$, $\gcd(4, 10) = 2$, $\gcd(6, 10) = 2$, $\gcd(8, 10) = 2$ (eliminate $\bar{2}, \bar{4}, \bar{6}, \bar{8}$),
- **Class $\bar{5}$:** $\gcd(5, 10) = 5$ (eliminate $\bar{5}$).

The remaining coprime residue classes form the group of units:

$$(\mathbb{Z}/10\mathbb{Z})^* = \{\bar{1}, \bar{3}, \bar{7}, \bar{9}\}.$$

This is an abelian group of order 4 under multiplication modulo 10. For instance, the inverse of $\bar{3}$ is $\bar{7}$, since:

$$\bar{3} \cdot \bar{7} = \overline{21} = \bar{1} \in \mathbb{Z}/10\mathbb{Z}.$$

III Euler's Totient Function

III Definition and Elementary Calculations

Spoken Transcript [00:08:52 - 00:10:35]

This is a more interesting object, and in fact you could ask: what is the order? This is a basic question in elementary number theory: what is the order of this group?

By order, this word means for groups the number of elements. So, when I say order of a group, if it's a finite group, I just mean the number of elements. What is the order of this? For example, sometimes this order is denoted just by straight lines, the bars. You should be familiar with this set-theoretic notation.

These two bars usually denote order. Is that clear? That's kind of standard set-theoretic notation. So, first of all, let's do 10: what is this? Well, the answer is 4; you just have to compute these.

Spoken Transcript [00:10:35 - 00:12:12]

The order of $(\mathbb{Z}/m\mathbb{Z})^*$, the group of units, has a name. Do you know the name of this function? Yes, it's ϕ — Euler's totient function. That's just the definition. That's the definition of this function: it's just the order.

And you could ask: can you compute this? There are different ways to compute it, but it has to do with the factorization, and we will get to that later. So, you could say this is maybe a question for you now: try to compute Euler's totient function.

Euler is like the most famous Swiss person that ever lived. A few years ago it might have been Federer, but now he's retired, so it's Euler. And it turns out that Euler and Federer were both from Basel, so you can check it.

You know, my first trip to Switzerland was when I was a kid in the eighties, and the Swiss still had Euler on the 10 Swiss Franc note. I don't know if you've ever seen that; it's a beautiful note. You should go look it up.

Okay, so let's do one example to compute this. Well, let's do two examples. Suppose you have a prime p , then what is this?

Student Answer

$p - 1$.

Spoken Transcript [00:12:13 - 00:12:15]

And why is that?

Student Answer

Because any number less than p is coprime to p .

Spoken Transcript [00:12:22 - 00:13:48]

In other words, if you write this, you'll start... well, you'll write p elements, and you'll cross out 0, and everybody else is relatively prime. So, it's $p - 1$.

And so what about this one: $\phi(p^2) = ?$ Do you know how to do this?

What you do — I mean, of course once you do this, you can do the general case for powers of a prime — you write all the elements (this is a really primitive way to do it, but let's just do it): all the elements up to $p^2 - 1$. That's what we did there, right? So, it starts out, and then you cross out all of the...

Here you're going to cross out zero, so it's $p^2 - 1$, but you have to take away... you know that for these classes, of course you don't want zero, but we have to take away all the multiples of p , right?

And so now you have to concentrate really hard: how many multiples of p ? Well, there's p here, there's $2p$ here, and what's the last one? The last one you have to cross out is $p(p - 1)$; that's the last one. Is that clear?

So, we have to take away $p - 1$, so the answer is $p^2 - p$. The answer to this is $p^2 - p$.

Okay, that's the beginning; you can play with it a bit. All right?

Euler's Totient Function and Prime Power Formulas

 **Definition 3.5 (Euler's Totient Function):** The “*Euler totient function*” (or Euler's ϕ -function) $\phi: \mathbb{Z}_{\geq 1} \rightarrow \mathbb{Z}_{\geq 1}$ is defined as the order of the group of units of $\mathbb{Z}/m\mathbb{Z}$:

$$\phi(m) := |(\mathbb{Z}/m\mathbb{Z})^*| = \#\{x \in \{0, 1, \dots, m-1\} \mid \gcd(x, m) = 1\}. \quad (3.5)$$

 **Proposition 3.6 (Totient Values for Primes and Prime Squares):** Let p be a prime

number. Then:

$$\phi(p) = p - 1, \quad (3.6)$$

$$\phi(p^2) = p^2 - p = p(p - 1). \quad (3.7)$$

Proof

We evaluate the number of coprime residue classes in each case:

- **For a prime p :** The representatives are $\{0, 1, 2, \dots, p - 1\}$. The only representative sharing a non-trivial factor with p is 0. Thus, the remaining $p - 1$ elements are all coprime to p , giving $\phi(p) = p - 1$.
- **For p^2 :** The representatives are $\{0, 1, 2, \dots, p^2 - 1\}$. An element x shares a non-trivial factor with p^2 if and only if $p \mid x$. The multiples of p in this range are:

$$0, p, 2p, \dots, (p - 1)p.$$

There are exactly p such multiples. Subtracting these from the total of p^2 elements yields:

$$\phi(p^2) = p^2 - p = p(p - 1).$$

Q.E.D.

Generalization to Prime Powers: For any prime power p^k with $k \geq 1$, an integer $x \in \{0, 1, \dots, p^k - 1\}$ shares a factor with p^k if and only if $p \mid x$. The multiples of p are $0, p, 2p, \dots, (p^{k-1} - 1)p$, totaling p^{k-1} elements. Hence, $\phi(p^k) = p^k - p^{k-1} = p^k(1 - 1/p)$.

Fields of Fractions

The Universal Property of the Rationals

Spoken Transcript [00:13:48 - 00:15:15]

The next topic in the course has to do with fields of fractions. This is a peculiar subject.

It starts with this idea that $\mathbb{Z}$ is inside $\mathbb{Q}$. $\mathbb{Z}$ is a ring, it's an integral domain, and $\mathbb{Q}$ is a field, and $\mathbb{Z}$ sits inside $\mathbb{Q}$. But their relationship is not just that of any ring inside any field. In some sense, $\mathbb{Q}$ is the smallest field that contains $\mathbb{Z}$.

This understanding will lead to the study of the field of fractions. So, let's try to first make precise the qualitative statement that...

Here's the statement: " $\mathbb{Q}$ is the smallest field which contains $\mathbb{Z}$ ". That's a statement that probably you feel is true. It's not a 100 percent mathematical statement, so let's try to make this precise. You don't contradict the meaning of this statement, right?

Spoken Transcript [00:15:15 - 00:17:24]

One thing that happens when one starts studying these fields of fractions and $\mathbb{Q}$ is

that you get this strange feeling that maybe you never checked everything in $\mathbb{Q}$ works. Have you checked everything in $\mathbb{Q}$ works? It's kind of a strange feeling, because you learn about the rational numbers in school, but it's not like the definitions there are checked to work. You just accept them, which is pretty reasonable — they are correct.

What I mean by that is that $\mathbb{Q}$ is a much more complicated thing than $\mathbb{Z}$ conceptually, because $\mathbb{Z}$ is really great: there are these objects and they have names, and everybody has one name. That's what's great about $\mathbb{Z}$.

The thing that's confusing about $\mathbb{Q}$ is that there are objects, but they have many names. You can have $5/7 = 60/84$, and that's kind of confusing: all these numbers have many, many different names.

And when I say “*have you ever checked it works*”, it means that whatever operations you do, you have to be able to do them with any name, and it has to give the same answer, so to speak. That's what I mean.

So, the question is: have you ever checked that? It's a weird question. But kind of the ideal math question in first grade or second grade, when you're adding fractions, is that you should say: “*But why is it well-defined?*” That's the question you should ask. And if you'd asked that question, then we wouldn't have to give this lecture today, actually.

Spoken Transcript [00:17:24 - 00:19:45]

Okay, so I will try to make precise the statement that $\mathbb{Q}$ is the smallest field which contains $\mathbb{Z}$. There are different ways to do it, but one way is to say it like this:

Suppose I have $f: \mathbb{Z} \rightarrow F$, where F is a field, so this big F is a field, and this little f is an injection of rings — an injective homomorphism of rings. Okay, so I start with this. I take any possible field, like maybe a field that I might think is smaller than $\mathbb{Q}$, or whatever field — take any field you want — and I start with a ring homomorphism which is injective. That means that I view this as $\mathbb{Z}$ sitting inside F , okay?

And then I claim... The typical way you write this is with a diagram: so this is F , and then I claim there exists a unique g — so this is $\mathbb{Z}$ sitting inside $\mathbb{Q}$, and there exists a unique g — such that g is an injective homomorphism of fields, and the diagram commutes. That means I can take an integer here and go here, or I can consider it in $\mathbb{Q}$ and go this way, so that for all $x \in \mathbb{Z}$, $f(x) = g(x)$.

We would say the diagram commutes because if you have x here, I can get to here by going straight across, or I can go inside $\mathbb{Q}$ and come around; that would say that $f(x) = g(x)$.

Spoken Transcript [00:19:45 - 00:20:08]

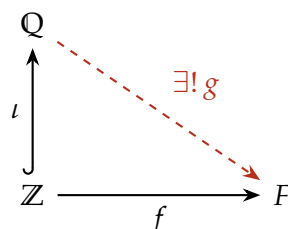
Okay. I'll give a little proof of this, but this is one way — this is now totally precise mathematics — that makes that statement, which is somehow slightly qualitative, into a mathematical statement.

Do you accept that? When I say $\mathbb{Q}$ is the smallest field which contains $\mathbb{Z}$, it means that if I have any other candidate field F , I can actually squeeze $\mathbb{Q}$ into it. And that means that $\mathbb{Q}$ is smaller than F . Does that make sense?

📖 Universal Property of the Rational Field

Qualitative Observation: $\mathbb{Q}$ is the “smallest” field containing the integral domain $\mathbb{Z}$.

💎 **Theorem 3.7 (Universal Property of $\mathbb{Q}$):** Let F be a field, and let $f: \mathbb{Z} \rightarrow F$ be an injective ring homomorphism (embedding of rings). Then there exists a unique field homomorphism $g: \mathbb{Q} \rightarrow F$ such that the following diagram commutes:



that is, $g|_{\mathbb{Z}} = f$, meaning $g(x) = f(x)$ for all $x \in \mathbb{Z}$. Furthermore, g is automatically injective.

≡ **Universal Property as Categorical Minimality:** In category theory, this universal mapping property identifies $\mathbb{Q}$ as the initial object among all field extensions receiving $\mathbb{Z}$. Any field F containing a copy of $\mathbb{Z}$ must necessarily contain a subfield isomorphic to $\mathbb{Q}$, showing that $\mathbb{Q}$ cannot be compressed or embedded into any strictly smaller field.

III Construction and Well-Definedness of the Field Extension

🗣 Spoken Transcript [00:20:08 - 00:21:23]

Okay, but I haven't proven it, so let's try to prove it. I won't prove the whole thing, because it's, of course, kind of painful.

Let us see: I have to prove this, which means I get to start with the data I suppose. That's what “suppose” is about. “Suppose” means I get to start with this data, which is here, all right? And then the whole claim is existence. So, to prove this, what I have to do is produce g .

Proof, more or less: I have to construct g . So, I have to write down... rational numbers: x over y , where x and y are integers, and $y \neq 0$. That's what a rational number is: x over y , x and y are integers, and y is not equal to zero. I hope you're happy with that; that's what a rational number is, okay?

And I have to define this — this is my responsibility — or we define: how am I going to define this?

👤 Student Answer

$f(x)/f(y)$.

Spoken Transcript [00:21:26 - 00:21:51]

Yeah. Formally, of course, we'll write it the way you said, but formally we write it like this: $f(x) \cdot f(y)^{-1}$, okay? Because $f(x)$ is in the field F , and $f(y)$ is in the field F , and I want to take its inverse. This is the kind of field language.

Why am I allowed to take its inverse? You cannot take the inverse of everything. Yeah?

Student Answer

Because F is a field and we chose, um... $y \neq 0$.

Spoken Transcript [00:21:58 - 00:23:12]

Yeah, but you have to check — and this is the whole thing — you have to check that this is legal so long as $f(y)$ is not zero. And why is $f(y)$ not zero? It's because f is injective and $y \neq 0$. That's actually using this thing here, okay? So, this is legal. Is that clear?

And as was implicit in the answer, we will often write this as $f(x)/f(y)$, because it's charming to do that. In a field, if you see this, all that means is that the fellow downstairs can be put upstairs by multiplicative inverse. To write this, you're only legally allowed to write it if you know this is not zero. Just like normally, you can't divide by zero. So, when we write this, it means in particular this one's not zero, and you can put it on either side — it doesn't matter, because our fields are commutative, okay?

Spoken Transcript [00:23:12 - 00:24:24]

Okay, so we managed to... but is this a definition? The answer, of course, is not! Because all you did was pick a particular name of this and give a definition. You have to check that all the names give the same definition. That's the business about the rational numbers which no one ever thinks about, and it's better not to think about it, but unfortunately today we have to think about it. *[The lecturer pauses to lower and erase the middle chalkboards]*

Candidate Definition of the Field Homomorphism g

 **Definition 3.8 (Candidate Map for Field Embedding):** Let $f: \mathbb{Z} \rightarrow F$ be an injective ring homomorphism into a field F . For a rational number represented by $\frac{x}{y} \in \mathbb{Q}$ with $x, y \in \mathbb{Z}$ and $y \neq 0$, we define:

$$g\left(\frac{x}{y}\right) := f(x) \cdot f(y)^{-1} = \frac{f(x)}{f(y)} \in F. \quad (3.8)$$

 **Legitimacy of Inversion in the Field F :** The inverse $f(y)^{-1}$ exists in F if and only if $f(y) \neq 0_F$:

(1) Non-Zero Denominator: In the rational representation $\frac{x}{y}$, we require $y \neq 0$ in $\mathbb{Z}$.

- (2) **Injectivity of Ring Embedding:** Since $f: \mathbb{Z} \rightarrow F$ is an injective ring homomorphism, its kernel is trivial: $\ker(f) = \{0\}$.
- (3) **Non-Vanishing Image:** It follows immediately that $y \neq 0 \implies f(y) \neq f(0) = 0_F$. Since F is a field, every non-zero element has a unique multiplicative inverse $f(y)^{-1} \in F$.

III Well-Definedness of the Field Extension

Spoken Transcript [00:24:25 - 00:26:11]

Okay, so what do we have to do? We aren't done with the definition, because we must further check that if x_1/y_1 is equal to x_2/y_2 , then $g(x_1/y_1)$ equals $g(x_2/y_2)$. You understand we have to check this, because that's the kind of slipperiness of the rational numbers: they don't have a unique name. All right. But you accept we have to check this, otherwise whatever was there was nonsense.

So, we have to check, and how are we going to check it? We have $g(x_1/y_1)$, and this is defined to be $f(x_1)/f(y_1)$, and $g(x_2/y_2)$ is $f(x_2)/f(y_2)$. We want to figure out whether these things are equal. So, we need to know: do we have this equality?

Spoken Transcript [00:26:11 - 00:27:58]

Using the fact that these are invertible and we can multiply by them, this question is equivalent to what's called clearing the denominators. By clearing the denominators, this is equivalent to $f(x_1)f(y_2) = f(x_2)f(y_1)$ by just multiplying both sides by $f(y_1)f(y_2)$, okay? That's a legal move in fields. And since these are invertible, this implies that, and it implies the other way, because they're both non-zero. Everything here is invertible.

We're using all the time that when we write this, neither y_1 nor y_2 is zero, which implies $f(y_1)$ and $f(y_2)$ are not zero, okay?

And now, these things are all integers, and everything's upstairs — there's no division. This is if and only if — using the homomorphism property — $f(x_1y_2) = f(x_2y_1)$, right? Because f is a homomorphism, okay?

And this is if and only if $f(x_1y_2 - x_2y_1) = 0$, okay? And why would this be the case? Yeah?

Student Answer

We could rearrange...

Spoken Transcript [00:27:59 - 00:29:02]

Yeah, we have to say what this means. What does this mean? This goes to the heart of the issue about whether you learned the rational numbers correctly, and the answer

is: this means that $x_1y_2 = x_2y_1$. It means nothing more and nothing less than that — it just actually means that, okay?

So, if we start with this, we have this equation, which is $x_1y_2 - x_2y_1 = 0$ in the integers. That's what this means.

And very fortunately, that's what we need: we need this to be zero. *[puts a checkmark on the board]* So, this is true. And so everything is true, and this is well-defined, all right?

That means I have defined a map. But there are a lot of things to check, which of course I'm not going to check. You have to do it, so to speak, on your own time.

Well-Definedness of the Extension Map g

 **Claim 3.9 (Well-Definedness of g):** The map $g: \mathbb{Q} \rightarrow F$ defined in (3.8) is well-defined; that is, if $\frac{x_1}{y_1} = \frac{x_2}{y_2}$ in $\mathbb{Q}$, then $g\left(\frac{x_1}{y_1}\right) = g\left(\frac{x_2}{y_2}\right)$ in F .

Proof

Let $\frac{x_1}{y_1}, \frac{x_2}{y_2} \in \mathbb{Q}$ with $y_1, y_2 \in \mathbb{Z} \setminus \{0\}$ such that $\frac{x_1}{y_1} = \frac{x_2}{y_2}$.

1. Meaning of Equality in $\mathbb{Q}$: By definition of the rational numbers as fractions, equality $\frac{x_1}{y_1} = \frac{x_2}{y_2}$ means precisely that the cross-multiplied relation holds in $\mathbb{Z}$:

$$x_1y_2 = x_2y_1 \iff x_1y_2 - x_2y_1 = 0 \in \mathbb{Z}. \quad (3.9)$$

2. Algebraic Chain in F : We want to establish whether:

$$g\left(\frac{x_1}{y_1}\right) \stackrel{?}{=} g\left(\frac{x_2}{y_2}\right) \iff \frac{f(x_1)}{f(y_1)} \stackrel{?}{=} \frac{f(x_2)}{f(y_2)}.$$

Since $y_1, y_2 \neq 0$ and f is injective, $f(y_1) \neq 0$ and $f(y_2) \neq 0$. Multiplying both sides by $f(y_1)f(y_2) \in F^\times$ (clearing denominators) yields an equivalent equality in F :

$$\begin{aligned} \frac{f(x_1)}{f(y_1)} = \frac{f(x_2)}{f(y_2)} &\iff f(x_1) \cdot f(y_2) = f(x_2) \cdot f(y_1) \\ &\iff f(x_1y_2) = f(x_2y_1) && \text{(since } f \text{ is a ring homomorphism)} \\ &\iff f(x_1y_2) - f(x_2y_1) = 0_F \\ &\iff f(x_1y_2 - x_2y_1) = 0_F. \end{aligned} \quad (3.10)$$

3. Conclusion: By (3.9), $x_1y_2 - x_2y_1 = 0 \in \mathbb{Z}$. Since any ring homomorphism maps 0 to 0:

$$f(x_1y_2 - x_2y_1) = f(0) = 0_F.$$

Thus, condition (3.10) holds, which proves that $g\left(\frac{x_1}{y_1}\right) = g\left(\frac{x_2}{y_2}\right)$. **Q.E.D.**

 **Core Pedagogical Principle: Non-Unique Names of Fractions:** A rational number has infinitely many equivalent names (e.g., $\frac{1}{2} = \frac{2}{4} = \frac{-3}{-6}$). Any function defined on rational numbers using numerator and denominator symbols must be proven invariant under replacement of representatives. As shown above, this reduces entirely to the cross-multiplication identity in the underlying ring $\mathbb{Z}$.

III Verification of Field Homomorphism Properties

Spoken Transcript [00:29:02 - 00:30:32]

I have now defined the map g , finally. Then you have to check that this map g is actually... I mean, all of the properties of g , that it's a homomorphism of fields. You have to check that it respects addition, and you have to check that it respects multiplication.

All of it is slightly painful, because at first blush it's obvious: you just pick representatives for $\mathbb{Q}$ as x_1/y_1 , and then you just check it. But then you have to check that all of that stuff respects the different choices, the different names of elements of $\mathbb{Q}$, just like we did here.

And the checks are all the same, so you can do it yourself. That's the end of what I'm going to give as the proof of this; you can check the rest. We're going to have to do many more things like this, so...

Spoken Transcript [00:30:32 - 00:32:18]

Okay, so I leave this to you. We're going to have to do things like this anyway, and it's all just unraveling the definitions. Meaning you have to check... *[writes on the board]* You have to check something like $g(x_1/y_1) + g(x_2/y_2) = g(x_1/y_1 + x_2/y_2)$.

You have to go check that; that's what you have to check. And you can see, when you write this, that this side... Okay, I'm going to do part of it for you. I don't know why I'm feeling compassionate today here.

This one is $f(x_1)/f(y_1) + f(x_2)/f(y_2)$; that's the definition here. And this has a different definition. First of all, how do we do it? We first have to add this fraction, right? Because our definition only applies to names of rational numbers, and this is a sum, so we first have to do this. Well, you know how to add fractions: this is equal to g of $(x_1y_2 + x_2y_1)/(y_1y_2)$. At least I hope you know how to add fractions. At least I hope I know how to add fractions, so.

Okay, so that's the addition of the fraction. And then this, by definition, is $f(x_1y_2 + x_2y_1)/f(y_1y_2)$, okay?

Spoken Transcript [00:32:18 - 00:32:48]

So, that's the one side to check, and that was supposed to equal this side, all right? And how are you going to check this? Well, you use the homomorphism property, you move everything upstairs, and then finally you're going to use the thing I erased, okay?

So, I leave it to you. There's nothing really I can add to this; you have to do it. But I do say it's painful — painful for me to write it on the board even.

Verification of the Field Homomorphism Axioms

 **Claim 3.10 (Preservation of Field Operations):** The map $g: \mathbb{Q} \rightarrow F$ is a field homo-

morphism; that is, for all $q_1, q_2 \in \mathbb{Q}$:

$$g(q_1 + q_2) = g(q_1) + g(q_2), \quad (3.11)$$

$$g(q_1 \cdot q_2) = g(q_1) \cdot g(q_2), \quad (3.12)$$

$$g(1_{\mathbb{Q}}) = 1_F. \quad (3.13)$$

Verification of Additivity

Let $q_1 = \frac{x_1}{y_1}$ and $q_2 = \frac{x_2}{y_2}$ with $x_i, y_i \in \mathbb{Z}$ and $y_i \neq 0$.

1. Evaluating the Sum Inside $\mathbb{Q}$: Adding fractions according to standard rational arithmetic:

$$q_1 + q_2 = \frac{x_1}{y_1} + \frac{x_2}{y_2} = \frac{x_1 y_2 + x_2 y_1}{y_1 y_2}.$$

Applying the definition of g :

$$\begin{aligned} g\left(\frac{x_1}{y_1} + \frac{x_2}{y_2}\right) &= g\left(\frac{x_1 y_2 + x_2 y_1}{y_1 y_2}\right) \\ &= \frac{f(x_1 y_2 + x_2 y_1)}{f(y_1 y_2)} \\ &= \frac{f(x_1)f(y_2) + f(x_2)f(y_1)}{f(y_1)f(y_2)}. \end{aligned}$$

2. Evaluating the Sum in F : On the other hand, evaluating the sum of images in the field F :

$$\begin{aligned} g\left(\frac{x_1}{y_1}\right) + g\left(\frac{x_2}{y_2}\right) &= \frac{f(x_1)}{f(y_1)} + \frac{f(x_2)}{f(y_2)} \\ &= f(x_1)f(y_1)^{-1} + f(x_2)f(y_2)^{-1}. \end{aligned}$$

Bringing both terms over the common denominator $f(y_1)f(y_2)$ in the field F :

$$\frac{f(x_1)}{f(y_1)} + \frac{f(x_2)}{f(y_2)} = \frac{f(x_1)f(y_2) + f(x_2)f(y_1)}{f(y_1)f(y_2)}.$$

Both expressions coincide identically, establishing (3.11). The multiplicative identity (3.12) and unit preservation (3.13) follow analogously. Q.E.D.

III The Field of Fractions of an Integral Domain

III Motivation: Embedding an Integral Domain into its Smallest Field

Spoken Transcript [00:32:48 - 00:34:30]

Okay, so that's the story about $\mathbb{Z}$. Their fates are really closely tied, $\mathbb{Z}$ and $\mathbb{Q}$. $\mathbb{Q}$ is not just any field that $\mathbb{Z}$ is in; it's the smallest field it's in.

So, we could ask: can we construct this relationship for other rings? Because we really like this relationship. If you study the integers, it's very useful sometimes to be able to divide. So, that's the movement to the field. Let's try this with a ring in general.

Suppose R is an integral domain. It turns out that this is kind of extremely important, and we will see in the definition — it may not be clear from the discussion so far

— but let's say R is an integral domain. Can we find the “smallest” field containing R ?

Already in this language, if we find a field containing R , that means R has to be a domain. Because domain means that when two elements are non-zero, their product stays non-zero; so if you're inside a field, that's always the case, okay? This is one of the reasons why we're talking only about integral domains.

So, that's the question: if you find on the street an integral domain, can you put it in its smallest field? That's the question. And the answer is yes, and that construction is called the field of fractions. And I will show you how to do it.

Spoken Transcript [00:34:30 - 00:36:15]

In some ways, there's lots of conceptual nice things about it, but this pain, you can imagine, we're not going to get away from. It's just going to be more abstract.

One of the things you will see here is that when we define the field of fractions of a ring, it will have this property that the elements are not single objects, but equivalence classes. And you should expect that answer, because that already happens for the rational numbers.

By the way, for a rational number, the objection — and I'm surprised no one made it — is that you could say that you could fix the name by taking the numerator and denominator and canceling all common factors. There's still a sign issue, but we leave that aside. If you try to do arithmetic with that, it's super painful, because you have to do a lot of cancellation. So, that is the argument against it; that's not really how you do arithmetic in the rational numbers. But that is the argument against what I was saying about the rational numbers.

In a sense, if you do that, it's kind of even more painful, because then when you write these expressions, you can't write these expressions — you have to cancel everywhere, and you know...

Generalization from Integers to Integral Domains

i Remark 3.11 (Why Integral Domains?): If a commutative ring R embeds into a field F via an injective homomorphism $\iota: R \hookrightarrow F$, then R must necessarily have no zero divisors:

$$a, b \in R \setminus \{0\} \implies \iota(a), \iota(b) \neq 0 \implies \iota(ab) = \iota(a)\iota(b) \neq 0 \implies ab \neq 0.$$

Thus, the property of being an “integral domain” is a strictly necessary prerequisite for any ring to be embedded into a field of fractions.

The Goal: Field of Fractions

Given an arbitrary integral domain R , our goal is to construct the minimal field containing R , called the “field of fractions” (or quotient field) $\text{Frac}(R)$, generalizing the classical relationship:

$$\mathbb{Z} \hookrightarrow \mathbb{Q} = \text{Frac}(\mathbb{Z}).$$

III The Set of Pairs and Equivalence Relations

Spoken Transcript [00:36:15 - 00:38:08]

Let's think about this in the abstract. The main thing is just the definition. You can imagine that here's our ring R — we don't know what it is, it's just any ring, but it's an integral domain. We want to construct a field containing R .

The field should have elements that are like x/y , where $x, y \in R$ and $y \neq 0$. This should be the element of that field, because if it contains R , I should be able to take x and multiply it by the inverse of y in the field, so it should have these elements in it. Is that clear? That's what happened with the rational numbers. So, we can just define it that way.

So, let S be the set of pairs. This whole thing is going to be a big disappointment if you think something fun is going to happen. I just tell you ahead of time: this is going to be a big disappointment. But some things in life are like that.

Okay, that's the set of pairs. For the integers, if we started with integers, we'd just get the set of pairs of integers where the second one is not zero, okay? But we can think about the set of pairs for any R , is that clear? This is inside $R \times R$; it's just some subset of pairs.

Spoken Transcript [00:38:08 - 00:39:18]

All right, and now the thing that would come is that there is an equivalence relation on S . Maybe we call this equivalence relation three lines — you can use various symbols for equivalence, that's kind of standard. How about three lines?

Do you know what an equivalence relation is? This was in your foundations class. An equivalence relation on a set T tells you when two elements are related, and the rules for the equivalence relation are: x is always related to itself (this is the reflexive property); if x is related to y , then y is related to x (this is the symmetric property).

And if you know the definition, what's the third one? Yeah?

Student Answer

Transitive.

Spoken Transcript [00:39:20 - 00:40:21]

Transitive property: it says that if we have x related to y and y related to z , then x is related to z . That's the whole thing about equivalence relations.

It comes up many times in mathematics; it will come up again in this course, so if you have not seen it, think about it. And the main thing we use about this equivalence relation is that if I have T with this equivalence relation, then I have another set, which is the set of equivalence classes: $T/\equiv$ is the set of equivalence classes.

And that's going to be very important for us. Whenever we have a set with an

equivalence relation, that equivalence relation divides the set up into disjoint equivalence classes, okay? And this thing here is the set of equivalence classes.

There are three objects in this kind of abstract situation: there's the set that we start with, there's the equivalence relation, and then there's the set of equivalence classes.

The Set of Pairs and General Equivalence Relations

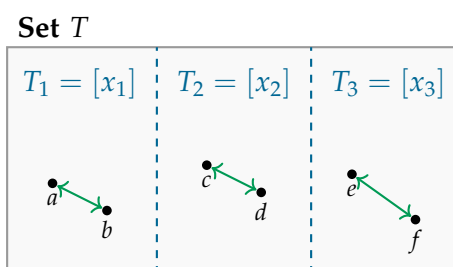
Definition 3.12 (*The Set of Formal Fractions S*): Let R be an integral domain. We define the set of formal numerator-denominator pairs S by:

$$S := \{(x, y) \in R \times R \mid y \neq 0\} \subset R \times R. \quad (3.14)$$

Remark 3.13 (*Review: Equivalence Relations and Partitions*): An “equivalence relation” $\equiv$ on a set T is a binary relation satisfying three foundational axioms:

- (i) **Reflexivity:** $\forall x \in T: x \equiv x$.
- (ii) **Symmetry:** $\forall x, y \in T: x \equiv y \implies y \equiv x$.
- (iii) **Transitivity:** $\forall x, y, z \in T: (x \equiv y \wedge y \equiv z) \implies x \equiv z$.

The quotient set $T/\equiv$ denotes the set of “equivalence classes”, which partitions T into mutually disjoint subsets.



Partition of T into disjoint equivalence classes $T/\equiv$

III The Equivalence Relation on Pairs

Spoken Transcript [00:40:21 - 00:41:03]

We have to define the equivalence relation. So, we now define this equivalence relation on our set S , which is the set of pairs. You should think about these pairs as numerator and denominator, but those are just words here.

We need to define: when is (x_1, y_1) equivalent to (x_2, y_2) ? We need a definition for this.

Okay, so what's the answer going to be? Yeah?

Student Answer

$$x_1 y_2 = x_2 y_1.$$

Spoken Transcript [00:41:06 - 00:42:06]

Very good. So, the definition is just exactly what you do with integer arithmetic: if you have a fraction, you say they're equal if $x_1y_2 = x_2y_1$, okay? That's the definition. There's nothing more or less than that; that's the whole definition.

And it's motivated by the integers. It's very important for this definition that this is only about R , right? This whole discussion starts with R , so we can only say things that are about R . We can make these pairs — that's only about R . We want to define this equivalence relation; at the end, these things are only in R , and the definition is only about some equation in R , okay? So, we haven't gotten out of R , in some sense, in this whole discussion. Is that clear?

Definition of the Equivalence Relation on S

 **Definition 3.14 (Equivalence Relation on S):** On the set $S = \{(x, y) \in R \times R \mid y \neq 0\}$, we define the relation $\equiv$ by:

$$(x_1, y_1) \equiv (x_2, y_2) : \Longleftrightarrow x_1y_2 = x_2y_1 \quad (\text{in } R). \quad (3.15)$$

 **Conceptual Motivation: Cross-Multiplication:** This relation is the formal algebraic counterpart to equating cross-products of fractions:

$$\frac{x_1}{y_1} = \frac{x_2}{y_2} \Longleftrightarrow x_1y_2 = x_2y_1.$$

Crucially, formula (3.15) involves only multiplication within the ring R , without invoking any external concept of division or fraction.

III Verification of Equivalence Relation Properties

Spoken Transcript [00:42:06 - 00:44:09]

Okay, so now we come to an issue: why is this an equivalence relation? This is the question that you should have asked your teacher in first grade or whatever it was. That's like the whole question about the rational numbers, in some sense — I mean the foundations.

[erases the lower chalkboard]

This is the first part where we have to do a little bit of algebra.

The first two properties: $x = x$, that's just identically true, okay? That's not a problem. We have to check three properties.

The first is if I take (x_1, y_1) , this is equivalent to itself *[equivalent to itself]*. Why is this? Because by definition, this is equal to $x_1y_1 = x_1y_1$, so we're very happy about that; there was nothing to do here.

You also have to check the symmetric property: that if we have $(x_1, y_1) \equiv (x_2, y_2)$, then the other way around is fine, and it'll just be the same equation. There's nothing to check about that.

Spoken Transcript [00:44:09 - 00:45:00]

So, the interesting one is the transitive property. This is the only fun one to check. We start with this — these are the inputs: $(x_1, y_1) \equiv (x_2, y_2)$ and $(x_2, y_2) \equiv (x_3, y_3)$. And we must show — this is for us to do — that $(x_1, y_1) \equiv (x_3, y_3)$.

The thing that's a little funny about this is that we have to write the first equation. This is what we have, that's what we've assumed, that's in the left column. What we have... *[bell chimes]*

Okay, you'll have to wait until after the break for this excitement to continue.

Verification of Reflexivity and Symmetry on S

 **Proposition 3.15** (*Reflexivity and Symmetry of $\equiv$*): The relation $\equiv$ defined on $S = \{(x, y) \in R \times R \mid y \neq 0\}$ by:

$$(x_1, y_1) \equiv (x_2, y_2) \iff x_1 y_2 = x_2 y_1 \quad (3.16)$$

is reflexive and symmetric.

Proof

Let $(x_1, y_1), (x_2, y_2) \in S$:

- **Reflexivity:** By commutativity of multiplication in the ring R , $x_1 y_1 = x_1 y_1$, which by (3.16) immediately implies:

$$(x_1, y_1) \equiv (x_1, y_1).$$

- **Symmetry:** Assume $(x_1, y_1) \equiv (x_2, y_2)$. Then $x_1 y_2 = x_2 y_1$. By symmetry of equality in R :

$$x_2 y_1 = x_1 y_2 \implies (x_2, y_2) \equiv (x_1, y_1).$$

Q.E.D.

Mid-Lecture Break

The lecturer announces the standard mid-lecture pause. The recording resumes after the break at timestamp 00:45:05.

Spoken Transcript [00:45:05 - 00:46:37]

Okay, so let's start. I should say that very few people have found my office, and it turns out I'll be in this visitor office for most of the semester, because they're clearing some kind of nuclear waste from my current office. So, my office now is HG G 65.1, if you want to find me there. *[writes "HG G 65.1" on the chalkboard]*

And yeah, I'm afraid I'll be there for most of the semester.

We come to this point that's actually kind of fun and interesting. We started with an integral domain, and led by the analogy with the rational numbers, we come to defining this field as some equivalence class of pairs, where — and I should have said this, and I

did say it, but I didn't write it, so the way to think about this... *[writes "Think of as x_1/y_1 " on the chalkboard]*

This is the mathematical text: think of it as x_1/y_1 , or x/y . You think of this pair in S as x/y , okay? Just like with the integers. But as an object, it's this formal object.

And then there was this proposal, which is correct, on how to define the equivalence relation. And now we need to actually check something: is it an equivalence relation? We have to prove the transitive property, and so we start with: what do we have?

Spoken Transcript [00:46:37 - 00:48:07]

What we have is that equivalence. If we translate that into equations, it says: x_1y_2 — and we have to be very careful of the subscripts, otherwise you'll confuse yourself — $x_1y_2 = x_2y_1$.

And then what we also have here is $x_2y_3 = x_3y_2$.

Okay, and what we want is this thing, which is $x_1y_3 = x_3y_1$.

The puzzling thing about this is that we have these quadratic equations and we need to get another quadratic equation. This seems a little bit unfortunate, because you might think you have to do some linear algebra with quadratic equations, and that's not going to work — you're not going to get a new quadratic equation from these.

So, the correct answer is that we're supposed to multiply here. The clever thing is to multiply the top equation by y_3 , which is what we want. Multiply by y_3 , and we get $x_1y_2y_3 = x_2y_1y_3$.

Okay, so this is kind of good: we see the x_1y_3 here. Can you see what we have to do next? This is actually algebra. What?

Student Answer

[suggests substituting from the second equation into the first]

Spoken Transcript [00:48:08 - 00:49:55]

Yeah, you could do that, but why don't we just take this y_2y_3 here and use this equation? It's the same thing. So, we take the y_2y_3 and we use this equation to flip them.

This is all about subscripts, and I should just tell you that subscripts have always been difficult for me.

Do you accept that? Because I just took this and flipped them using this equation.

So, now what do we do? We have here x_1y_3 , that looks pretty good, and we have y_3y_1 , but they're both multiplied by y_2 , right?

Now we use the fact that in these pairs, the second one is always born non-zero; that's part of being in this set: the second one has to be non-zero.

And this is an integral domain, so then we use the integral domain property.

As I said, this is a place where it's extremely important that it's an integral domain.

Since $y_2 \neq 0$, we can cancel, okay? Because being an integral domain says that's the cancellation property. Is that okay?

If y_2 is not zero, we can cancel it. So, we cancel it. This is kind of dramatic, but we cancel it here. *[cancels y_2 from both sides on the chalkboard]*

And that's what we want! We have now proven transitivity. And you see there was a little bit of cleverness here; it's easy to confuse yourself.

Proof of Transitivity and the Role of the Integral Domain

 **Proposition 3.16 (Transitivity of $\equiv$ on S):** Let R be an integral domain. The relation $\equiv$ on S defined by $(x_1, y_1) \equiv (x_2, y_2) \iff x_1 y_2 = x_2 y_1$ is transitive. Consequently, $\equiv$ is an equivalence relation on S .

Proof

Let $(x_1, y_1), (x_2, y_2), (x_3, y_3) \in S$, which implies $y_1, y_2, y_3 \in R \setminus \{0\}$.

1. Assumptions: We assume the relations:

$$(x_1, y_1) \equiv (x_2, y_2) \implies x_1 y_2 = x_2 y_1, \quad (3.17)$$

$$(x_2, y_2) \equiv (x_3, y_3) \implies x_2 y_3 = x_3 y_2. \quad (3.18)$$

2. Target: We must prove that $(x_1, y_1) \equiv (x_3, y_3)$, which means showing that:

$$x_1 y_3 = x_3 y_1. \quad (3.19)$$

3. Derivation via Multiplication and Substitution: Multiplying equation (3.17) on both sides by $y_3 \in R$:

$$(x_1 y_2) y_3 = (x_2 y_1) y_3 \iff (x_1 y_3) y_2 = (x_2 y_3) y_1. \quad (3.20)$$

Substituting $x_2 y_3 = x_3 y_2$ from (3.18) into the right-hand side of (3.20):

$$(x_1 y_3) y_2 = (x_3 y_2) y_1 = (x_3 y_1) y_2. \quad (3.21)$$

Subtracting the right-hand side gives:

$$(x_1 y_3 - x_3 y_1) y_2 = 0. \quad (3.22)$$

4. Cancellation via the Integral Domain Property: By definition of S , $y_2 \neq 0$. Because R is an **integral domain**, it contains no zero divisors:

$$ab = 0 \text{ with } b \neq 0 \implies a = 0.$$

Applying this cancellation property to (3.22) with $b = y_2 \neq 0$:

$$x_1 y_3 - x_3 y_1 = 0 \implies x_1 y_3 = x_3 y_1.$$

Thus, $(x_1, y_1) \equiv (x_3, y_3)$, which completes the proof. Q.E.D.

≡ Why Integral Domains Are Essential for Transitivity: Notice where the ring axioms alone would fail: equation (3.22) holds in any commutative ring. However, to deduce $x_1 y_3 = x_3 y_1$ from $(x_1 y_3 - x_3 y_1) y_2 = 0$, we must divide out and cancel the non-zero element y_2 . If R had zero divisors (for instance, in $\mathbb{Z}/6\mathbb{Z}$ with $y_2 = 2$ and $x_1 y_3 - x_3 y_1 = 3$), the relation $\equiv$ would **fail** to be transitive.

III Definition of the Field of Fractions and Canonical Embedding

Spoken Transcript [00:49:55 - 00:51:06]

We have come so far as to define S as a set with an equivalence relation. So, now we can define the points of the field that we're constructing, which will be the equivalence classes...

[The lecturer begins erasing the lower chalkboards]

This was almost like the first piece of algebra in this class, I think. I don't remember what we did; did we have any algebra before?

By that I mean playing with equations. Which, in the end... you know, I'm saying a lot of words, but in the end the subject is about equations, that's...

Spoken Transcript [00:51:06 - 00:52:37]

Okay, so where are we? We started with R as an integral domain. We constructed S with an equivalence relation, and from this we define F .

The name of this is the field of fractions of R — “field of fractions of R ”, that's the name of this object. F is equal to the set of equivalence classes of S . Is it completely clear what this thing is?

We have defined S , we have defined an equivalence relation, we took great pains to check all of its properties, and so we get the set of equivalence classes, and that is F .

Then we need to do a lot of things. But let's start by defining the map which is the inclusion from R to F , because we said this F should be a field that contains R . That means we have to have some way of getting points from R to F . And what's that way going to be?

This inclusion i : we have to put R into F . In other words, if I have an element $r \in R$ — normally people don't even write it, but I just... we have to put it into F . That means we have to assign it an equivalence class in this set. So, what are we going to do?

Yeah?

Student Answer

$(r, 1)$?

Spoken Transcript [00:52:39 - 00:53:41]

Yeah, that's a good idea: $(r, 1)$.

It has some advantages: $1 \neq 0$, because an integral domain is never the zero ring. So, this is legal, this is an element of S , and then we can take its equivalence class.

I'm not going to write equivalence classes like $[(r, 1)]$ everywhere because it becomes too complicated. You could say formally take its equivalence class like this, but if you start doing this, you get committed to writing things like $[(3, 4)]$ for rational numbers, and no one ever does that. So, we don't want to do it, but this would be kind

of more logically correct.

When I write this, it's just the equivalence class. I looked at what the book does, and the book says that $(r, 1)$ is an element of S , and that is its equivalence class. That's a very clever solution. I wouldn't say it's totally standard, but anyway; standard is just to ignore the issue.

So, this is the equivalence class here. Is this reasonable? Well, it better at least be... yeah...

Student Question

So, the set S , is it the whole $R \times R$, or is it just any subset of $R \times R$?

Spoken Transcript [00:53:50 - 00:55:47]

No, the set S was very precisely defined on the previous board: S is equal to those elements in $R \times R$ where $y \neq 0$. That's S . *[writes $S = \{(x, y) \in R \times R \mid y \neq 0\}$ on the board]*

And this is an element of S because $1 \neq 0$; and $1 \neq 0$ because R is an integral domain, and an integral domain is by definition not the zero ring. That's the full logical certificate, if you want it.

Still, we'd like more: we'd like this to be an injection. *[writes "Is i injective?" on the board]*

This means: when are $(r, 1)$ and $(\hat{r}, 1)$ in the same equivalence class of S , right? *[writes on the board]* That's what injectivity is about.

How do we check this? The answer is that these are in the same equivalence class if and only if that equivalence class identity holds: if and only if $r \cdot 1 = \hat{r} \cdot 1$, which is if and only if $r = \hat{r}$. This is great for us, because it shows that this is an injection. *[writes on the board and puts a checkmark]*

We've constructed... there's a lot of things to do in this, like a big house where you have to build all the pieces of it. This shows that now we have not only defined F , we've defined i , and we've put R inside F . And this placement is injective, okay?

Definition of the Field of Fractions and Canonical Embedding

Definition 3.17 (Field of Fractions): Let R be an integral domain, and let $S := \{(x, y) \in R \times R \mid y \neq 0\}$. The "field of fractions" (or quotient field) of R , denoted by $\text{Frac}(R)$ or F , is defined as the quotient set:

$$F := S / \equiv = \{[(x, y)] \mid (x, y) \in S\}. \quad (3.23)$$

The equivalence class $[(x, y)]$ is commonly written as $\frac{x}{y}$.

Definition 3.18 (Canonical Inclusion Map): We define the canonical embedding $i: R \rightarrow F$ by assigning to each $r \in R$ the equivalence class of the pair $(r, 1)$:

$$i(r) := [(r, 1)] \in F. \quad (3.24)$$

Proposition 3.19 (Injectivity of the Canonical Inclusion): The canonical map $i: R \rightarrow F$

is injective.

Proof

Let $r, \hat{r} \in R$. By definition of equality in the quotient set $F = S/\equiv$:

$$\begin{aligned} i(r) = i(\hat{r}) &\iff [(r, 1)] = [(\hat{r}, 1)] \\ &\iff (r, 1) \equiv (\hat{r}, 1) \\ &\iff r \cdot 1 = \hat{r} \cdot 1 \\ &\iff r = \hat{r}. \end{aligned}$$

Therefore, i is injective.

Q.E.D.

≡ Why $1_R \neq 0_R$ Is Guaranteed: For the pair $(r, 1)$ to belong to S , the denominator must satisfy $1 \neq 0$. By standard ring-theoretic convention, an integral domain must satisfy $1_R \neq 0_R$ (i.e., it cannot be the trivial zero ring). Hence, $(r, 1) \in S$ is unconditionally valid for every $r \in R$.

III Field Operations on the Field of Fractions: Zero, Unit, and Addition

Spoken Transcript [00:55:50 - 00:57:04]

As I said, this is just the beginning of the story. R already is born with all of the data: $(R, +, \cdot, 0, 1)$. We need to define all the structures on F and show this is a homomorphism also. *[The lecturer pushes the upper chalkboards up and erases the middle boards]*

Spoken Transcript [00:57:04 - 00:58:39]

We have to start talking about all the structures for F as a field. And for many of them, we don't have any choice: the zero in F has to correspond to the zero in R for it to be a homomorphism. So, the zero in F has to be $(0, 1)$. *[writes on the board]*

And also the unit: the unit in F has to be $(1, 1)$, because of course it has to correspond to the unit in R by the rule of getting R to F . *[writes on the board]*

We don't really have any choice in this matter. But we have to define plus and multiplication on F . *[writes on the board]* That's something that we have to construct: we have to define it, check it's compatible with R , check it satisfies the field axioms — and it's a lot of long and boring work.

The interesting part to start with is that we have to define it. If we have an element here, $(x_1, y_1) \in S$, and we have another one, $(x_2, y_2) \in S$, then we have to define this operation. This is up to us; we must define it. *[writes on the board]*

That's the problem. So, what's the solution? How are we going to define this? What do you think?

Tell me.

Student Answer

Take $x_1y_2 + x_2y_1$, and y_1y_2 .

Spoken Transcript [00:58:43 - 01:00:00]

Yeah, so we just use the standard definition for how you add fractions. *[writes $(x_1y_2 + x_2y_1, y_1y_2)$ on the board]* Meaning you pretend that it's this problem, which it kind of is: *[writes $\frac{x_1}{y_1} + \frac{x_2}{y_2} = \frac{x_1y_2 + x_2y_1}{y_1y_2}$ on the right]* You know how to do this.

I hope I got the subscripts right; you should tell me if I get the subscripts wrong.

This is the definition, nothing more, nothing less.

Now, is it a legal definition? There's an easy part: for it to be legal, the denominator better not be zero, because in our set S , the one in the basement always has to be non-zero.

We have here that this guy's non-zero, that guy's non-zero, and that implies their product is non-zero, because it is an integral domain. *[writes on the board]* Is that clear? Otherwise, if this were zero, this would not be legal.

Spoken Transcript [01:00:00 - 01:01:45]

That's the easy part. The hard part is you have to check that... this is a definition on S , but we have to check that it's well-defined on equivalence classes.

That means if I trade this person for another one in the same equivalence class, and I trade this for another person in the same equivalence class, the result has to be in the same equivalence class. Is that clear?

That's the check of rational arithmetic. That's the check that I claim, to understand the rational numbers, you should have done in school. Otherwise, what are you doing? You're doing some random manipulation.

So, you have to check this. *[writes "You have to check the indep. of choices of equiv. class" on the board]*

We've already done this kind of thing twice in the lecture. If you just do it and follow your nose, you'll see it all unwind: you just use the definition, and you have to say formally that this result respects our definition of the equivalence class.

In various books they do it; mostly it's left as an exercise. It's really important, so you should do it. If I do it, it's just a bunch of indices — it doesn't really help. You have to go through it yourself.

Is it clear what the problem is? I recommend trying this yourself, and if you get stuck, look at the book. The book does it. It'll give you a line of things, and you can imagine it's probably true.

Neutral Elements and Addition on the Field of Fractions

 **Definition 3.20** (*Neutral Elements in F*): The zero element and multiplicative unit

in F are defined by:

$$0_F := i(0_R) = [(0, 1)], \quad (3.25)$$

$$1_F := i(1_R) = [(1, 1)]. \quad (3.26)$$

Definition 3.21 (Addition on F): For any two equivalence classes $[(x_1, y_1)], [(x_2, y_2)] \in F$, addition is defined by:

$$[(x_1, y_1)] + [(x_2, y_2)] := [(x_1 y_2 + x_2 y_1, y_1 y_2)]. \quad (3.27)$$

Proposition 3.22 (Legitimacy and Well-Definedness of Addition): Addition on F as given in (3.27) is well-defined.

Proof

We verify two essential requirements:

- (1) **Legitimacy of the Denominator:** For $(x_1 y_2 + x_2 y_1, y_1 y_2)$ to lie in S , we must have $y_1 y_2 \neq 0$. Since $(x_1, y_1), (x_2, y_2) \in S$, we know $y_1 \neq 0$ and $y_2 \neq 0$. Because R is an **integral domain**, the product of two non-zero elements is non-zero:

$$y_1 \neq 0 \wedge y_2 \neq 0 \implies y_1 y_2 \neq 0.$$

Thus, $(x_1 y_2 + x_2 y_1, y_1 y_2) \in S$.

- (2) **Independence of Representatives:** Suppose $(x_1, y_1) \equiv (\hat{x}_1, \hat{y}_1)$ and $(x_2, y_2) \equiv (\hat{x}_2, \hat{y}_2)$, meaning:

$$x_1 \hat{y}_1 = \hat{x}_1 y_1 \quad \text{and} \quad x_2 \hat{y}_2 = \hat{x}_2 y_2. \quad (3.28)$$

We must show that $(x_1 y_2 + x_2 y_1, y_1 y_2) \equiv (\hat{x}_1 \hat{y}_2 + \hat{x}_2 \hat{y}_1, \hat{y}_1 \hat{y}_2)$, which requires:

$$(x_1 y_2 + x_2 y_1)(\hat{y}_1 \hat{y}_2) \stackrel{?}{=} (\hat{x}_1 \hat{y}_2 + \hat{x}_2 \hat{y}_1)(y_1 y_2).$$

Expanding the left-hand side and applying (3.28):

$$\begin{aligned} (x_1 y_2 + x_2 y_1) \hat{y}_1 \hat{y}_2 &= (x_1 \hat{y}_1)(y_2 \hat{y}_2) + (x_2 \hat{y}_2)(y_1 \hat{y}_1) \\ &= (\hat{x}_1 y_1)(y_2 \hat{y}_2) + (\hat{x}_2 y_2)(y_1 \hat{y}_1) \\ &= (\hat{x}_1 \hat{y}_2 + \hat{x}_2 \hat{y}_1) y_1 y_2. \end{aligned}$$

This establishes equivalence, confirming that addition is independent of the choice of representative.

Q.E.D.

III Multiplication and Ring Embedding Properties

Spoken Transcript [01:01:45 - 01:03:17]

[The lecturer erases the left chalkboard]

You also have to do it with multiplication. Multiplication is a bit easier; addition is the most painful one. Also define multiplication:

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2, y_1 y_2),$$

which will surprise no one, okay? *[writes on the board]* That's the definition, right?

And again, next is the check of compatibility with the equivalence classes. This check is easy, so I'll do it since it's easy.

Spoken Transcript [01:03:17 - 01:05:40]

Suppose we have that (x_1, y_1) is in the same equivalence class as $(\hat{x}_1, \hat{y}_1)$, and (x_2, y_2) is in the same equivalence class as $(\hat{x}_2, \hat{y}_2)$. *[writes on the board]*

Then we have to show — that's the input, right? — we must show that (x_1x_2, y_1y_2) is in the same equivalence class as if we chose the other representatives: $(\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2)$. *[writes on the board]* That's what we have to do.

What we know is this: the equation that these are in the same equivalence class is $x_1y_1 = \hat{x}_1y_1$, and this equation is $x_2y_2 = \hat{x}_2y_2$, all very confusing. *[writes on the board]*

And now we just multiply them together! These are what we know, and then we have these two equations: we can multiply them all together, and we get $x_1x_2y_1y_2 = \hat{x}_1\hat{x}_2y_1y_2$. *[writes on the board]*

That equation is exactly the certificate that these things are in the same equivalence class, okay? This implies that.

This shows that if I pick different equivalence classes here, the result doesn't matter. This one I did completely, because it's easy: all you do is multiply. For the other one, you have to do the kind of tricks we were doing before, where you have to clear the denominators, etc., so you can do that, so to speak, on your own time.

Spoken Transcript [01:05:40 - 01:06:50]

Then you have to check that these definitions are compatible with R . I mean, they're just all defined that way, meaning we have to check that this inclusion i is a homomorphism of rings. *[writes on the board]*

And again, there's really nothing to this: it means that if you take $r + \hat{r}$, and then you go by i , you get $(r + \hat{r}, 1)$. *[writes on the board]*

Then you have to check that this thing is equal to $(r, 1) + (\hat{r}, 1)$, and that's the same by the definition that we used there. But you have to check it, and you have to also check it for multiplication. So, it's just a bunch of checking. Here, all the cards are played, and you just have to check everything, okay?

Multiplication on F and Ring Homomorphism Properties

 **Definition 3.23 (Multiplication on F):** For any $[(x_1, y_1)], [(x_2, y_2)] \in F$, multiplication is defined by:

$$[(x_1, y_1)] \cdot [(x_2, y_2)] := [(x_1x_2, y_1y_2)]. \quad (3.29)$$

 **Proposition 3.24 (Well-Definedness of Multiplication):** Multiplication on F is well-defined and independent of the choice of representative.

Proof

Let $(x_1, y_1) \equiv (\hat{x}_1, \hat{y}_1)$ and $(x_2, y_2) \equiv (\hat{x}_2, \hat{y}_2)$ in S . By definition of $\equiv$:

$$x_1y_1 = \hat{x}_1y_1 \quad \text{and} \quad x_2y_2 = \hat{x}_2y_2. \quad (3.30)$$

Multiplying these two equations in the commutative ring R :

$$(x_1y_1)(x_2y_2) = (\hat{x}_1y_1)(\hat{x}_2y_2) \implies (x_1x_2)(y_1y_2) = (\hat{x}_1\hat{x}_2)(y_1y_2).$$

By (3.16), this algebraic identity is precisely the condition for equivalence:

$$(x_1x_2, y_1y_2) \equiv (\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2).$$

Thus, $[(x_1x_2, y_1y_2)] = [(\hat{x}_1\hat{x}_2, \hat{y}_1\hat{y}_2)]$, proving well-definedness.

Q.E.D.

Proposition 3.25 (*i is an Injective Ring Homomorphism*): The canonical inclusion $i: R \hookrightarrow F$ is an injective ring homomorphism:

$$i(r + \hat{r}) = i(r) + i(\hat{r}), \quad (3.31)$$

$$i(r \cdot \hat{r}) = i(r) \cdot i(\hat{r}), \quad (3.32)$$

$$i(1_R) = 1_F. \quad (3.33)$$

Proof

Evaluating the definitions directly:

- **Additivity:**

$$i(r) + i(\hat{r}) = [(r, 1)] + [(\hat{r}, 1)] = [(r \cdot 1 + \hat{r} \cdot 1, 1 \cdot 1)] = [(r + \hat{r}, 1)] = i(r + \hat{r}).$$

- **Multiplicativity:**

$$i(r) \cdot i(\hat{r}) = [(r, 1)] \cdot [(\hat{r}, 1)] = [(r\hat{r}, 1 \cdot 1)] = [(r\hat{r}, 1)] = i(r\hat{r}).$$

- **Identity:**

$$i(1_R) = [(1, 1)] = 1_F.$$

Injectivity was already established in Proposition 3.19.

Q.E.D.

≡ Multiplicative Inverses in F : To confirm that F is indeed a field, let $[(x, y)] \in F$ with $[(x, y)] \neq 0_F = [(0, 1)]$. Then $(x, y) \not\equiv (0, 1) \implies x \cdot 1 \neq 0 \cdot y \implies x \neq 0$. Thus, $(y, x) \in S$, and:

$$[(x, y)] \cdot [(y, x)] = [(xy, yx)] = [(1, 1)] = 1_F.$$

Therefore, every non-zero element in F possesses a multiplicative inverse $[(x, y)]^{-1} = [(y, x)]$.

III Summary of the Explicit Construction

Spoken Transcript [01:06:50 - 01:08:46]

That is the construction.

As I said, when you understand it, you'll realize that actually you knew the whole thing anyway in your understanding of the rational numbers. You were subconsciously doing all of this — not explicitly, but underneath the surface, you were assuming all of these properties were true; it made rational number arithmetic work.

There are different ways to try to learn this material, but the way that's foundational is that you actually go yourself and check all the things that you're supposed to check. That takes a little bit of time, and it's somewhat painful, but you only have to do it once in your life.

You can read the book, too. I think that doesn't help so much, because then you just

read what the author has written, and it's just a bunch of things with equalities. You can convince yourself of anything if you read what someone else did. So, the best thing is to go through all of it yourself; then you'll understand it once in your life.

And that's the construction, so we're done with this construction. I'll just summarize here: *[writes on the lower left chalkboard]*

We start with R as an integral domain, and then we construct explicitly — that's as explicit a mathematical construction as you get. You just take R , and you make some sets and equivalence classes and relations; it's totally explicit. We construct this F , which is the field of fractions.

And R sits inside F by this map: $r \mapsto (r, 1)$.

Summary of the Construction of the Field of Fractions

 **Theorem 3.26 (Existence and Structure of the Field of Fractions):** Let R be an integral domain. There exists a field F , called the “*field of fractions*” of R (denoted $\text{Frac}(R)$), and an injective ring homomorphism:

$$i: R \hookrightarrow F, \quad r \mapsto [(r, 1)],$$

such that:

- (1) **Field Structure:** F is a field with zero $0_F = [(0, 1)]$ and multiplicative identity $1_F = [(1, 1)]$.
- (2) **Quotient Representation:** Every element of F can be written as a quotient of elements in the image of R :

$$\forall q \in F \exists r, s \in R (s \neq 0) \quad \text{such that} \quad q = i(r) \cdot i(s)^{-1} = \frac{i(r)}{i(s)}. \quad (3.34)$$

III Algebraic Perspective: Fractions in an Integral Domain

Spoken Transcript [01:08:46 - 01:09:44]

When you get used to it, the way an algebraist always thinks about this is: you have elements in R , and the elements of F are just r/s ; you just view them as fractions in R .

This is a perfectly reasonable way to think about it. In our language, when someone writes this, what they really mean is the equivalence class of (r, s) in this set S . That's what it actually means. And it's only legal if the bottom one is not zero, which means you're not allowed to write this anyway if the bottom one's not zero. Then you can add fractions and multiply fractions just as you would normally do, and they correspond exactly to the rules that were prescribed here. That's the natural way to think about it.

Once all of this is done, then you just start using it like you normally would anyway. But this is the underlying foundation.

Are there any conceptual questions about that?

Algebraic View: Working with Fractions

In mathematical practice, once the formal construction of $F = S/\equiv$ is established, we identify R with its image $i(R) \subseteq F$ and adopt standard fraction notation:

$$\frac{r}{s} := [(r, s)] \in \text{Frac}(R) \quad (r, s \in R, s \neq 0). \quad (3.35)$$

Arithmetic in the Field of Fractions: Under this notation, the field operations on $\text{Frac}(R)$ take the standard form familiar from elementary arithmetic:

$$\text{Equality:} \quad \frac{r_1}{s_1} = \frac{r_2}{s_2} \iff r_1 s_2 = r_2 s_1, \quad (3.36)$$

$$\text{Addition:} \quad \frac{r_1}{s_1} + \frac{r_2}{s_2} = \frac{r_1 s_2 + r_2 s_1}{s_1 s_2}, \quad (3.37)$$

$$\text{Multiplication:} \quad \frac{r_1}{s_1} \cdot \frac{r_2}{s_2} = \frac{r_1 r_2}{s_1 s_2}, \quad (3.38)$$

$$\text{Inverse:} \quad \left(\frac{r}{s}\right)^{-1} = \frac{s}{r} \quad (\text{provided } r \neq 0). \quad (3.39)$$

III Examples of Fields of Fractions

III The Rationals, Gaussian Rationals, and Subrings of $\mathbb{Q}$

Spoken Transcript [01:09:44 - 01:10:44]

If I take $\mathbb{Z}$, you can ask: what is the field of fractions?

The answer is going to be $\mathbb{Q}$, which means that if you start with $\mathbb{Z}$ and you do this crazy construction, you get the rational numbers. I hope that's clear by now.

All right, so what happens if I do something else? What happens if I start with the ring of Gaussian integers? What is the field of fractions?

That's a strange question, because I've just explained what the field of fractions is: you take the Gaussian integers, you do this construction, and that's the field of fractions. So, when I say "*what is the field of fractions*", what the question is actually asking is: what's another way of thinking about the answer?

Do you have a proposal for what that is?

Yeah?

Student Answer

Just $\mathbb{Q}$ adjoined with i .

Spoken Transcript [01:10:46 - 01:12:32]

Yeah, so the answer is this: $\mathbb{Q}(i)$, which is equal to the set of $a + bi$, where $a, b \in \mathbb{Q}$.

It's correct, but now to check this is correct, you have to do a bunch of things: one is you have to show this is a field (you can invert), and then you have to show that this

sits inside, and everything here is a ratio of that. It's all true, but that's the answer here, and it requires some checking. All right, this is correct.

What about this? We saw last week that there are some funny rings that sit inside $\mathbb{Z}$ and $\mathbb{Q}$. This ring is the set of $a/3^b$, where $a \in \mathbb{Z}$ and b is some integer greater than or equal to 0, right? We discussed this last week: the subrings that sit in between. I hope you remember.

Do you remember?

This is an integral domain also, because it sits inside $\mathbb{Q}$. So, what is the field of fractions of... I just gave it a name, $\mathbb{Z}[1/3]$, because you're allowed to have threes in the bottom. There are two standard names for this: one is $\mathbb{Z}[1/3]$, and the other is... yeah, that's a better name, okay.

What is the field of fractions of this integral domain?

Student Answer

Also $\mathbb{Q}$.

Spoken Transcript [01:12:34 - 01:13:33]

Also $\mathbb{Q}$! Nothing says it can't be $\mathbb{Q}$. It certainly sits inside $\mathbb{Q}$, and everything in $\mathbb{Q}$ is already a ratio of numbers here. So, it's just $\mathbb{Q}$. If you go through this definition, you get $\mathbb{Q}$.

As I said, you think about this field of fractions as the smallest field that contains the integral domain. You can't really contain something smaller than $\mathbb{Q}$, because it already has $\mathbb{Z}$ in it, and $\mathbb{Z}$ already gives you everything in $\mathbb{Q}$.

So, what is the field of fractions of this? The answer is also $\mathbb{Q}$.

In fact, if you take any ring, any integral domain that sits in between $\mathbb{Z}$ and $\mathbb{Q}$, the field of fractions is just $\mathbb{Q}$.

Examples: Fractions of the Integers, Gaussian Integers, and Subrings of $\mathbb{Q}$

 **Example 3.27 (The Integers):** For the ring of integers $\mathbb{Z}$:

$$\text{Frac}(\mathbb{Z}) = \mathbb{Q}.$$

 **Example 3.28 (The Gaussian Integers):** For the ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$:

$$\text{Frac}(\mathbb{Z}[i]) = \mathbb{Q}(i) := \{a + bi \mid a, b \in \mathbb{Q}\}.$$

The field $\mathbb{Q}(i)$ is called the field of “*Gaussian rationals*”.

 **Example 3.29 (Dyadic and Triadic Rationals):** Consider the ring of triadic fractions $\mathbb{Z}[\frac{1}{3}]$, defined by:

$$\mathbb{Z}[\frac{1}{3}] := \left\{ \frac{a}{3^b} \mid a \in \mathbb{Z}, b \in \mathbb{Z}_{\geq 0} \right\} \subset \mathbb{Q}.$$

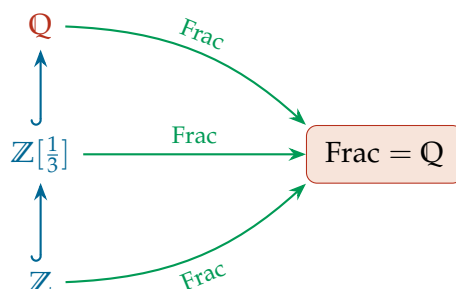
Since $\mathbb{Z} \subset \mathbb{Z}[\frac{1}{3}] \subset \mathbb{Q}$, any fraction of elements from $\mathbb{Z}[\frac{1}{3}]$ is already a rational number, and every rational number $\frac{p}{q}$ can be written as a quotient of elements in $\mathbb{Z} \subset \mathbb{Z}[\frac{1}{3}]$.

Thus:

$$\text{Frac} \left(\mathbb{Z} \left[\frac{1}{3} \right] \right) = \mathbb{Q}.$$

◆ **Proposition 3.30 (Intermediate Subrings of $\mathbb{Q}$):** Let R be any subring of $\mathbb{Q}$ with $\mathbb{Z} \subseteq R \subseteq \mathbb{Q}$. Then:

$$\text{Frac}(R) = \mathbb{Q}. \quad (3.40)$$



≡ **Why All Intermediate Subrings Yield $\mathbb{Q}$:** By categorical minimality, $\text{Frac}(R)$ is the smallest field containing R . Since $\mathbb{Z} \subseteq R$, any field containing R must contain $\mathbb{Z}$, and hence must contain $\text{Frac}(\mathbb{Z}) = \mathbb{Q}$. Since $R \subseteq \mathbb{Q}$, $\mathbb{Q}$ is already a field containing R . Therefore, $\text{Frac}(R)$ cannot be strictly larger or smaller than $\mathbb{Q}$, forcing $\text{Frac}(R) = \mathbb{Q}$.

III Field of Fractions of an Existing Field

🎤 Spoken Transcript [01:13:33 - 01:14:49]

Okay, so now a stranger question is: what happens if... well, yeah.

You know, on the business of guessing: once I was at a conference, and the conference had some math researchers and some high school teachers. At dinner I was sitting next to some teachers, and I was trying to explain to them my theory of how you should teach children math. And the outcome was that they wanted me disbarred from academics.

The theory, by the way, was that on exams you should make the kids just guess the answers: no calculation, just guess everything. That's, you know, maybe a little excessive, but you know...

Maybe a compromise: on half the exam they should just guess everything. But I think that would be really, somehow, more in the spirit of mathematics than...

🎤 Spoken Transcript [01:14:49 - 01:15:25]

A kind of stranger question is: suppose I start with a field. Suppose R is a field. *[writes on the board]*

Then, of course, it's still an integral domain, because every field is an integral domain. Then what is the field of fractions? *[writes on the board]*

What do you think the field of fractions is going to be here?
Yeah?

Student Answer

It's R itself.

Spoken Transcript [01:15:28 - 01:16:00]

Yes, R itself! *[writes " R itself" on the board]*

And the answer is R itself.

To prove that: again, this fits in with this view that it's the smallest field. If it's already a field, how could there be a smaller field that contains it? It's already a field.

To prove it, you have to go through this construction and show that in the case where R is a field, that construction just reconstructs R . To do that, you have to go through that construction and see it, so that's also an exercise for you.

Those are really the basic facts.

 **Proposition 3.31 (Field of Fractions of a Field):** Let K be a field. Then the canonical inclusion $i: K \hookrightarrow \text{Frac}(K)$ is an isomorphism of fields:

$$\text{Frac}(K) \cong K. \quad (3.41)$$

Proof

By [Proposition 3.25](#), $i: K \hookrightarrow \text{Frac}(K)$ is an injective field homomorphism. To establish surjectivity, let $[(x, y)] \in \text{Frac}(K)$ with $x, y \in K$ and $y \neq 0$. Since K is a field, y has a multiplicative inverse $y^{-1} \in K$. Then:

$$(x, y) \equiv (xy^{-1}, 1) \iff x \cdot 1 = (xy^{-1}) \cdot y = x.$$

Consequently:

$$[(x, y)] = [(xy^{-1}, 1)] = i(xy^{-1}).$$

Every element of $\text{Frac}(K)$ is the image of the element $xy^{-1} \in K$. Hence, i is surjective and constitutes a field isomorphism. **Q.E.D.**

Polynomial Rings and Power Series Rings

Definition of the Polynomial Ring in One Indeterminate

Spoken Transcript [01:16:00 - 01:18:16]

The next topic this week is about polynomial rings. These are really fun. We'll talk about polynomial rings *[writes on the board]*, and in some sense even more fun than polynomial rings are power series rings.

Again, this material, in some sense, you've been using all of this already in the first year, but here we take a formal approach.

The idea is: if I take R as a ring — and now, no assumptions, meaning not necessarily an integral domain, so this just means commutative for us: a commutative ring with unit. That's our basic ring: when we say ring with no assumptions, we mean a commutative ring with unit. In particular, for everything today so far, R was an integral domain. Now, I don't care if it's an integral domain; it's just a commutative ring with unit.

Then I can make a new ring here: $R[x]$.

Here, x is the variable, or sometimes it's called the indeterminate. That's kind of a fancy name, but it's just x , right? The indeterminate or the variable.

The name of this ring is the polynomial ring with coefficients in R . Or if you want to be even more precise: the polynomial ring in one variable with coefficients in R . That's kind of a long sentence, but if you meet a mathematician and say, *"I'm going to consider the polynomial ring in one variable with coefficients in R "*, as long as you agree on what a ring is, that's absolutely unambiguous — the person will know exactly what you're talking about.

What is this? Well, I have to tell you what it is as a set. So, the first thing is: what's the set?

Spoken Transcript [01:18:16 - 01:20:17]

Here the definition is that it's the set of polynomials in x with coefficients in R . Polynomials look like this: $a_0 + a_1x + a_2x^2 + \dots$ [writes $R[x] = \{a_0 + a_1x + a_2x^2 + \dots + a_nx^n \mid a_i \in R, n \geq 0\}$ on the board]

It's these symbols, up to eventually... One of the properties of a polynomial is that it's not infinite; a polynomial is finite. That's always part of the definition in mathematics of a polynomial. That means we stop somewhere, at some n , with a_nx^n . The elements of this ring are such symbols: formal combinations of a_ix^i , where these a_i are in R , and possibly different n , where $n \in \mathbb{Z}_{\geq 0}$.

If I have $\mathbb{Z}[x]$, the elements in here are sometimes written $f(x)$, because we like to write polynomials with the variable. It could be, for example, $f(x) = 3 + 2x + 15x^2 + 345x^{200}$. This is a totally legitimate element of it. Other times, you'll see people just write this as f and suppress the variable. Those are two different ways you can write polynomials.

For $R[x]$, it's just the same thing: you write an element here, $f(x)$, by picking some finite number of elements of R — the coefficients, all the way to a_n — and then you just put the powers of x here.

Here, x is a formal variable: there's no relation; x is in some sense just a placeholder. Those are the elements of $R[x]$.

Spoken Transcript [01:20:17 - 01:21:26]

$R[x]$ has a confusing element, just like the integers: there's a confusing polynomial,

like the constant polynomial. If I put 17 here, that is also in $R[x]$.

That's the polynomial that only has a constant term; it just happens that there's no x in that. It's weird to write this polynomial as a function of x , but it is a function of x — it's a constant function. In this ring, there are also the constant polynomials, the ones with no x .

Is it clear what this set is? You take R , and then you take an x that has nothing to do with R — it's just some kind of Platonic ideal indeterminate — and then you consider these formal linear combinations. That's what a polynomial is.

In particular, I don't really mean anything by this plus: it's just a formal linear combination. Some authors don't like to write this because they think you might think that this plus here means something, which is a legitimate thing to think, actually.

The Polynomial Ring $R[x]$: Informal Formulation

 **Definition 3.32 (Polynomial Ring over a Commutative Ring):** Let R be a commutative ring with unit 1_R . The “polynomial ring in one variable” (or indeterminate) x with coefficients in R , denoted $R[x]$, is defined as the set of formal expressions:

$$R[x] := \left\{ \sum_{i=0}^n a_i x^i = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \mid n \in \mathbb{N}_0, a_i \in R \right\}. \quad (3.42)$$

 **Example 3.33 (Polynomials in $\mathbb{Z}[x]$):** In the ring $\mathbb{Z}[x]$, an element may be denoted $f(x)$ or simply f :

$$f(x) = 3 + 2x + 15x^2 + 345x^{200} \in \mathbb{Z}[x].$$

Constant elements such as $17 \in \mathbb{Z}$ correspond to polynomials of degree 0 with $n = 0$:

$$17 = 17 \cdot x^0 \in \mathbb{Z}[x].$$

 **The Indeterminate as a Formal Placeholder:** Crucially, x is not an element of R , nor does it represent an unknown numerical value in this context. It is an “indeterminate” (formal variable) acting as a positional placeholder. The symbols $+$ and x^i in (3.42) are purely typographical markers separating the coefficients $a_i \in R$, rather than evaluations of operations.

III Rigorous Foundation: Polynomials as Almost-Zero Sequences

Spoken Transcript [01:21:26 - 01:22:28]

[The lecturer pushes the upper chalkboard up and begins erasing the lower board] If you look at books where they're really formally correct, they will do some weird things just so you don't get confused. I will explain it for a moment, but I don't usually do that because I think it's a little dishonest.

When I give these lectures, I try to actually give the lectures in the way I think about the subject. There's some kind of underlying honesty in the lecture.

If I were to be dishonest and formally correct, the way you define these polynomials is the following way.

Spoken Transcript [01:22:28 - 01:24:00]

You say a polynomial, an element of $R[x]$, is a tuple of elements of R , $(a_0, a_1, a_2, \dots)$ — an infinite tuple — such that there exists some N such that for all $m > N$, the element in this tuple is zero, okay? *[writes on the board]*

That's a formal definition of a polynomial in one variable, where I just give you the coefficients. The coefficients are $a_0, a_1, a_2, \dots$; they're all elements of R .

I just give you the coefficients, and there could be lots of them — any finite number — but eventually this sequence of coefficients is all zero. That's the formal definition of a polynomial.

It's actually the same thing that I'm saying, but it's much more reasonable to think about it with the x 's. This way is just so that you don't get confused formally.

That's the set. I hope there's no confusion about the set.

Spoken Transcript [01:24:00 - 01:24:24]

In this abstract way, what is the variable x ? *[pause]*

You have five minutes to figure it out.

What's the variable x ? In my notation there's an x ; here there's no x . So, what's x here?

Yeah?

Student Answer

$(0, 1, 0, \dots)$

Spoken Transcript [01:24:27 - 01:24:40]

Yeah. So, x is $(0, 1, 0, 0, \dots)$; that's x if you think about it this way, okay? *[writes $x = (0, 1, 0, \dots)$ on the board]*

Formal Definition of Polynomials as Sequences with Finite Support

 **Definition 3.34 (Rigorous Definition of $R[x]$ via Coefficient Tuples):** Let R be a commutative ring with unit. The polynomial ring $R[x]$ is formally defined as the set of sequences $(a_k)_{k \in \mathbb{N}_0}$ of elements in R with “finite support” (almost-zero sequences):

$$R[x] := \left\{ (a_0, a_1, a_2, \dots) \in R^{\mathbb{N}_0} \mid \exists N \in \mathbb{N}_0 \forall m > N: a_m = 0 \right\}. \quad (3.43)$$

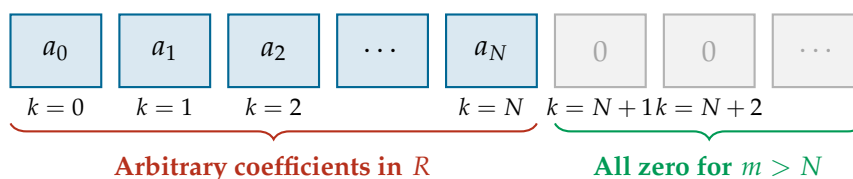
 **Proposition 3.35 (Identification of the Indeterminate x and Constants):** Under the sequence model (3.43):

- (1) **Indeterminate x :** The indeterminate x corresponds to the elementary sequence with 1 in position 1:

$$x := (0, 1, 0, 0, \dots). \quad (3.44)$$

(2) **Constant Elements:** For any $a \in R$, the constant polynomial corresponds to the sequence:

$$a := (a, 0, 0, 0, \dots). \quad (3.45)$$



III Ring Operations on $R[x]$: Addition and the Cauchy Product

Spoken Transcript [01:24:40 - 01:25:20]

That's the set. Once you agree on what the set is — and I gave you two ways to think about it: one way which I think is very simple, and this way which is more formally correct.

I need to give you the addition rule, the multiplication rule, 0, and 1. This is going to be easy: 0 is just the zero polynomial. In my case, it's just the polynomial 0; in this sequence case, you have to write an infinite number of zeros, which takes a long time.

And 1 is just the constant one polynomial; here it's $(1, 0, 0, 0, \dots)$.

Student Question

Is the plus in that definition not the plus from the ring?

Spoken Transcript [01:25:25 - 01:26:28]

It is eventually, but I haven't defined the plus for the ring yet. In this definition, I'm defining $R[x]$ as a set first, and then I'll define addition. When I finish, it will be. But when it's born, it's just a set. Is that clear?

It's a good question, and that's the reason why thinking about it this way is slightly confusing. Although I encourage you to think about it this way, because you have a lot of experience with polynomials, and in the end, nothing will go wrong.

You have to agree on what the set is, and I gave you two different ways to think about the set. Then you have to define addition.

You can define addition in that informal way as the addition of polynomials, which is very good because I don't have to write it out, since you already know how to add polynomials. If I write it here, then I have to decide how to add polynomials, and that's not hard. How do you add polynomials here?

Yeah?

Student Answer

Componentwise addition.

Spoken Transcript [01:26:30 - 01:26:48]

Componentwise addition, which is also how you add polynomials there: componentwise addition. And it will be the case that if I take the constant polynomial and I take this polynomial and add them, I get the thing I call $a_0 + x$, which is also kind of nice. *[writes $a_0 + x$ on the chalkboard]*

All right, addition is easy.

Spoken Transcript [01:26:48 - 01:27:43]

Then we have to define multiplication. Multiplication is more of a pain. If you think about it informally, you already know how to multiply polynomials: you do this thing where you distribute and whatever — you know how to do it.

How do you define polynomial multiplication here? Let's try to do that.

If I use this list of coefficients: here's my first polynomial f , and my second polynomial g ... *[writes $f = (a_0, a_1, a_2, \dots)$ and $g = (b_0, b_1, b_2, \dots)$ on the board]*

Then I have to define this polynomial which is $f \times g$. How do I do it? What's this first coefficient going to be?

What do you think?

Yeah?

Student Answer

a_0b_0 .

Spoken Transcript [01:27:45 - 01:28:54]

Yeah, because it's just the constant term. The secret to doing this is that you write this out as a polynomial. That's the easy way to think about it: write this as a polynomial, do polynomial multiplication, and the constant term is a_0b_0 .

What is this next term? Well, it's going to be $a_1b_0 + a_0b_1$. The second term is going to be $a_2b_0 + a_1b_1 + a_0b_2$. *[writes $fg = (a_0b_0, a_1b_0 + a_0b_1, a_2b_0 + a_1b_1 + a_0b_2, \dots)$ on the chalkboard]*

Formally, this is the definition of polynomial multiplication.

Is that clear?

If you want to be on the formal side: polynomials are these lists of coefficients that eventually become all zero; they contain the 0 and 1 by the elements here and all those zeros; you can add them by componentwise addition; and the interesting thing is multiplying them, where you use this rule.

Student Question

This is essentially like if you do, like, in... what do you call that, the Cauchy product or convolution...

Spoken Transcript [01:29:06 - 01:30:00]

Yeah, you do this thing. That's what you're saying? Yeah, that's what you do: that's how you multiply polynomials. This is the same rule as you'd get here. Look: if you take the linear term, how do you get an x here? You have to take x times a constant; those are the two ways to get the x .

So, you get this rule.

Okay, that's the ring. We're going to talk about it more next time. But yeah, it's a little bit of a subtle matter, and it's good to make sure you completely understand it: given a ring, there's a polynomial ring.

A polynomial is actually just a list of coefficients. It's very helpful to think about the polynomial that way, because that's how you normally think about it, and you won't go wrong thinking about it that way.

Okay, stop. *[applause; video fades out]*

The Ring Structure of $R[x]$

Definition 3.36 (Ring Operations on $R[x]$): Let R be a commutative ring with unit 1_R . The polynomial ring $(R[x], +, \cdot, 0_{R[x]}, 1_{R[x]})$ is equipped with the following structures:

(1) Neutral Elements:

$$0_{R[x]} := (0_R, 0_R, 0_R, \dots), \quad (3.46)$$

$$1_{R[x]} := (1_R, 0_R, 0_R, \dots). \quad (3.47)$$

(2) Addition (Componentwise): For $f = (a_k)_{k \geq 0}$ and $g = (b_k)_{k \geq 0}$:

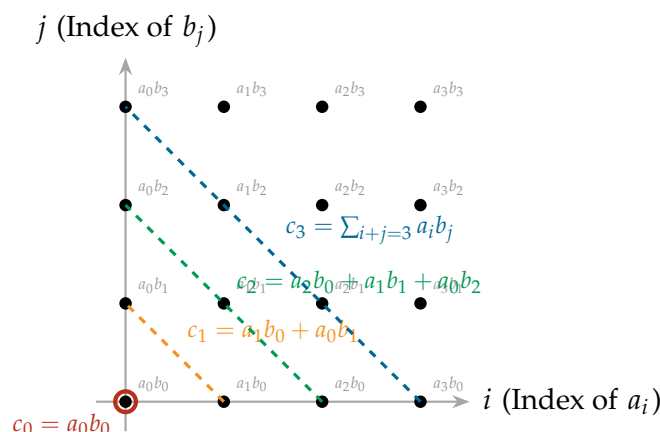
$$(a_0, a_1, a_2, \dots) + (b_0, b_1, b_2, \dots) := (a_0 + b_0, a_1 + b_1, a_2 + b_2, \dots). \quad (3.48)$$

(3) Multiplication (Cauchy Product / Convolution): The product $f \cdot g = (c_k)_{k \geq 0}$ is defined by:

$$c_k := \sum_{i=0}^k a_i b_{k-i} = a_0 b_k + a_1 b_{k-1} + \dots + a_k b_0. \quad (3.49)$$

Explicitly:

$$f \cdot g = (a_0 b_0, a_0 b_1 + a_1 b_0, a_0 b_2 + a_1 b_1 + a_2 b_0, \dots). \quad (3.50)$$



≡ **Connection to the Cauchy Product and Discrete Convolution:** The formula $c_k = \sum_{i=0}^k a_i b_{k-i}$ exactly mirrors:

(1) **Distributive Expansion:** In formal series notation:

$$\left(\sum_{i=0}^n a_i x^i \right) \left(\sum_{j=0}^m b_j x^j \right) = \sum_{i=0}^n \sum_{j=0}^m a_i b_j x^{i+j} = \sum_{k=0}^{n+m} \left(\sum_{i+j=k} a_i b_j \right) x^k.$$

The powers of the indeterminate x naturally collect all terms where the sum of indices equals k .

(2) **Cauchy Product of Power Series:** In analysis, the product of two absolutely convergent power series $\sum a_n x^n$ and $\sum b_n x^n$ is given by the exact same convolution formula.

Summary of Key Concepts

- **The Group of Units $\text{Unit}(R)$:** The set of multiplicatively invertible elements $R^* = \{x \in R \mid \exists y \in R: xy = 1\}$ forms an abelian group under ring multiplication. Key examples include $\mathbb{Z}^* = \{\pm 1\}$, $\mathbb{Z}[i]^* = \{\pm 1, \pm i\}$ (characterized via the norm $N(a + bi) = a^2 + b^2 = 1$), and $(\mathbb{Z}/m\mathbb{Z})^* = \{\bar{x} \mid \gcd(x, m) = 1\}$.
- **Euler's Totient Function $\phi(m)$:** Defined as the order $|(\mathbb{Z}/m\mathbb{Z})^*|$, with explicit values $\phi(p) = p - 1$ and $\phi(p^k) = p^k - p^{k-1} = p^k(1 - 1/p)$ for prime powers.
- **Universal Property of $\mathbb{Q}$:** As the minimal field containing the integral domain $\mathbb{Z}$, any injective ring homomorphism $f: \mathbb{Z} \rightarrow F$ into a field F factors uniquely through an injective field homomorphism $g: \mathbb{Q} \rightarrow F$ defined by $g(x/y) = f(x)f(y)^{-1}$.
- **Field of Fractions $\text{Frac}(R)$:** For any integral domain R , the field of fractions is constructed as the quotient of formal pairs $S = R \times (R \setminus \{0\})$ by the cross-multiplication equivalence relation $(x_1, y_1) \equiv (x_2, y_2) \iff x_1 y_2 = x_2 y_1$. Transitivity and well-definedness of addition and multiplication crucially rely on the zero-divisor-free property of R .
- **Canonical Inclusion and Examples:** The map $i: R \hookrightarrow \text{Frac}(R)$, $r \mapsto [(r, 1)]$, is an injective ring embedding. Examples include $\text{Frac}(\mathbb{Z}) = \mathbb{Q}$, $\text{Frac}(\mathbb{Z}[i]) = \mathbb{Q}(i)$, $\text{Frac}(\mathbb{Z}[\frac{1}{3}]) = \mathbb{Q}$, and $\text{Frac}(K) \cong K$ for any existing field K .

- **Polynomial Rings $R[x]$:** For a commutative ring with unity R , the polynomial ring $R[x]$ is formally defined as the ring of almost-zero sequences $(a_k)_{k \in \mathbb{N}_0}$ in $R^{\mathbb{N}_0}$. It is equipped with componentwise addition and Cauchy product (convolution) multiplication $c_k = \sum_{i=0}^k a_i b_{k-i}$, with indeterminate $x = (0, 1, 0, 0, \dots)$.